MEMO NO: _____
DATE: 1 2 / 26

Chapter 1 Ring Theory

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Lemma A : ring $\quad A=0 \iff 0=1$

$\{n1_A : n\in\mathbb{Z}\}$: smallest subring of A

Prop A: ring $\quad X\subseteq A$ subset $\quad$ Then

$\mathbb{Z}[X] := \{ f(a_1,\cdots,a_n) : f(x_1,\cdots,x_n)\in\mathbb{Z}[x_1,\cdots,x_n]\ a_i\in X,\ n\geq 0\}$

is the smallest subring which contains X (generated by X)

Prop $(X) = \{\sum_{i=1}^{n} a_i x_i : a_i\in A,\ x_i\in X,\ n\geq 0\}$

is the smallest ideal of A which contains X (generated by X)

Proper ideal $\mathfrak{a}\subsetneq A \iff 1\notin\mathfrak{a}$

There is a bijection of sets.

$\{$ideals of A containing $\mathfrak{a}\} \xleftrightarrow{1:1} \{$ideals of $A/\mathfrak{a}\}$

$\mathfrak{b} \longmapsto \mathfrak{b}/\mathfrak{a}$

ring map : ring homomorphism

$A'\subseteq A$ subring $\quad j: A'\to A \quad x\mapsto x$ injective

$\mathfrak{a}\subseteq A$ ideal $\quad \pi: A\to A/\mathfrak{a} \quad x\mapsto \bar{x}$ surjective

Prop $f: A\to B$ map ring Then

(1) $\ker f$ is an ideal of A $\quad f$ injective $\iff \ker f = 0$

(2) $\mathrm{Im} f$ is a subring of B $\quad f$ surjective $\iff \mathrm{Im} f = B$

(3) f induces a canonical isom of rings $\bar{f}: A/\ker f \to \mathrm{Im} f$ $\quad \bar{x}\mapsto f(x)$

and the diagram $A \xrightarrow{f} B$ commutes

$\pi \downarrow \quad \uparrow j$

$A/\ker f \xrightarrow[\cong]{\bar{f}} \mathrm{Im} f$

Special cases f injective

$A \xhookrightarrow{f} B$

$\bar{f} \searrow \ \cong\uparrow j$

$\mathrm{Im} f$

f surjective

$A \xrightarrow{f} B$

$\pi\downarrow \quad \circlearrowleft \quad \nearrow\cong$

$A/\ker f$

Prop $\mathfrak{a} \subseteq \mathfrak{b}$ ideals Then $\frac{A/\mathfrak{a}}{\mathfrak{b}/\mathfrak{a}} \cong \frac{A}{\mathfrak{b}}$

Operations of ideals

(1) sum of $\{\mathfrak{a}_i\}_{i\in A}$: $\sum_{i\in I} \mathfrak{a}_i = \{\sum_{\text{finite sum}} x_i : x_i \in \mathfrak{a}_i\}$

It is the smallest ideal of A which contains all $\mathfrak{a}_i$

NB[1] The union $\bigcup_{i\in I} \mathfrak{a}_i$ is in general not an ideal of A

However if $\forall i,j \in I$ $\mathfrak{a}_i \subseteq \mathfrak{a}_j$ or $\mathfrak{a}_j \subseteq \mathfrak{a}_i$ then $\bigcup_{i\in I} \mathfrak{a}_i = \sum_{i\in I} \mathfrak{a}_i$ is an ideal

(2) intersection $\bigcap_{i\in I} \mathfrak{a}_i$ is an ideal

(3) product of $\mathfrak{a}, \mathfrak{b}$: $\mathfrak{a}\mathfrak{b} = \{\sum_{i=1}^{n} x_i y_i : x_i \in \mathfrak{a}, y_i \in \mathfrak{b}\ n \geq 0\}$

similar $\mathfrak{a}_1 \mathfrak{a}_2 \cdots \mathfrak{a}_m$ generated by $x_1 x_2 \cdots x_m$ $x_i \in \mathfrak{a}_i$

(4) quotient of $\mathfrak{a}, \mathfrak{b}$: $(\mathfrak{a} : \mathfrak{b}) := \{x \in A : x\mathfrak{b} \in \mathfrak{a}\}$

If $\mathfrak{a} = 0$ $(0:\mathfrak{b}) =: \mathrm{Ann}(\mathfrak{b})$ annihilator of $\mathfrak{b}$

If $\mathfrak{b} = (y)$ then we write $(\mathfrak{a} : y)$ in replace of $(\mathfrak{a} : (y))$

annihilator of y. $\mathrm{Ann}(y)$ $\mathrm{Ann}((y))$

Ex $(\mathfrak{a} : \mathfrak{b}) \supseteq \mathfrak{a}$ $(\mathfrak{a} : \mathfrak{b})/\mathfrak{a} = \mathrm{Ann}_{A/\mathfrak{a}}(\mathfrak{a}+\mathfrak{b}/\mathfrak{a})$

eg (1) $\mathfrak{a}_i = (X_i) \Rightarrow \sum_{i\in I} \mathfrak{a}_i = (\bigcup_{i\in I} X_i)$

$(x_1, \cdots, x_m) + (y_1, \cdots, y_n) = (x_1, \cdots, x_m, y_1, \cdots, y_n)$

$\mathfrak{a} = (X)$ $\mathfrak{b} = (Y)$ then $\mathfrak{a}\mathfrak{b} = (XY)$ where $XY = \{xy : x \in X, y \in Y\}$

$(x_1, \cdots, x_m)(y_1, \cdots, y_n) = (x_i y_j : 1 \leq i \leq m, 1 \leq j \leq n)$

$m_1\mathbb{Z} + \cdots + m_k\mathbb{Z} = \gcd(m_1, \cdots, m_k)\mathbb{Z}$ $m_1\mathbb{Z} \cap \cdots \cap m_k\mathbb{Z} = \mathrm{lcm}(m_1, \cdots, m_k)\mathbb{Z}$

$m_1\mathbb{Z} \cdots m_k\mathbb{Z} = m_1 \cdots m_k \mathbb{Z}$ $(m\mathbb{Z} : n\mathbb{Z}) = \frac{m}{\gcd(m,n)}\mathbb{Z}$

Def $\mathfrak{a}, \mathfrak{b}$ coprime : $\mathfrak{a} + \mathfrak{b} = (1) \iff \exists\, x \in \mathfrak{a}\ y \in \mathfrak{b}\ \ x + y = 1$

Prop $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b}$ equality holds iff $\mathfrak{a}, \mathfrak{b}$ are coprime

$(\mathfrak{a} + \mathfrak{b}) + \mathfrak{c} = \mathfrak{a} + (\mathfrak{b} + \mathfrak{c})$ $\mathfrak{a} + \mathfrak{b} = \mathfrak{b} + \mathfrak{a}$

$\mathfrak{a}(\mathfrak{b} + \mathfrak{c}) = \mathfrak{a}\mathfrak{b} + \mathfrak{a}\mathfrak{c}$

$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) \supseteq \mathfrak{a}\cap\mathfrak{b} + \mathfrak{a}\cap\mathfrak{c}$

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MEMO NO: _____
DATE: / 2 / 28

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Def $a \in A$ — a is a unit or invertible if $\exists\, b \in A$ s.t. $ab = 1$ $\boxed{A^\times}$

a is a zero-divisor if $\exists\, b \in A$ $b \neq 0$ s.t. $ab = 0$ $\boxed{\mathrm{Zdiv}(A)}$

non-zero-divisor (opposite) $\Leftrightarrow$ $a: A \to A$ $x \mapsto ax$ injective

a is nilpotent if $\exists\, n \geq 1$ s.t. $a^n = 0$ $\boxed{\mathrm{nil}(A)}$

Prop (1) $(A^\times, \cdot)$ is a commutative gp

(2) $\mathrm{nil}(A)$ is an ideal of A — nilradical of A

(3) $ab \in \mathrm{Zdiv}(A) \Leftrightarrow a \in \mathrm{Zdiv}(A)$ or $b \in \mathrm{Zdiv}(A)$

(4) units are non-zero-divisors

(5) $\mathrm{nil}(A) \subseteq \mathrm{Zdiv}(A)$ unless $A = 0$

eg. $\mathbb{Z}^\times = \{\pm 1\}$ $\mathrm{nil}(\mathbb{Z}) = 0$ $\mathrm{Zdiv}(\mathbb{Z}) = \{0\}$

$n \geq 2$ $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{m} : (m,n) = 1\}$ $\mathrm{nil}(\mathbb{Z}/n\mathbb{Z}) = \{\bar{m} : m$ is divisible by every prime divisor of $n\}$

$\mathrm{Zdiv}(\mathbb{Z}/n\mathbb{Z}) = \{\bar{m} : (m,n) \neq 1\}$

Def A is reduced if $\mathrm{nil}(A) = 0$

integral domain : a nonzero ring with no nonzero zero-divisors

field : a nonzero ring in which every nonzero ele is invertible

$\Leftrightarrow$ $A \neq 0$ and A has no ~~non-tra~~ nontrivial ideals

{fields} $\subsetneq$ {integral domains} $\subsetneq$ {reduced rings}

eg $\mathbb{Z}/n\mathbb{Z}$ is reduced $\Leftrightarrow$ n is square-free

$\mathbb{Z}/n\mathbb{Z}$ is an integral domain $\Leftrightarrow$ $\mathbb{Z}/n\mathbb{Z}$ is a field $\Leftrightarrow$ n is prime

Ex A proper ideal is prime iff $\alpha\beta \subseteq \mathfrak{P} \Rightarrow \alpha \subseteq \mathfrak{P}$ or $\beta \subseteq \mathfrak{P}$

Prop Let A be a ring, $\alpha, \mathfrak{P}, \mathfrak{m}$ be ideals. Then

(1) $\mathfrak{P}$ is a prime ideal $\Leftrightarrow$ $A/\mathfrak{P}$ is an integral domain

(2) $\mathfrak{m}$ is a maximal ideal $\Leftrightarrow$ $A/\mathfrak{m}$ is a field

(3) maximal ideals are prime ideals

(4) Under the 1:1 correspondence {ideals of A containing α} $\overset{1:1}{\longleftrightarrow}$

{ideals of $A/\mathfrak{a}$} prime (maximal) ideals containing $\mathfrak{a}$ correspond to prime (maximal) ideals of $A/\mathfrak{a}$

Prop Let A be a PID but not a field. Then

(1) The prime ideals of A are 0, (p) where p is a prime elt

(2) maximal (p)

e.g. the prime elts in $\mathbb{Z}[i]$ are $\pm 1 \pm i$, p ($p \equiv 3 \pmod 4$), $\pm m \pm ni$ ($m^2+n^2 = p \equiv 1 \pmod$

Thm Every nonzero ring has a maximal ideal, hence a prime ideal.

Use Zorn's Lemma

Cor Every proper ideal of $\mathfrak{a}$ of A is contained in a maximal ideal.

pf Apply Thm to $A/\mathfrak{a}$ ($\neq 0$)

Cor Every non-unit of A is contained in a maximal ideal.

pf. $(x) \subsetneq A$

Hence $A - A^\times = \bigcup_{\mathfrak{m} \text{ maximal}} \mathfrak{m}$

Prop $\mathrm{nil}(A/\mathrm{nil}(A)) = 0$ i.e. $A/\mathrm{nil}(A)$ is reduced.

Thm The nilradical of A is the intersection of all prime ideals of A

i.e. $\mathrm{nil}(A) = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$

pf. $\mathrm{nil}(A) \subseteq \bigcap_{\mathfrak{p}} \mathfrak{p}$ triv.

$\mathrm{nil}(A) \supseteq \bigcap_{\mathfrak{p}} \mathfrak{p}$: For x is not nilpotent, we shall find prime $\mathfrak{p} \not\ni x$

Let $S = \{x^i : i \geq 0\}$ $\Sigma = \{\mathfrak{a} : \mathfrak{a} \text{ ideal of } A \text{ and } \mathfrak{a} \cap S = \phi\}$

Order Σ by inclusion. x not nilpotent $\Leftrightarrow 0 \notin S \Rightarrow 0 \in \Sigma \Rightarrow \Sigma \neq \phi$

If $\{\mathfrak{a}_i\}_{i \in I}$ is a chain in Σ then clearly $\bigcup_{i \in I} \mathfrak{a}_i$ is an upper bound of the chain in Σ. By Zorn's lemma Σ has a maximal elt $\mathfrak{p}$

It remains to show $\mathfrak{p}$ is prime. Obviously $1 \notin \mathfrak{p} \Rightarrow \mathfrak{p} \subsetneq A$ ($1 \in S \Rightarrow$)

If $\exists\, a, b \notin \mathfrak{p}$ s.t. $ab \in \mathfrak{p} \Rightarrow \mathfrak{p} \subsetneq (\mathfrak{p}, a)$ $\mathfrak{p} \subsetneq (\mathfrak{p}, b)$

then by the maximality of $\mathfrak{p}$ $\exists\, i \geq 0$ s.t. $x^i \in \mathfrak{p}+(a)$ $\exists\, j \geq 0$ s.t. $x^j \in \mathfrak{p}+(b)$

Hence $x^{i+j} \in (\mathfrak{p}+(a))(\mathfrak{p}+(b)) \subseteq \mathfrak{p}+(ab) = \mathfrak{p}$. contradicting $\mathfrak{p} \in \Sigma$.

Def Let $\mathfrak{a}$ be an ideal of A. The radical of $\mathfrak{a}$ is defined to be

$$\sqrt{\mathfrak{a}} = r(\mathfrak{a}) := \{x \in A : \exists\, n \geq 1,\ x^n \in \mathfrak{a}\}$$

e.g. $\sqrt{0} = \mathrm{nil}(A)$. $\sqrt{\mathfrak{a}}$ is an ideal containing $\mathfrak{a}$ and $\sqrt{\mathfrak{a}}/\mathfrak{a} = \mathrm{nil}(A/\mathfrak{a})$

An ideal $\mathfrak{a}$ is a ~~rad~~ radical ideal if $\sqrt{\mathfrak{a}} = \mathfrak{a}$. or equiv. $A/\mathfrak{a}$ is reduced

Prop (1) $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$

(2) If $\mathfrak{p}$ is a prime ideal and $n \geq 1$, then $\sqrt{\mathfrak{p}^n} = \mathfrak{p}$

(prime ideals are radical ideals)

pf. (2) First note that $\sqrt{\mathfrak{p}} = \mathfrak{p}$. We have $\mathfrak{p} \subseteq \sqrt{\mathfrak{p}^n} \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p} \Rightarrow \sqrt{\mathfrak{p}^n} = \mathfrak{p}$

Cor Let $\mathfrak{a}$ be an ideal of A. Then $\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{a} \subseteq \mathfrak{p} \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}$

pf. Apply them to $A/\mathfrak{a}$

$$\sqrt{\mathfrak{a}}/\mathfrak{a} = \mathrm{nil}(A/\mathfrak{a}) = \bigcap_{\substack{\mathfrak{a} \subseteq \mathfrak{p} \\ \mathfrak{p} \text{ prime}}} (\mathfrak{p}/\mathfrak{a}) = \Big(\bigcap_{\substack{\mathfrak{a} \subseteq \mathfrak{p} \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}\Big)/\mathfrak{a} \Rightarrow \sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{a} \subseteq \mathfrak{p} \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}$$

Def The Jacobson radical of A denoted by $\mathrm{rad}(A)$ is the intersection of all the maximal ideals in A. That is $\mathrm{rad}(A) = \bigcap_{\mathfrak{m} \text{ maximal}} \mathfrak{m}$

Lemma Let $\mathfrak{a}$ be an ideal of A. Then TFAE

(1) $\mathfrak{a} \subseteq \mathrm{rad}(A)$

(2) $\forall x \in \mathfrak{a}$ $1-x$ is a unit

pf (1) $\Rightarrow$ (2) If $1-x$ is not a unit, then $1-x \in \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$

But $x \in \mathfrak{a} \subseteq \mathrm{rad}(A) \subseteq \mathfrak{m} \Rightarrow 1 \in \mathfrak{m}$ absurd!

(2) $\Rightarrow$ (1) If $\exists$ maximal ideal $\mathfrak{m}$ s.t. $\mathfrak{a} \nsubseteq \mathfrak{m}$ then $\mathfrak{a} + \mathfrak{m} = (1)$

$\Rightarrow \exists\, x \in \mathfrak{a}$ $y \in \mathfrak{m}$ s.t. $x + y = 1 \Rightarrow y = 1 - x$ is a unit by (2) contradicting $y \in \mathfrak{m}$

Prop Let A be a ring then $\mathrm{rad}(A) = \{x \in A : 1 - xy \text{ is invertible } \forall\, y \in A\}$

pf. $x \in \mathrm{rad}(A) \Leftrightarrow (x) \subseteq \mathrm{rad}(A)$

$\Leftrightarrow \forall\, y \in A$ $1 - xy$ is a unit

NB generally, $\mathrm{rad}(A) \neq \{x \in A : 1 - x \text{ is a unit}\}$

Prop $\mathrm{nil}(A) \subseteq \mathrm{rad}(A)$

Direct product of A, B : $A \times B$

Set $R = A \times B$ $\mathfrak{a} = \{(a,0) : a \in A\} = A \times 0$ (not subring of R)

$\mathfrak{b} = \{(0,b) : b \in B\} = 0 \times B$

Then $\mathfrak{a}$ $\mathfrak{b}$ are ideals of R and $R = \mathfrak{a} \oplus \mathfrak{b}$

$\pi_1 : A \times B \to A$; $(a,b) \mapsto a$ $\pi_2 : A \times B \to B$; $(a,b) \mapsto b$

are surjective ring homomorphisms and $\ker \pi_1 = \mathfrak{b}$ $\ker \pi_2 = \mathfrak{a}$

Thus $A \times B / \mathfrak{b} \cong A$ $A \times B / \mathfrak{a} \cong B$

Prop Let R be a ring. Suppose $\mathfrak{a}, \mathfrak{b}$ are ideals of R s.t. $R = \mathfrak{a} \oplus \mathfrak{b}$

Then (1) $\exists!$ $e \in \mathfrak{a}$ $f \in \mathfrak{b}$ s.t. $e + f = 1$

(2) e, f also satisfy $e^2 = e$, $f^2 = f$, $ef = 0$

(3) $\mathfrak{a} = Re$ $\mathfrak{b} = Rf$

(4) $\mathfrak{a}$ is a ring under the addition and mult of R

with identity ett e

$\mathfrak{b}$ --- ett f

(5) $\mathfrak{a} \times \mathfrak{b} \to R$; $(a,b) \mapsto a+b$ is a ring isomorphism

Def Let R be a ring. An ett $e \in R$ is an idempotent if $e^2 = e$

($\Rightarrow 1-e$ is an idempotent)

A complete set of orthogonal idempotents is a set $\{e_1, \cdots, e_n\} \subseteq R$

s.t. 1) $e_i^2 = e_i$ $\forall i$ 2) $e_i e_j = 0$ $\forall i \neq j$ 3) $1 = e_1 + \cdots + e_n$.

Prop If $\{e_1, \cdots, e_n\}$ is a complete set of orthogonal idempotents in a ring R then (1) $R = Re_1 \oplus \cdots \oplus Re_n$

with identity ett e_i

(2) Re_i is a ring under the addition and mult of R

(3) $R \cong Re_1 \times \cdots \times Re_n$ as rings

Thm (CRT) Let A be a ring. $\mathfrak{a}_1, \cdots, \mathfrak{a}_n$ be ideals of A that are pairwise coprime. Then $\bigcap_{i=1}^{n} \mathfrak{a}_i = \prod_{i=1}^{n} \mathfrak{a}_i$ and $A/\mathfrak{a}_1 \cdots \mathfrak{a}_n \cong A/\mathfrak{a}_1 \times \cdots \times A/\mathfrak{a}_n$

In particular if A is a PID $x = p_1^{e_1} \cdots p_r^{e_r}$ where p_i are distinct ett (up to units)

prime elts. $e_i \geq 1$ then $A/(p_1^{e_1}\cdots p_r^{e_r}) \simeq A/(p_1^{e_1}) \times \cdots \times A/(p_r^{e_r})$

$A/(p_i^{e_i})$ has a unique prime ideal $(p_i)/(p_i^{e_i})$

$\Rightarrow A/(p_1^{e_1}\cdots p_r^{e_r})$ has r prime ideals $\prod_{j\neq i} A/(p_j^{e_j}) \times (p_i)/(p_i^{e_i})$

e.g. Find all the prime elts in $\mathbb{Z}[i]$

Local rings

Def. A ring A with exactly one maximal ideal $\mathfrak{m}$ is called a local ring. $A/\mathfrak{m}$ is called the residue field of A. We usually denote a local ring by $(A, \mathfrak{m})$

Lemma (1) If $(A,\mathfrak{m})$ is a local ring then $\mathrm{rad}(A)=\mathfrak{m}$, $A = \mathfrak{m} \amalg A^\times$ and $\forall x \in \mathfrak{m}$ $1-x$ is invertible

(2) A ring A is local iff $A - A^\times$ is an ideal

pf (2): $\Rightarrow$ If $\mathfrak{m} := A - A^\times$ is an ideal, then $\mathfrak{m}$ contains every proper ideal of A, so it must be the unique maximal ideal of A

e.g. (1) A field is a local ring with maximal ideal 0

(2) $X = \{(U,f) : U$ is an open nbhd of 0 in $\mathbb{R}$, $f: U \to \mathbb{R}$ continuous$\}$

$(U,f) \sim (V,g) \Leftrightarrow \exists$ open nbhd W of 0 s.t. $0 \in W \subseteq U \cap V$ and $f|_W = g|_W$

Set $A = X/\sim$; the equivalence class of (U,f) by $[(U,f)]$ (germ)

Claim. A is a local ring

$[(U,f)]$ is invertible in $A \Leftrightarrow f(0) \neq 0$

Actually, $[(U,f)]^{-1} = [((-\varepsilon,\varepsilon), \frac{1}{f})]$ where $\varepsilon > 0$ is small enough so that $f(x) \neq 0 \ \forall x \in (-\varepsilon,\varepsilon)$

$A - A^\times = \{[(U,f)] : f(0) = 0\}$ is an ideal $\Rightarrow A$ is local

(3) $A = \{\frac{f(x)}{g(x)} : f(x), g(x) \in k[x], g(0) \neq 0\}$ (k field)

$A^\times = \{\frac{f(x)}{g(x)} : f(x), g(x) \in k[x], x \nmid f(x), x \nmid g(x)\}$ ($\Leftrightarrow x \nmid g(x)$)

$A - A^\times = (x)$ is an ideal

$\Rightarrow A$ is local

(4) $A=\{\frac{m}{n}: m,n\in\mathbb{Z}\ 2\nmid n\}$ $A^\times=\{\frac{m}{n}: m,n\in\mathbb{Z}\ 2\nmid m, 2\nmid n\}$

$A-A^\times=(2)\Rightarrow A$ is local

(5) (rings with a unique prime ideal)

Let A be a ring $\mathfrak{m}$ a maximal ideal of A $i\geq 1$

Then $A/\mathfrak{m}^i$ has a unique prime ideal $\mathfrak{m}/\mathfrak{m}^i$ hence local

($\mathfrak{m}^i\subseteq\mathfrak{p}\Rightarrow\mathfrak{m}=\sqrt{\mathfrak{m}^i}\subseteq\sqrt{\mathfrak{p}}=\mathfrak{p}\Rightarrow\mathfrak{m}=\mathfrak{p}$)

such as $(\mathbb{Z}/p^n\mathbb{Z}, p\mathbb{Z}/p^n\mathbb{Z})$ $(k[x]/(p^{e(x)}), (p(x))/(p^{e(x)}))$ $p(x)$ irred over k $e\geq 1$

(6) If A is a local ring, then so is $A[[x]]$

Extensions and Contractions

Def Let $f:A\to B$ be a ring map $\mathfrak{a}$, $\mathfrak{b}$ be ideals of A, B respectively.

Then extension of $\mathfrak{a}$ is the ideal generated by $f(\mathfrak{a})$, usually denoted by $f(\mathfrak{a})B$, $\mathfrak{a}B$, $\mathfrak{a}^e$. It consists of elts of the form $\sum_{i=1}^{n} f(x_i)b_i$ where $n\geq 0$, $x_i\in\mathfrak{a}$, $b_i\in\mathfrak{b}$

The contraction of $\mathfrak{b}$ is $f^{-1}(\mathfrak{b})=\{x\in A: f(x)\in\mathfrak{b}\}$, which is an ideal of A and denoted by $\mathfrak{b}^c$, $\mathfrak{b}\cap A$

Lemma If $\mathfrak{a}$ is generated by $\{x_i\}_{i\in I}$ as an ideal of A, then $\mathfrak{a}^e$ is generated by $\{f(x_i)\}_{i\in I}$

Lemma The contraction of a prime ideal is prime.

pf. $\mathfrak{q}\subsetneq B$ prime $\mathfrak{p}:=f^{-1}(\mathfrak{q})$ $f\rightsquigarrow\bar{f}: A/\mathfrak{p}\hookrightarrow B/\mathfrak{q}$ $\Rightarrow A/\mathfrak{p}$ integral domain

$B/\mathfrak{q}$ integral domain $\Rightarrow\mathfrak{p}$ prime

Rmk The extension of a prime ideal need not to be prime

e.g. $\mathbb{Z}\subseteq\mathbb{Z}[i]$ $5\mathbb{Z}$ is prime in $\mathbb{Z}$ $5\mathbb{Z}[i]$ is not prime in $\mathbb{Z}[i]$

Def Let $f:A\to B$ be a ring map. An ideal $\mathfrak{a}$ of A is a contracted ideal if $\exists$ an ideal $\mathfrak{b}$ of B s.t. $\mathfrak{a}=\mathfrak{b}^c$

An ideal $\mathfrak{b}$ of B is an extended ideal if $\mathfrak{b}=\mathfrak{a}^e$ for some ideal $\mathfrak{a}$ of A

Prop Let $f:A\to B$ be a ring map. $\mathfrak{a}$ an ideal of A $\mathfrak{b}$ an ideal of B

Then (1) $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$ $\mathfrak{b} \supseteq \mathfrak{b}^{ce}$

(2) $\mathfrak{a}^e = \mathfrak{a}^{ece}$ $\mathfrak{b}^c = \mathfrak{b}^{cec}$

(3) If C is the set of contracted ideals of in A and E is the set of extended ideals in B then

$C = \{\mathfrak{a} : \mathfrak{a} = \mathfrak{a}^{ec}\}$

$E = \{\mathfrak{b} : \mathfrak{b} = \mathfrak{b}^{ce}\}$ and $\mathfrak{a} \mapsto \mathfrak{a}^e$ gives a bijective map from C to E whose inverse is $\mathfrak{b} \mapsto \mathfrak{b}^c$

e.g. Let A be a ring $\mathfrak{a}$ an ideal of A; then the extension of $\mathfrak{a}$ in $A[x]$ is

$\mathfrak{a}[x] = \{\sum_{i=0}^{n} a_i x^i : a_i \in \mathfrak{a}, n \geq 0\}$

$\mathfrak{a}[x] \cap A = \mathfrak{a} \Rightarrow$ every ideal $\mathfrak{a}$ of A is the contraction of an ideal in $A[x]$

$(\mathfrak{a}, x) = \{\sum_{i=0}^{n} a_i x^i : a_0 \in \mathfrak{a}, n \geq 0\} \supseteq \mathfrak{a}[x]$ $\mathfrak{a} = \mathfrak{a}[x] \cap A = (\mathfrak{a}, x) \cap A$

$A[x]/\mathfrak{a}[x] \simeq (A/\mathfrak{a})[x]$ $A[x]/(\mathfrak{a}, x) \simeq A/\mathfrak{a}$

$\mathfrak{p}$ prime $\Rightarrow A/\mathfrak{p}$ integral domain $\Rightarrow (A/\mathfrak{p})[x]$ integral domain

$\Rightarrow \mathfrak{p}[x]$, $(\mathfrak{p}, x)$ prime in $A[x]$ and $\mathfrak{p} = \mathfrak{p}[x] \cap A = (\mathfrak{p}, x) \cap A$

NB the extension of $\mathfrak{a}$ in $A[[x]]$ is not $\mathfrak{a}[[x]]$

& $\mathfrak{a}^e \subseteq \mathfrak{a}[[x]]$ equivality holds if $\mathfrak{a}$ is a finitely generated.

Chapter 2 Module Theory

§1. modules and module homomorphisms

Def Let A be a ring. An **A-module** is an abelian group $(M, +)$ equipped with a map $\mu : A \times M \to M$; $(a, x) \mapsto ax$ which satisfies the following axioms.

(1) $a(x+y) = ax + ay$

(2) $(a+b)x = ax + bx$

(3) $a(bx) = (ab)x$

(4) $1x = x$ $\quad \forall a, b \in A,\ x, y \in M$

$\mu: A\times M\to M \rightsquigarrow \rho: A\to \mathrm{Map}(M,M)$

$a \mapsto (\mu(a,-): M\to M,\ x\mapsto \mu(a,x))$

(1) $\Leftrightarrow$ $\mu(a,-): M\to M$ is a homomorphism of abelian gps

$(M,+)$ abelian gp $\mathrm{End}_{\mathbb{Z}}(M)$: set of endomorphisms of M

Then $\mathrm{End}_{\mathbb{Z}}(M)$ is a (not necessarily commutative) ring with addition and mult given by $(f+g)(x)=f(x)+g(x)$ $(fg)(x)=f(g(x))$

The zero ett is $0: M\to M,\ x\mapsto 0$

The identity ett is $\mathrm{id}: M\to M,\ x\mapsto x$

i.e. (1) $\Leftrightarrow$ $\mu(a,-): M\to M \in \mathrm{End}_{\mathbb{Z}}(M)$

$\rho: A\to \mathrm{End}_{\mathbb{Z}}(M),\ a\mapsto \mu(a,-)$

(2) $\Leftrightarrow$ $\mu(a+b,-)=\mu(a,-)+\mu(b,-) \Leftrightarrow \rho(a+b)=\rho(a)+\rho(b)$

(3) $\Leftrightarrow$ $\mu(a,-)\mu(b,-)=\mu(ab,-) \Leftrightarrow \rho(a)\rho(b)=\rho(ab)$

(4) $\Leftrightarrow$ $\mu(1,-)=\mathrm{id}_M \Leftrightarrow \rho(1)=1$

Def (Equiv) An **A-module** is an abelian gp $(M,+)$ together with a ring homomorphism $\rho: A\to \mathrm{End}_{\mathbb{Z}}(M)$

Rmk $a0=0$ $\forall a\in A$

$0x=0$ $\forall x\in M$

$(-a)x=-ax$ $\forall a\in A, x\in M$

Lemma (restriction of scalars) If $f: A\to B$ is a ring map and N is a B-module, then N is naturally an A-module via $ay=f(a)y$ for $a\in A$ $y\in N$

pf $A\xrightarrow{f} B\xrightarrow{\rho} \mathrm{End}_{\mathbb{Z}}(N)$

e.g. (1) Let A be a ring. Then A is an A-module via the mult map $m: A\times A\to A$

(2) $A=\mathbb{Z}$ $\mathbb{Z}$-modules $=$ abelian gps

(3) $A=k$ a field k-modules $=$ k-vector spaces

(4) k field $k[x]$-modules $=$ k vector space with a linear transformation

If V is a $k[x]$-module, then V is a k-module, hence a k vector space

Since $ax = xa$ in $k[x]$ for all $a \in k$, $x: V \to V$, $v \mapsto xv$ is a k-linear transf.

Conversely if V is a k-vector space endowed with a k-linear transf. φ.

then V is a $k[x]$-module via $f(x)v = f(\varphi)v$ $\forall f(x) \in k[x]$ $v \in V$

Def. Let M, N be A-module. A map $f: M \to N$ is a homomorphism of A-module or A-linear map if (1) $f(x+y) = f(x) + f(y)$ $\forall x, y \in M$ ($\Leftrightarrow$ f is a homo of abelian gps)

(2) $f(ax) = af(x)$ $\forall x \in M$, $a \in A$

Equivalently $f(ax+by) = af(x) + bf(y)$ $\forall a, b \in A$ $x, y \in M$

We denote the set of all A-module homos from M to N by $\mathrm{Hom}_A(M,N)$

$\mathrm{End}_A(M) := \mathrm{Hom}_A(M,M)$

e.g. (1) k-module homo = k-linear map (k field)

(2) $\mathbb{Z}$-module homo = homo of abelian gps

(3) $V_\varphi = (V, \varphi)$ $W_\psi = (W, \psi)$ ~~$k=[x]$~~ $k[x]$-modules

$\mathrm{Hom}_{k[x]}(V,W) = \{f \in \mathrm{Hom}_k(V,W) : f\varphi = \psi f\}$

Recall V, W k-vector spaces $\mathrm{Hom}_k(V,W)$ is a k-vector space

M, N abelian gps $\mathrm{Hom}_{\mathbb{Z}}(M,N)$ is an abelian gp

Prop Let A be a ring. M, N be A-modules. Then $\mathrm{Hom}_A(M,N)$ is an A module with addition and scalar mult given as follows.

$(f+g)(x) = f(x) + g(x)$

$(af)(x) = af(x)$ $\forall f, g \in \mathrm{Hom}_A(M,N)$, $x \in M$, $a \in A$

Pf. $\mathrm{Hom}_A(M,N) \subseteq \mathrm{Hom}_{\mathbb{Z}}(M,N)$

Verify 1) $\mathrm{Hom}_A(M,N)$ is an abelian ~~subg~~ subgp of $\mathrm{Hom}_{\mathbb{Z}}(M,N)$

2) $\forall a \in A$ $f \in \mathrm{Hom}_A(M,N)$ $af \in \mathrm{Hom}_A(M,N)$

$(af)(bx) = af(bx) = abf(x) = baf(x) = b(af)(x)$

3) $\mathrm{Hom}_A(M,N)$ is an A-module

The composition of A-mod homos is again an A-mod homo

$\Rightarrow$ $\mathrm{Hom}_A(M_1,M_2) \times \mathrm{Hom}_A(M_2,M_3) \longrightarrow \mathrm{Hom}_A(M_1,M_3)$

$(f, g) \longmapsto gf$

The above map is A-bilinear i.e. $\forall f$ $g \mapsto gf$ is A-linear, $\forall g$ $f \mapsto gf$ is A-linear

Prop Let M be an A-module. Let ~~$\rho_A \to \text{End}_{\mathbb{Z}}$~~ $\rho: A \to \text{End}_{\mathbb{Z}}(M)$ be the ring homo which defines the A-mod structure of M. Then

(1) ~~$\text{End}_{\mathbb{Z}}(M)$~~ is the centralizer of $\text{Im}\,\rho$ in $\text{End}_{\mathbb{Z}}(M)$ $\text{End}_A(M)$. In particular, $\text{End}_A(M)$ is a (not necessarily commutative) ring.

(2) $\text{Im}\,\rho \subseteq Z(\text{End}_A(M)) \subseteq \text{End}_A(M)$ so ρ induces a ring homo $\rho': A \to \text{End}_A(M)$ and $\text{Im}\,\rho'$ is contained in the center of $\text{End}_A(M)$

(3) The A-module structure of $\text{End}_A(M)$ is induced by ρ' in the sense that $af = \rho'(a)f$ for $f \in \text{End}_A(M)$, $a \in A$
(↑ scalar mult, ↖ mult in $\text{End}_A(M)$)

pf (1) $\text{End}_A(M) = \{f \in \text{End}_{\mathbb{Z}}(M) : f(ax) = af(x)\ \forall a \in A, x \in M\}$
$= \{f \in \text{End}_{\mathbb{Z}}(M) : f\rho(a) = \rho(a)f\ \forall a \in A\}$
= centralizer of $\text{Im}\,\rho$ in $\text{End}_{\mathbb{Z}}(M)$

(2) A commutative $\Rightarrow \text{Im}\,\rho$ comm $\Rightarrow \text{Im}\,\rho \subseteq Z_{\text{End}_{\mathbb{Z}}(M)}(\text{Im}\,\rho)$ ~~$= Z\text{End}_A(M)$~~
$\text{Im}\,\rho$ commutes with $\text{End}_A(M) \Rightarrow \text{Im}\,\rho \subseteq Z(\text{End}_A(M))$

Lemma Let $\rho: A \to B$ be a ring homo ~~from~~ from a comm ring A to a not necessarily comm ring B. Then there is a bijection between the set of ring homos $P: A[x] \to B$ ~~with~~ which extends ρ and the centralizer of $\rho(A)$ in B, given by $P \mapsto P(x)$

pf If $P: A[x] \to B$ is a ring homo s.t. $P|_A = \rho$ Then $\forall a \in A$

$\rho: A \to B$, $A \to A[x] \to B$

$ax = xa$ in $A[x] \Rightarrow \rho(a)P(x) = P(x)\rho(a)$ in B
i.e. $\rho(a)P(x) = P(x)\rho(a)$ in $B \Rightarrow P(x) \in Z_B(\rho(A))$

Conversely, if $b \in B$ satisfies $\rho(a)b = b\rho(a)$ $\forall a \in A$ then $P: A[x] \to B$
$\sum_{i=0}^{n} a_i x^i \mapsto \sum_{i=0}^{n} \rho(a_i) b^i$ is a ring homo extending ρ.
Verify that the two processes are inverse to each other

MEMO NO: _____
DATE: / /

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Prop Let A be a ring, x an indeterminate. Then the data of an $A[x]$-module is equivalent to that of an A-module equipped with an A-linear endomorphism. Explicitly, if M is an $A[x]$-module, then M is an A-module by restrictions of scalars and $x: M \to M$ is A-linear. Conversely, if M is an A-module equipped with an A-linear endomorphism φ then M is an $A[x]$-module via

$$f(x)\,m = f(\varphi)\,m \quad \forall f(x) \in A[x],\ m \in M$$

pf. $\rho: A \longrightarrow \mathrm{End}_{\mathbb{Z}}(M)$ given ρ to give an extension of ρ to $A[x]$

$A \downarrow \nearrow A[x]$ is equivalent to give a $\varphi \in Z_{\mathrm{End}_{\mathbb{Z}}(M)}(\mathrm{Im}\,\rho) = \mathrm{End}_A(M)$

e.g. Let M be an A-module. Then there is a canonical isom of A-modules

$$\mathrm{Hom}_A(A, M) \simeq M;\ f \mapsto f(1) \quad (a \mapsto ax) \hookleftarrow x$$

§2. Submodules and quotient modules

Def Let M be an A-module. An <u>A-submodule</u> of M is an abelian subgp $N \subseteq M$ s.t. $\forall a \in A, y \in N, ay \in N$

Prop Let N be an A-submodule of M. Then

(1) N is an A-module and the inclusion $j: N \to M;\ y \mapsto y$ is an injective A-module homo

(2) The additive quotient gp $M/N = \{\bar{x} = x + N : x \in M\}$ is an A-module via $a\bar{x} = \overline{ax}$ for $a \in A$ $\bar{x} \in M/N$. We call M/N the <u>quotient module</u>

(3) $\Pi: M \to M/N$ i $x \mapsto \bar{x}$ is a surjective A-mod homo

(4) There is a 1-1 correspondence between the submodules of M containing N and the submodules of M/N, given by $L \to L/N$

e.g. (1) $0, M$ are the <u>trivial submodules</u> of M

(2) $\mathbb{Z}$-submodule of an abelian gp = abelian subgp ~~quotient $\mathbb{Z}$-module~~ quotient $\mathbb{Z}$-module = quotient abelian gp

(3) k-submodule of a k-vector space = k-linear subspace

quotient k-module = quotient k-vector space

(4) $k[x]$-submodule of V_φ = φ invariant k-linear subspace of V

If U is an φ-invariant subspace of V, then the quotient $k[x]$-mod is $(V/U, \bar{\varphi})$ where $\bar{\varphi}: V/U \to V/U$ $\bar{v} \mapsto \overline{\varphi(v)}$

(5) A-submodules of A = ideals of A

Let $\mathfrak{a}$ be an ideal of A. There are a priori two possible A-mod structures of on $A/\mathfrak{a}$: One comes from realizing $A/\mathfrak{a}$ as the quotient of the A-module A by the submodule $\mathfrak{a}$. The other is induced by restriction of scalars of the $A/\mathfrak{a}$-module $A/\mathfrak{a}$ via the ring map $A \to A/\mathfrak{a}$. It turns out that these two A-mod structures coincide.

$$f: M \longrightarrow N$$
$$\downarrow \quad \tilde{f} \quad \uparrow$$
$$M/\ker f \xrightarrow{\cong} \operatorname{Im} f$$

Prop Let $N \subseteq L \subseteq M$ be A-modules, Then $(M/N)/(L/N) \cong M/L$

pf. $M/N \xrightarrow{\pi} M/L$; $\bar{x} \mapsto \bar{x}$ $\ker \pi = L/N$

Let $N \subseteq M$ be a submodule Then $\pi: M \to M/N$ has the following universal property:

$\forall$ A-linear map $f: M \to P$ s.t. $f|_N = 0$ $\exists!$ A-linear map $f': M/N \to P$ s.t. $f = f'\pi$

$$\begin{array}{ccc} M & \xrightarrow{\pi} & M/N \\ {\scriptstyle f}\downarrow & \circlearrowright & \swarrow {\scriptstyle \exists! f'} \\ P & & \end{array}$$

Prop Let $N \subseteq M$ be a submodule $\pi: M \to M/N$ quotient map

Then for any A-module P there is an injective A-linear map

$$\operatorname{Hom}_A(M/N, P) \longrightarrow \operatorname{Hom}_A(M, P) \; ; \; g \mapsto g\pi$$

whose image consists of those A-linear maps from M to P vanishing on N

Hence $\mathrm{Hom}_A(M/N, P) \cong \{f \in \mathrm{Hom}_A(M,P) : f|_N = 0\}$

Ex $\mathrm{Hom}_A(A, M) \cong M$; $f \mapsto f(1)$

$\mathfrak{a} \subseteq A$ ideal $\mathrm{Hom}_A(A/\mathfrak{a}, M) \cong \{x \in M : \mathfrak{a}x = 0\}$

§ Operations on submodules

Prop Let M be an A-module $\{M_i\}_{i\in I}$ a family of submodules of M

Then (1) $\sum_{i\in I} M_i = \{\sum_{\substack{i\in I \\ \text{finite sum}}} x_i : x_i \in M_i\}$ is the smallest submod of M which contains all the M_i

Furthermore if $\forall i,j \in I$ $M_i \subseteq M_j$ or $M_i \supseteq M_j$ then $\sum_{i\in I} M_i = \bigcup_{i\in I} M_i$

(2) $\bigcap_{i\in I} M_i$ is a submodule of M

(3) Given a subset $X \subseteq M$ $\langle X \rangle = \{\sum_{i=1}^{n} a_i x_i : a_i \in A,\ x_i \in X,\ n \geq 0\}$ is the smallest submodule of M containing X

We call it the <u>submodule generated by</u> X

Def Let $\{M_i\}_{i\in I}$ be a family of submodules of M. If every elt in $\sum_{i\in I} M_i$ can be uniquely written as a finite sum $\sum_{i\in I} x_i$ with $x_i \in M_i$

or equiv. $\forall i \in I$ $M_i \cap (\sum_{j\neq i} M_j) = 0$

Then we say $\sum_{i\in I} M_i$ is a <u>direct sum</u> of submodules and denoted by $\bigoplus_{i\in I} M_i$

Prop Let M_1, M_2 be submodules of M. Then $M_1 + M_2/M_2 \cong M_1/M_1 \cap M_2$

pf. $M_1 \longrightarrow M_1+M_2/M_2$; $x \mapsto \bar{x}$

Def An A-module M is <u>finitely generated</u> if $\exists$ finite subset $\{x_1, \cdots, x_n\} \subseteq M$ s.t. $M = \langle x_1, \cdots, x_n \rangle = Ax_1 + \cdots + Ax_n$

M is <u>cyclic</u> if M is generated by one elt

M is <u>simple</u> if $M \neq 0$ and M has no nontrivial submodules

Def Let M be an A-module $x \in M$

The <u>annihilator</u> of x is $\mathrm{Ann}(x) = \{a \in A : ax = 0\}$

The annihilator of M is $\mathrm{Ann}(M)=\{a\in A : aM=0\}$

$\mathrm{Ann}(M)=\bigcap_{x\in X}\mathrm{Ann}(x)$ for any generating set X of M

Prop Let A be a ring. Then

(1) every cyclic A-module is isom to $A/\mathfrak{a}$ for some ideal $\mathfrak{a}$ of A

(2) simple $A/\mathfrak{m}$ maximal ideal $\mathfrak{m}$

pf (1) Let $M=Ax$ be a cyclic A-module

Then $f: A\to Ax,\ a\mapsto ax$ is a surj A-linear map

$\ker f=\mathrm{Ann}(x)$ Hence $Ax\cong A/\mathrm{Ann}(x)$

(2) Let M be a simple module. Any nonzero elt x of M generates a nonzero submodule Ax hence $M=Ax\cong A/\mathfrak{a}$ $\mathfrak{a}=\mathrm{Ann}(x)$

Since A-submodules of M are in one-to-one correspondence with ideals of A containing $\mathfrak{a}$. Simplicity of M implies that $\mathfrak{a}$ is maximal

e.g. Let k be a field V finite dimension k vector space φ linear transform of V

$V_\varphi=(V,\varphi)$ is a $k[x]$-module

Let $v\in V$ then the $k[x]$-submodule generated by v is

$k[x]v=\{f(x)v : f(x)\in k[x]\}=\{f(\varphi)v : f(x)\in k[x]\}$

$=\mathrm{span}_k\{v,\varphi(v),\varphi^2(v),\cdots,\varphi^n(v),\cdots\}$

The annihilator of v is $\{f(x)\in k[x] \mid f(\varphi)v=0\}$

The annihilator of V_φ is $\{f(x)\in k[x] \mid f(\varphi)v=0\ \forall v\in V\}$

$=\{f(x)\in k[x] \mid f(\varphi)=0\}$

Prop ($A/\mathfrak{a}$-module) Let A be a ring, $\mathfrak{a}$ an ideal.

(1) If M is an A-module s.t. $\mathfrak{a}\subseteq\mathrm{Ann}(M)$, then M is naturally an $A/\mathfrak{a}$-module via $\bar{a}x=ax$ for $a\in A$ $x\in M$

(2) If M is an $A/\mathfrak{a}$-module then M is an A-module by restriction of scalars and $\mathfrak{a}\subseteq\mathrm{Ann}_A(M)$

The above constructions are inverse to each other.

In this sense, an $A/\mathfrak{a}$-module is the same thing as an A-module on which $\mathfrak{a}$ acts by 0

- If M is such a module then $A/\mathfrak{a}$-submodule of M is the same thing as an A-submodules of M
- If M, N are two such modules then $f: M \to N$ is an $A/\mathfrak{a}$-module homo iff f is an A-module homo

Prop Let A be a ring, $\mathfrak{a}$ an ideal, M an A-module

Then $\mathfrak{a}M = \{\sum_{i=1}^{n} a_i x_i : a_i \in \mathfrak{a}, x_i \in M, n \geq 0\}$ is an A-submodule of M

Moreover $M/\mathfrak{a}M$ is naturally an $A/\mathfrak{a}$-module

$\mathrm{Hom}_A(A/\mathfrak{a}, M) \cong \{x \in M : \mathfrak{a}x = 0\} \subseteq M$

$f \mapsto f(\bar{1})$

§4. direct product and direct sum of modules

Prop Let $\{M_i\}_{i \in I}$ be a family of A-modules

Then $\prod_{i \in I} M_i = \{(x_i)_i : x_i \in M_i\}$ is an A-module

via $(x_i)_i + (y_i)_i = (x_i + y_i)_i$

$a(x_i)_i = (ax_i)_i$

Moreover it contains a submodule $\bigoplus_{i \in I} M_i = \{(x_i)_i \in \prod_{i \in I} M_i \mid x_i = 0 \text{ for almost all } i\}$

Def $\prod_{i \in I} M_i$, $\bigoplus_{i \in I} M_i$ are called the direct product and direct sum of the family $\{M_i\}_{i \in I}$ of A-modules

Rmk $\forall i \in I$ set $M_i' = \{(x_j) \in \bigoplus_{j \in I} M_j \mid x_j = 0 \ \forall j \neq i\}$

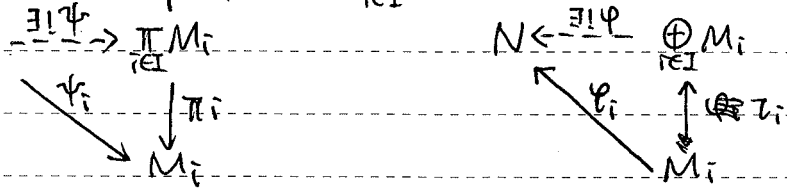
Then M_i' is a submodule of $\bigoplus_{j \in I} M_j$ and $\bigoplus_{j \in I} M_j = \bigoplus_{i \in I} M_i'$

$M_i' \cong M_i$ as A-modules

Prof For each $i \in I$ $\pi_i : \prod_{j \in I} M_j \to M_i$, $(x_j)_j \mapsto x_i$

is an A-linear map. Moreover $(\prod_{i \in I} M_i, \{\pi_i\}_{i \in I})$ satisfies the following property that

$\forall$ A-module N and a family of A-linear maps $\{\psi_i : N \to M_i\}_{i\in I}$
$\exists!$ A-linear map $\psi : N \to \prod_{i\in I} M_i$ s.t. $\pi_i\psi = \psi_i$ $\forall i$



Prop For each i $\tau_i : M_i \to \bigoplus_{j\in I} M_j$; $x_i \mapsto (x_j)_j$ where $x_j = 0$ $(j \neq i)$ is an A-linear map. Moreover $(\bigoplus_{i\in I} M_i, \{\tau_i\}_{i\in I})$ satisfies the u.p. that for any A-module N and a family of A-linear maps $\{\varphi_i : M_i \to N\}_{i\in I}$, $\exists!$ A-linear map $\varphi : \bigoplus_{i\in I} M_i \to N$ s.t. $\varphi\tau_i = \varphi_i$ $\forall i \in I$

pf. $\varphi((x_i)_i) = \sum_{i\in I} \varphi_i(x_i)$ ($x_i = 0$ for almost all i)
$\varphi(\sum_i \tau_i(x_i)) = \sum_i \varphi(\tau_i(x_i))$

Prop Let $\{M_i\}_{i\in I}$ be a family of A-modules, N an A-module. Then we have canonical isoms of A-modules

$$\mathrm{Hom}_A(\bigoplus_{i\in I} M_i, N) \cong \prod_{i\in I} \mathrm{Hom}_A(M_i, N)$$
$$\varphi \mapsto (\varphi\tau_i)_i$$

$$\mathrm{Hom}_A(N, \prod_{i\in I} M_i) \cong \prod_{i\in I} \mathrm{Hom}_A(N, M_i)$$
$$\psi \mapsto (\pi_i\psi)_i$$

$$\Rightarrow \mathrm{Hom}_A(\bigoplus_{i\in I} M_i, \prod_{j\in J} N_j) \cong \prod_{i\in I}\prod_{j\in J} \mathrm{Hom}_A(M_i, N_j)$$

$|I| < \infty$ $\prod_{i\in I} M_i = \bigoplus_{i\in I} M_i$

$$\mathrm{Hom}_A(\bigoplus_{i=1}^m M_i, \bigoplus_{j=1}^n N_j) \cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \mathrm{Hom}_A(M_i, N_j)$$

$$\mathrm{Hom}_A(\bigoplus_{i=1}^m M_i, \bigoplus_{j=1}^n N_j) \times \mathrm{Hom}_A(\bigoplus_{j=1}^n N_j, \bigoplus_{k=1}^l P_k) \longrightarrow \mathrm{Hom}_A(\bigoplus_{i=1}^m M_i, \bigoplus_{k=1}^l P_k)$$
$$((f_{ij}), (g_{jk})) \mapsto \sum_{j=1}^n g_{jk} f_{ij}$$

MEMO NO: _____
DATE: / 3 / 13

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§5. free modules

Let A be a ring, I a set. Denote $A^I = \prod_{i\in I} A$, $A^{(I)} = \bigoplus_{i\in I} A$

$A^n = \bigoplus_{i=1}^n A \simeq \prod_{i=1}^n A$

Def An A-module M is free if $M \cong A^{(I)}$ for some set I

Def Let M be an A-module. $\{x_i\}_{i\in I}$ a family of elts in M ($a_i = 0$ for all i almost)

We say $\{x_i\}_{i\in I}$ is linearly independent if over A if $\sum_{i\in I} a_i x_i = 0$ with $a_i \in A$

$\Rightarrow a_i = 0 \ \forall i$

$\mathrm{Hom}_A(A^{(I)}, M) \cong \prod_{i\in I} \mathrm{Hom}_A(A, M) \cong \prod_{i\in I} M = M^I$ — e_i is the identity

$f: A^{(I)} \to M \longmapsto (f\tau_i)_i \longmapsto (f\tau_i(1)) = (f(e_i))$

$((a_i) \mapsto \sum_{i\in I} a_i x_i) \longleftarrow (x_i)$

Prop Let A be a ring, I a set, M an A-module $\{x_i\}_{i\in I}$ a family of elts in M

Then $\exists!$ A-linear map $f: A^{(I)} \to M$ s.t. $f(e_i) = x_i \ \forall i$

Moreover f injective $\Leftrightarrow$ $\{x_i\}_{i\in I}$ linearly indep over A

f surjective $\Leftrightarrow$ $\{x_i\}_{i\in I}$ generate M

Prop Let M be an A-module. Then

M is free $\Leftrightarrow$ $\exists$ a family of elts $\{x_i\}_{i\in I}$ in M which is linearly indep over A and is a generating set of M.

Such a family of elts is called a basis of the free module M

e.g. All modules over a field are free

A nonzero ideal $\mathfrak{A}$ of A is a free A-module $\Leftrightarrow$ $\mathfrak{A}$ is generated by a nonzerodivisor

Prop. (1) Every A-module is ~~isomorphism~~ isomorphic to the quotient of a free A-module

(2) Every f.g. A-module is isomorphic to the quotient of A^n for some $n \geq 1$

pf (1) Choose a generating set Ω for M (e.g. $\Omega = M$)

Then $\exists$ a surjective A-linear map $f: A^{(\Omega)} \to M$; $e_w \mapsto w$ $\Rightarrow M \cong A^{(\Omega)}/\ker f$

(2) Similar

Theorem Let A be a nonzero ring. If a free A-module M has two basis $\{x_i\}_{i\in I}$ and $\{y_j\}_{j\in J}$ then $|I|=|J|$. Equivalently, $A^{(I)}\cong A^{(J)} \iff |I|=|J|$

Theorem Let A be a nonzero ring, $m,n\geq 1$

(1) If there is an injective A-linear map $A^m\to A^n$ then $m\leq n$

(2) If there is a surjective A-linear map $A^m\to A^n$, then $m\geq n$

(3) If $A^m\cong A^n$ as A-modules, then $m=n$

Prop Let A be a ring $m,n\geq 1$. Then there is a canonical isom of A-modules

$$\mathrm{Hom}_A(A^m,A^n)\cong M_{n\times m}(A)$$
$$f \mapsto (f(e_1),f(e_2),\cdots,f(e_m))$$
$$(x\mapsto Sx) \longleftarrow S$$

where $M_{n\times m}(A)$ is an A-module via $(a_{ij})+(b_{ij})=(a_{ij}+b_{ij})$
(is A^{mn}) $\quad c(a_{ij})=(ca_{ij})$

Moreover under the above Isom. the composition map

$$\mathrm{Hom}_A(A^m,A^n)\times\mathrm{Hom}_A(A^n,A^k)\longrightarrow\mathrm{Hom}_A(A^m,A^k)$$
$$(f,g)\qquad\qquad\mapsto gf$$

can be identified with $M_{n\times m}(A)\times M_{k\times n}(A)\to M_{k\times m}(A)$

$$(S,T)\mapsto TS$$

Pf. $\mathrm{Hom}_A(A^m,A^n)\simeq\bigoplus_{i=1}^{m}\bigoplus_{j=1}^{n}\underbrace{\mathrm{Hom}_A(A,A)}_{\simeq A}\simeq A^{mn}$

pf of Thm2. (1)$\Leftrightarrow$ If $S\in M_{n\times m}(A)$ with $m>n$ then $Sx=0$ has nontrivial sol^n in A^m (HW)

(1)$\Rightarrow$(2) If $f:A^m\to A^n$ is surjective A-linear then $\exists$ a linear map

$g:A^n\to A^m$ s.t. $fg=id_{A^n}$ (For each $j=1,\cdots,n$ choose $v_j\in A^m$ s.t. $f(v_j)=e_j$. Define g to be the unique A-linear map s.t. $g(e_j)=v_j$ $1\leq j\leq n$)

In particular g is injective $\overset{(1)}{\Rightarrow} n\leq m$

(2)$\Rightarrow$(3) ✓

MEMO NO: _____
DATE: / /

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Another pf of (a): Let $f: A^m \to A^n$ be a surjective A-linear map

Then $\exists$ an injective A-linear map $g: A^n \to A^m$ s.t. $fg = id_{A^n}$

By prop f and g are represented by $S \in M_{n\times m}(A)$ $T \in M_{m\times n}(A)$ and $ST = I_n$

Suppose for contradiction that $m < n$ Set $\widetilde{S} = (S\ 0)$ $\widetilde{T} = \begin{pmatrix} T \\ 0 \end{pmatrix}$ $n\times n$

then $\widetilde{S}\widetilde{T} = (S\ 0)\begin{pmatrix} T \\ 0 \end{pmatrix} = ST = I_n \Rightarrow \det(\widetilde{S})\det(\widetilde{T}) = 1$

$\Rightarrow 0 = 1$ contradicting $A \neq 0$

pf of Thm1 Suppose $A^{(I)} \cong A^{(J)}$.

If I is finite $A^{(I)}$ is f.g. $\Rightarrow A^{(J)}$ is f.g. $\Rightarrow J$ is finite

I, J finite Thm 2(3) $\Rightarrow |I| = |J|$

Now suppose M is a free A-module with basis $\{x_i\}_{i\in I}$ and $\{y_j\}_{j\in J}$ where both I and J are infinite

For each $i \in I$ $\exists$ a finite subset $J_i \subset J$ s.t. x_i is a linear combination of the y_j for $j \in J_i$ $\Rightarrow$ M is generated by $\{y_j : j \in \bigcup_{i\in I} J_i\}$

$\Rightarrow J = \bigcup_{i\in I} J_i \Rightarrow |J| \le \sum_{i\in I} |J_i| \le \sum_{i\in I} \omega = |I|\omega = |I|$ $\omega = |\mathbb{Z}|$

By symmetry $|I| \le |J| \Rightarrow |I| = |J|$

Def The rank of a free module is the cardinality of any basis of M

Q: Is A^I free?

Theorem $\prod_{i=1}^{\infty} \mathbb{Z}$ is not a free $\mathbb{Z}$-module. Moreover there is a canonical isom of $\mathbb{Z}$-modules.

$$\mathrm{Hom}_{\mathbb{Z}}\left(\prod_{i=1}^{\infty}\mathbb{Z}, \mathbb{Z}\right) \cong \bigoplus_{i=1}^{\infty}\mathbb{Z}$$

$$(n_i) \mapsto \sum_{\text{finite } i} m_i n_i \longleftarrow (m_i)$$

Note $\mathrm{Hom}_{\mathbb{Z}}(\bigoplus_{i=1}^{\infty}\mathbb{Z}, \mathbb{Z}) \cong \prod_{i=1}^{\infty}\mathbb{Z}$

$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^{(I)}, \mathbb{Z}) \cong \mathbb{Z}^I$

I infinite $|\mathbb{Z}^{(I)}| = |I|$ $|\mathbb{Z}^I| = 2^{|I|}$

If $\prod_{i=1}^{\infty}\mathbb{Z}$ is free then $\prod_{i=1}^{\infty}\mathbb{Z} \cong \mathbb{Z}^{(I)}$ for some infinite set I

$|\mathrm{Hom}_{\mathbb{Z}}(\prod_{i=1}^{\infty}\mathbb{Z}, \mathbb{Z})| = 2^{|\prod_{i=1}^{\infty}\mathbb{Z}|} = 2^{2^{\omega}} \neq \omega = |\bigoplus_{i=1}^{\infty}\mathbb{Z}|$

---

*. $A^{(I)} \cong A^{(J)} \rightsquigarrow A^{(I)}/\mathfrak{m}A^{(I)} \cong A^{(J)}/\mathfrak{m}A^{(J)}$ i.e. $(A/\mathfrak{m})^{(I)} \cong (A/\mathfrak{m})^{(J)}$ $\mathfrak{m}A^{(I)} = \mathfrak{m}^{(I)}$ $A^{(I)}/\mathfrak{m}^{(I)} \cong (A/\mathfrak{m})^{(I)}$

§6. Finitely generated modules over a PID

If A is a PID, then every nonzero ideal of A is a free A-module of rank 1

Thm If A is a PID, then every submodule of M of a free A module L is itself free and $\operatorname{rk} M \le \operatorname{rk} L$

pf. Suppose L is a free A-module with basis $\{x_i\}_{i\in I}$

Fix a well-order $\le$ on I $\quad \forall i\in I \ L_i := \langle x_k : k \le i\rangle$

$L_i' := \langle x_k : k < i\rangle$ $\quad M_i = M\cap L_i$ $\quad M_i' = M\cap L_i'$

We have $L = \bigcup_{i\in I} L_i$ $\quad \forall x\in L$ $\mu(x)$ = least index i s.t. $x\in L_i$

Let $\pi_i : L \to A$ be the ith coordinate map $\quad \sum_{j\in I} a_j x_j \mapsto a_j$

Consider $\pi_i : M_i \to A$ Then $\ker \pi_i = M_i'$

A PID $\Rightarrow$ $\operatorname{Im}\pi_i$ is an ideal of A say $\operatorname{Im}\pi_i = (a_i)$

Let $I_0 = \{i\in I : a_i \neq 0\}$ and for each $i\in I_0$ choose a $y_i \in M_i$

s.t. $\pi_i(y_i) = a_i$

$a_i = 0 \Leftrightarrow M_i = M_i'$

If $a_i \neq 0$ then $M_i = M_i' \oplus \langle y_i\rangle$ and $\langle y_i\rangle \simeq A$ (A is an integral domain)

In other words $\forall y\in M_i$ $\exists!\ y'\in M_i'$ and $c\in A$ s.t. $y = y' + cy_i$

Claim. $\{y_i : i\in I_0\}$ linearly indep over A

If not $\exists k_1, \dots, k_n \in I_0$ $(n\ge 1)$ $c_1, \dots, c_n \in A \neq 0$ s.t. $c_1 y_{k_1} + \cdots + c_n y_{k_n} = 0$

Arrange the terms so that $k_1 < k_2 < \cdots < k_n$ Then $c_1 y_{k_1} + \cdots + c_{n-1} y_{k_{n-1}} \in M_{k_n}'$

and $c_n y_{k_n} \in \langle y_{k_n}\rangle$ $\Rightarrow$ $c_n = 0$ contradiction

Claim. $\{y_i : i\in I_0\}$ generates M

Let $M' = \langle y_i : i\in I_0\rangle$ Suppose for contradiction that $M' \subsetneq M$

Then the nonempty set $\{\mu(y) : y\in M,\ y\notin M'\}$ has a least elt j

Choose a $y\in M$ s.t. $y\notin M'$ and $\mu(y) = j$ $\quad y = y' + cy_j$ where $y'\in M_j'$ $c\in A$

$y\in M'$ $y_j \in M'$ $\Rightarrow$ $y'\notin M'$ $\quad y'\in M_j'$ $\Rightarrow$ $\mu(y') < j$ contradicts the def of j

MEMO NO: _____
DATE: / /

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Thm Let A be a PID, M a f.g. A-module. Then

(1) $M \cong A^r \oplus A/(a_1) \oplus \cdots \oplus A/(a_t)$ where $r \geq 0$, $a_1, \cdots, a_t$ are nonzero nonunit elts of A and $a_1 | a_2 | \cdots | a_t$

(2) $M \cong A^r \oplus A/(p_1^{e_1}) \oplus \cdots \oplus A/(p_s^{e_s})$ where $r \geq 0$, $p_1, \cdots, p_s$ are prime elts of A and $e_i \geq 1$ $(1 \leq i \leq s)$

(3) The r is uniquely determined by M

The a_i and $p_j^{e_j}$ are uniquely determined be M up to units

Lemma Let A be a PID, $S \in M_{m\times n}(A)$

Then $\exists\ P \in GL_m(A)$, $Q \in GL_n(A)$ s.t. $PSQ = \begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_r & \\ & & & \text{zero} \end{pmatrix}$ where $a_1, \cdots, a_r \in A \neq 0$, $a_1 | a_2 | \cdots | a_r$

In fact $r = \operatorname{rank}(S)$

$a_i = \frac{D_i}{D_{i-1}}$, $D_i = \gcd(i\times i \text{ minors of } S)$

pf of Thm (1): M f.g. $\Rightarrow M \cong A^n/N$ for some $n \geq 1$ and $N \subseteq A^n$

By Thm 1, N is free of rank $\leq n$. Choose a basis $v_1, \cdots, v_m$ of N $(m \leq n)$

Set $S = (v_1, \cdots, v_m) \in M_{n\times m}(A)$. By Lemma $\exists\ P \in GL_n(A)$, $Q \in GL_m(A)$ s.t.

$D = PSQ = \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_m \\ & \text{zero} & \end{pmatrix}$ where $a_1 | \cdots | a_m$, $(a_1, \cdots, a_m \neq 0)$

$$\begin{array}{ccc} A^m & \xrightarrow{S} & A^n \\ Q \uparrow \cong & & \cong \uparrow P \\ A^m & \xrightarrow{D} & A^n \end{array} \qquad \begin{array}{l} x \mapsto Sx \\ \operatorname{Im} S = N \\ (x_1, \cdots, x_m) \longmapsto (a_1x_1, \cdots, a_mx_m, 0, \cdots, 0) \end{array}$$

$M \cong A^n/N = A^n/\operatorname{Im} S \cong A^n/\operatorname{Im} D$

$= A \oplus \cdots \oplus A / a_1A \oplus \cdots \oplus a_mA \oplus 0 \oplus \cdots \oplus 0$

$\cong A/a_1A \oplus \cdots \oplus A/a_mA \oplus A^{n-m}$

e.g. k field, V n-dim'l k vector space, $\varphi \in \operatorname{End}_k(V)$, $v_1, \cdots, v_n$ k-basis of V

$\varphi(v_1, \cdots, v_n) = (v_1, \cdots, v_n)A$, $A = (a_{ij}) \in M_n(k)$, V_φ $k[x]$-module

$\Pi: k[x]^n \twoheadrightarrow V_\varphi$; $(f_1(x), \cdots, f_n(x)) \mapsto f_1(\varphi)v_1 + \cdots + f_n(\varphi)v_n$

$\varphi(v_j) = \sum_{i=1}^n a_{ij}v_i \Rightarrow \sum_{i=1}^n (x\delta_{ij} - a_{ij})v_i = 0$ (j-th column of $xI_n - A$)

Fact: the columns of $xI_n - A$ form a free basis of $\ker \Pi$

$\exists\ P(x), Q(x) \in GL_n(k[x])$ s.t. $P(x)(xI_n - A)Q(x) = \begin{pmatrix} m_1(x) & & \\ & \ddots & \\ & & m_n(x) \end{pmatrix}$ $\begin{pmatrix} 1 & & \\ & \ddots & \\ & & m_1(x) \\ & & & \ddots \\ & & & & m_2(x) \end{pmatrix}$

where $m_1(x) \mid \cdots \mid m_k(x) \; m_\ell(x)$

$\Rightarrow V_\varphi \cong k[x]^n / \ker\pi \simeq k[x]/(m_1(x)) \oplus \cdots \oplus k[x]/(m_\ell(x)) \oplus k[x]^{n-\ell}$

§7. tensor product of modules

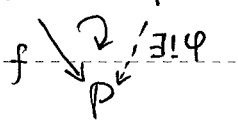
Def Let M, N, P be A-modules

A map $f: M \times N \to P$ ($(x,y) \mapsto f(x,y)$) is A-bilinear if $\forall x \in M$ $f(x,-): N \to P$ is A-linear and $\forall y \in N$ $f(-,y): M \to P$ is A-linear

Fact if $f: M \times N \to P$ is A-bilinear and $\varphi: P \to P'$ is A-linear then $\varphi f: M \times N \to P'$ is A-bilinear

Q: Does there exist an A-bilinear map $g: M \times N \to T$ s.t. every A-bilinear map $f: M \times N \to P$ is the composition of g and some A-linear map $T \to P$?

Def Let M, N be A-modules. A pair (T, g) consisting of an A-module T and an A-bilinear map $g: M \times N \to T$ is a tensor product of M and N if it satisfies the universal property that $\forall$ A-module P and A-bilinear map $f: M \times N \to P$ $\exists!$ A-linear map $\varphi: T \to P$ s.t. $f = \varphi g$



Thm. The tensor product of M and N exists and is unique up to unique isomorphism.

More precisely if (T_1, g_1) and (T_2, g_2) are both tensor products of M and N, $j: T_1 \to T_2$ is the unique A-linear map s.t. $g_2 = j g_1$. $j': T_2 \to T_1$ is the unique A-linear map s.t. $g_1 = j' g_2$ then $j'j = id_{T_1}$ and $jj' = id_{T_2}$

pf. (existence)

Let $A^{(M \times N)}$ be the free A-module with basis $\{e_{(x,y)} : x \in M, y \in N\}$

Let B be the submodule of $A^{(M \times N)}$ generated by elts of the form:

MEMO NO: _____
DATE: 1 3 / 20

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$e_{(x_1+x_2,y)} - e_{(x_1,y)} - e_{(x_2,y)}$ $e_{(ax,y)} - a e_{(x,y)}$

$e_{(x,y_1+y_2)} - e_{(x,y_1)} - e_{(x,y_2)}$ $e_{(x,ay)} - a e_{(x,y)}$ $\forall x, x_1, x_2 \in M, y, y_1, y_2 \in N$ $a \in A$

Let $T = A^{(M\times N)}/B$ Denote the image of $e_{(x,y)}$ in T by $x \otimes y$

From our def. of T, we have $(x_1+x_2)\otimes y = x_1\otimes y + x_2 \otimes y$

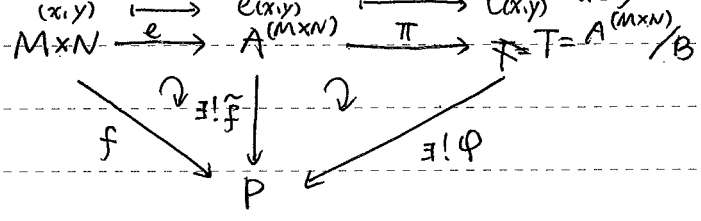
$(ax)\otimes y = a(x\otimes y) = x\otimes(ay)$ $x\otimes(y_1+y_2) = x\otimes y_1 + x\otimes y_2$

Define $g: M\times N \to T = A^{(M\times N)}/B$; $(x,y) \mapsto x\otimes y$ g is clearly A-bilinear

Now suppose $f: M\times N \to P$ is a A-bilinear map

Then by the u.p of free module $\exists!$ A-linear map $\tilde{f}: A^{(M\times N)} \to P$ s.t.

$\tilde{f}(e_{(x,y)}) = f(x,y)$



bilinearity of f implies $\tilde{f}$ vanishes on the generators of B hence vanish on B

$\Rightarrow \exists!$ A-linear map $\varphi: T \to P$ s.t. $\tilde{f} = \varphi\pi$ $f = \tilde{f}e = \varphi(\pi e) = \varphi g$

Check that φ is the unique A-linear map s.t. $\varphi g = f$

Rmk We denote $A^{(M\times N)}/B$ by $M \underset{A}{\otimes} N$ and denote $g: M\times N \to A^{(M,N)}/B$

by $t_{M,N}: M\times N \to M\underset{A}{\otimes}N$; $(x,y) \mapsto x\otimes y$

Every elt in $M\underset{A}{\otimes}N$ can be written as a finite sum $\sum_{i=1}^{n} x_i \otimes y_i$ however such an expression is not unique.

More generally, if $\{x_i\}_{i\in I}$ generates M $\{y_j\}_{j\in J}$ generates N then $\{x_i\otimes y_j\}_{\substack{i\in I\\ j\in J}}$ generates $M\underset{A}{\otimes}N$

Prop Let $M_1, \cdots, M_n$ be A-modules Then $\exists$ a pair $(M_1\underset{A}{\otimes}\cdots\underset{A}{\otimes}M_n, t)$ consisting of an A-module $M_1\underset{A}{\otimes}\cdots\underset{A}{\otimes}M_n$ and an A-multilinear map

$t: M_1\times\cdots\times M_n \to M_1\underset{A}{\otimes}\cdots\underset{A}{\otimes}M_n$; $(x_1,\cdots,x_n) \mapsto x_1\otimes\cdots\otimes x_n$ with the property that

$\forall$ A-multilinear map $f: M_1\times\cdots\times M_n \to P$ $\exists!$ A-linear map $\varphi: M_1\underset{A}{\otimes}\cdots\underset{A}{\otimes}M_n \to P$

$x_1\otimes\cdots\otimes x_n \mapsto f(x_1,\cdots,x_n)$

s.t. $f = \varphi t$. Such a pair is unique up to unique isom.

Prop (1) If $f: M \to N$ and $f': M' \to N'$ are A-linear maps then $\exists!$ A-linear map $T(f,f') : M \otimes_A M' \to N \otimes_A N'$ s.t.

$$T(f,f')(x \otimes x') = f(x) \otimes f'(x')$$

(2) If $f: M \to N$ $g: N \to P$ $f': M' \to N'$ $g: N' \to P'$ are A-linear maps then $T(gf, g'f') = T(g,g')\,T(f,f')$

(3) There exists an A-linear map

$$\mathrm{Hom}_A(M,N) \otimes_A \mathrm{Hom}_A(M',N') \longrightarrow \mathrm{Hom}_A(M \otimes_A M', N \otimes_A N')$$
$$f \otimes f' \longmapsto T(f,f')$$

(4) There exists an A-linear map

$$\mathrm{Hom}_A(M,N) \otimes_A \mathrm{Hom}_A(N,P) \longrightarrow \mathrm{Hom}_A(M,P)$$
$$f \otimes g \mapsto gf$$

Pf. (1) Check that $M \times M' \to N \otimes_A N'$; $(x,x') \mapsto f(x) \otimes f'(x')$ is A-bilinear

(2) $x \otimes x' \xmapsto{T(f,f')} f(x) \otimes f'(x') \xmapsto{T(g,g')} g(f(x)) \otimes g'(f'(x'))$

(3) Check that $\mathrm{Hom}_A(M,N) \times \mathrm{Hom}_A(M',N') \to \mathrm{Hom}_A(M \otimes_A M', N \otimes_A N')$;
$(f,f') \mapsto T(f,f')$ is A-bilinear

Prop Let M, N, P, M_i be A-modules. Then there are A-module isomorphisms

(1) $M \otimes_A N \to N \otimes_A M$; $x \otimes y \mapsto y \otimes x$

(2) $(M \otimes_A N) \otimes_A P \to M \otimes_A N \otimes_A P \leftarrow\!\shortmid\!\to M \otimes_A (N \otimes_A P)$

$(x \otimes y) \otimes z \mapsto x \otimes y \otimes z \longmapsfrom x \otimes (y \otimes z)$

(3) $\left(\bigoplus_{i \in I} M_i\right) \otimes_A N \to \bigoplus_{i \in I} (M_i \otimes_A N)$; $(x_i) \otimes y \mapsto (x_i \otimes y)$

(4) $A \otimes_A M \to M$; $a \otimes x \mapsto ax$; $1 \otimes x \longmapsfrom x$

pf of (2a). The strategy is that we construct an A-linear map

$f: (M \otimes_A N) \otimes_A P \to M \otimes_A N \otimes_A P$ s.t. $f((x \otimes y) \otimes z) = x \otimes y \otimes z$

and an A-linear map $g: M \otimes_A N \otimes_A P \to (M \otimes_A N) \otimes_A P$ s.t. $g(x \otimes y \otimes z) = g((x \otimes y) \otimes z)$

$\Rightarrow gf = id$ and $fg = id$

MEMO NO: _____
DATE: / /

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Check that $M\times N\times P \to (M\underset{A}{\otimes}N)\underset{A}{\otimes}P$ $(x,y,z)\mapsto (x\otimes y)\otimes z$ is A-triple linear

u.p. $\Rightarrow \exists\, g: M\underset{A}{\otimes}N\underset{A}{\otimes}P \to (M\underset{A}{\otimes}N)\underset{A}{\otimes}P$ $g(x\otimes y\otimes z)=(x\otimes y)\otimes z$

Fix $z\in P$ $M\times N \to M\underset{A}{\otimes}N\underset{A}{\otimes}P$, $(x,y)\mapsto x\otimes y\otimes z$ is A-bilinear

u.p. $\Rightarrow \exists\, f_z: M\underset{A}{\otimes}N \to M\underset{A}{\otimes}N\underset{A}{\otimes}P$ s.t. $f_z(x\otimes y)=x\otimes y\otimes z$

$(M\underset{A}{\otimes}N)\times P \to M\underset{A}{\otimes}N\underset{A}{\otimes}P$, $(t,z)\mapsto f_z(t)$ A-bilinear

u.p. $\Rightarrow \exists f: (M\underset{A}{\otimes}N)\underset{A}{\otimes}P \to M\underset{A}{\otimes}N\underset{A}{\otimes}P$ s.t. $f(t\otimes z)=f_z(t)$

$f((x\otimes y)\otimes z)=x\otimes y\otimes z$

(3) $(\underset{i\in I}{\oplus}M_i)\times N \to \underset{i\in I}{\oplus}(M_i\underset{A}{\otimes}N)$ $((x_i),y)\mapsto (x_i\otimes y)$ A-bilinear

$\Rightarrow \exists f: (\underset{i\in I}{\oplus}M_i)\underset{A}{\otimes}N \to \underset{i\in I}{\oplus}(M_i\underset{A}{\otimes}N)$ $(x_i)\otimes y\mapsto (x_i\otimes y)$

$\iota_i: M_i\to \underset{j\in I}{\oplus}M_j \Rightarrow T(\iota_i, id_N): M_i\underset{A}{\otimes}N \to (\underset{j\in I}{\oplus}M_j)\underset{A}{\otimes}N$

u.p of $\oplus$ $\Rightarrow \exists g: \underset{i\in I}{\oplus}(M_i\underset{A}{\otimes}N)\to(\underset{i\in I}{\oplus}M_i)\underset{A}{\otimes}N$ Check $gf=id$ $fg=id$

Rmk tensor product does not commute with direct products

$(\underset{i\in I}{\prod}M_i)\underset{A}{\otimes}N \ncong \underset{i\in I}{\prod}(M_i\underset{A}{\otimes}N)$

In fact $\exists$ A-linear map $\alpha: (\underset{i\in I}{\prod}M_i)\underset{A}{\otimes}N \to \underset{i\in I}{\prod}(M_i\underset{A}{\otimes}N)$

α need not be injective nor surjective $(x_i)\otimes y\mapsto (x_i\otimes y)$

We do have (1) N f.g $\Rightarrow \alpha$ is surjective

(2) N finitely presented $\Rightarrow \alpha$ is an isom.

e.g. (1) $A^{(I)}\underset{A}{\otimes}A^{(J)} = (\underset{i\in I}{\oplus}A)\underset{A}{\otimes}(\underset{j\in J}{\oplus}A)\cong \underset{i\in I}{\oplus}\underset{j\in J}{\oplus}A\underset{A}{\otimes}A\cong \underset{i\in I}{\oplus}\underset{j\in J}{\oplus}A = A^{(I\times J)}$

$A^m\underset{A}{\otimes}A^n\cong A^{mn}$

M free A-module with basis $\{x_i\}_{i\in I}$, $(N$ $\{y_j\}_{j\in J})$ $\Rightarrow M\underset{A}{\otimes}N$ free A-module with basis $\{x_i\otimes y_j\}_{i\in I, j\in J}$

(2) $\mathbb{Q}\underset{\mathbb{Z}}{\otimes}\mathbb{Z}/n\mathbb{Z}=0$ $n\geq 2$ $q\otimes\bar{m}=n(\frac{q}{n}\otimes\bar{m})=\frac{q}{n}\otimes\overline{nm}=\frac{q}{n}\otimes\bar{0}=0$

$\gcd(m,n)=1$ $\mathbb{Z}/m\mathbb{Z}\underset{\mathbb{Z}}{\otimes}\mathbb{Z}/n\mathbb{Z}=0$

(3) $A/\mathfrak{a}\underset{A}{\otimes}A/\mathfrak{b}\cong A/(\mathfrak{a}+\mathfrak{b})$ $\mathbb{Z}/m\mathbb{Z}\underset{\mathbb{Z}}{\otimes}\mathbb{Z}/n\mathbb{Z}\cong\mathbb{Z}/d\mathbb{Z}$ where $d=\gcd(m,n)$

(4) $\mathbb{Q}\underset{\mathbb{Z}}{\otimes}\mathbb{Q}\cong\mathbb{Q}$ $\mathbb{Q}\underset{\mathbb{Z}}{\otimes}(\mathbb{Z}^r\oplus\mathbb{Z}/(m_1)\oplus\cdots\oplus\mathbb{Z}/(m_t))$

$\cong\mathbb{Q}\underset{\mathbb{Z}}{\otimes}\mathbb{Z}^r\cong\underset{i=1}{\overset{r}{\oplus}}(\mathbb{Q}\underset{\mathbb{Z}}{\otimes}\mathbb{Z})\cong\underset{i=1}{\overset{r}{\oplus}}\mathbb{Q}=\mathbb{Q}^r$

§ 8. tensor-hom adjunction

Thm Let M, N, P be A-modules. Then there is an A-module isom

$$\mathrm{Hom}_A(M \underset{A}{\otimes} N, P) \simeq \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P))$$

$$\varphi \longmapsto (x \mapsto (y \mapsto \varphi(x \otimes y)))$$

$$((x \otimes y) \mapsto \psi(x)(y)) \longmapsto \psi$$

$$\mathrm{Hom}_A(M \underset{A}{\otimes} N, P) \simeq \mathrm{Bil}_A(M \times N, P) \simeq \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P))$$

$$\cap \qquad\qquad\qquad\qquad \cap$$

$$\mathrm{Map}(M \times N, P) \qquad \mathrm{Map}(M, \mathrm{Map}(N, P))$$

Def Let A, B be rings. An (A,B)-bimodule M is an abelian gp $(M, +)$ which is simultaneously an A-module and a B-module and the two structures commutes in the sense that $a(bx) = b(ax)$ $\forall a \in A$ $b \in B$ $x \in M$

$\Rightarrow$ $a: M \to M$; $x \mapsto ax$ is a B-module homo $\Rightarrow$ $A \to \mathrm{End}_B(M) \subseteq \mathrm{End}_{\mathbb{Z}}(M)$

$b: M \to M$ $x \mapsto bx$ $\quad A \to \mathrm{End}_A(M)$

The data of an (A,B)-bimodule is equivalent to that of a B-module M together with a ring homo $A \to \mathrm{End}_B(M)$

or that of an A-module M together with a ring homo $B \to \mathrm{End}_A(M)$

Prop Let A, B be rings. M an A-module. L an (A,B)-bimodule

Then (1) $M \underset{A}{\otimes} L$ is an (A,B)-bimodule where the B-module structure is given by $b(x \otimes y) = x \otimes (by)$ $\forall x \in M$ $y \in L$ $b \in B$

(2) $\mathrm{Hom}_A(L, M)$ is an (A,B)-bimodule where the B-module structure is given by $(bf)(x) = f(bx)$ $\forall f \in \mathrm{Hom}_A(L, M)$ $x \in L$, $b \in B$

(3) $\mathrm{Hom}_A(M, L)$ is an (A,B)-bimodule where the B-module structure is given by $(bf')(y) = bf'(y)$ $\forall f' \in \mathrm{Hom}_A(M, L)$, $y \in M$, $b \in B$

Rmk L (A,B)-bimodule N B-module

$L \underset{B}{\otimes} N$ $\mathrm{Hom}_B(L, N)$, $\mathrm{Hom}_B(N, L)$ are all (A,B)-bimodule

Thm Let A, B be rings, M an A-module N a B-module. L an (A,B)-bimodule

Then we have a canonical isom of (A,B)-bimodules:

$$\mathrm{Hom}_B(M \underset{A}{\otimes} L, N) \cong \mathrm{Hom}_A(M, \mathrm{Hom}_B(L, N))$$

$$f \longmapsto (x \mapsto (y \mapsto f(x \otimes y)))$$

$$(x \otimes y \mapsto \varphi(x)(y)) \longleftarrow \varphi$$

$\mathrm{Hom}_B(M \underset{A}{\otimes} L, N) \hookrightarrow$ This is not trivial

$$\mathrm{Hom}_A(N \underset{B}{\otimes} L, M) \cong \mathrm{Hom}_B(N, \mathrm{Hom}_A(L, M))$$

Prop Let A, B be rings, M an A-module N a B-module L an (A,B)-bimodule. Then we have a canonical isom of (A,B)-bimodules

$$(M \underset{A}{\otimes} L) \underset{B}{\otimes} N \cong M \underset{A}{\otimes} (L \underset{B}{\otimes} N)$$

$$(x \otimes y) \otimes z \mapsto x \otimes (y \otimes z)$$

§9. (co)extension and restriction of scalars

Recall that if $f: A \to B$ is a ring map and N is a B-module then N is an A-module via $ay = f(a)y \quad \forall a \in A \quad y \in N$. This A-module $N|_A$ is said to be obtained ~~fe~~ from N by restriction of scalars.

In particular B is an A-module via $ab = f(a)b$

e.g. $K \subseteq L$ field extension $[L:K] = \dim_K L$

Any L-vector space V is a K-vector space and $\dim_K V = [L:K]\dim_L V$

e.g. Any A-module M is a $\mathbb{Z}$-module via the unique ring map $\mathbb{Z} \to A$, $n \mapsto n1_A$

This $\mathbb{Z}$-module is the underlying abelian gp $(M, +)$

e.g. $A \to A/\mathfrak{a}$ Every $A/\mathfrak{a}$-module is an A-module

Rmk Not every A-module is obtained from a B-module via restriction of scalars

e.g. an A-module M is the restriction of an $A/\mathfrak{a}$-module iff $\mathfrak{a} \subseteq \mathrm{Ann}_A(M)$.

e.g. a real vector space V is the restriction of a complex vector space iff its real dim is even.

Prop. Let $f: A \to B$ be a ring map. M an A-module N a B-module

Then (1) N is an (A,B)-bimodule. In particular B is an (A,B)-bimodule.

(2) $M_B := B \underset{A}{\otimes} M$ is a B-module via $b(b' \otimes x) = (bb') \otimes x$

This B-module is said to be obtained from M by extension of scalars

(3) $\mathrm{Hom}_A(B, M)$ is a B-module via $(bf)(b') = f(bb')$ coextension of scalars

Thm. Let $f: A \to B$ be a ring map, M an A-module, N a B-module. Then we have the following canonical isoms of (A, B)-bimodules

$$\mathrm{Hom}_B(M \underset{A}{\otimes} B, N) \cong \mathrm{Hom}_A(M, N|_A)$$

$$\mathrm{Hom}_A(N|_A, M) \cong \mathrm{Hom}_B(N, \mathrm{Hom}_A(B, M))$$

pf. $\mathrm{Hom}_B(M \underset{A}{\otimes} B, N) \cong \mathrm{Hom}_A(M, \mathrm{Hom}_B(B, N)) \cong N|_A$

$\mathrm{Hom}_A(N \underset{}{\otimes} B, M) \cong \mathrm{Hom}_B(N, \mathrm{Hom}_A(B, M)) \cong N|_A$

$\pi: A \to A/\mathfrak{a}$

M, A-module $\quad \mathrm{Hom}_A(A/\mathfrak{a}, M)$ "$\subseteq$" $\mathrm{Hom}_A(A, M)$

$\quad\quad\quad

MEMO NO: _____
DATE: 1 3 1 27

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Prop Let $f: A \to B$ be a ring map, M, M' A-modules, N B-module

Then (1) $M \otimes_B N \cong M \otimes_A N$ as B-modules

(2) $M \otimes_B M'_B \cong (M \otimes_A M')_B$

pf. (1) $M \otimes_B N = (M \otimes_A B) \otimes_B N \cong M \otimes_A (B \otimes_B N) \cong M \otimes_A N$

(2) $M \otimes_B M'_B \cong M \otimes_A M'_B = M \otimes_A (M' \otimes_A B) = (M \otimes_A M') \otimes_A B = (M \otimes_A M')_B$

Chapter 3 Categories and Functors

Def A category $\mathcal{C}$ consists of

- a collection $Ob(\mathcal{C})$ of objects
- a set $Hom_{\mathcal{C}}(X, Y)$ of morphisms for every ordered pair of objects (X, Y) in $\mathcal{C}$
- an identity morphism $id_X \in Hom_{\mathcal{C}}(X, X)$ for every object X

and a composition fcn $Hom_{\mathcal{C}}(X, Y) \times Hom_{\mathcal{C}}(Y, Z) \to Hom_{\mathcal{C}}(X, Z)$

$(f, g) \mapsto gf$

for every ordered triple (X, Y, Z) of objects s.t.

(1) $(hg)f = h(gf)$ (composition is associative)

(2) $id_Y f = f = f id_X$ $\forall f \in Hom_{\mathcal{C}}(X, Y)$

e.g. Set Grp Ring CRing A-Mod Top DiffMfd

$f \in Hom_{\mathcal{C}}(X, Y) \Leftrightarrow f: X \to Y$ object $X \in Ob(\mathcal{C})$ or $X \in \mathcal{C}$

Def a category $\mathcal{C}$ is small if $Ob(\mathcal{C})$ is a set

For $X \in \mathcal{C}$ $Hom_{\mathcal{C}}(X, X)$ is a monoid (i.e. semigroup with identity)

Def a category $\mathcal{C}$ is discrete if every morphism is an identity morphism

A discrete small category is the same thing as a set $X \circlearrowright id_X$ $Y \circlearrowright id_Y$

Def A preordered set $(P, \leq)$ is a set P equipped with a reflexive

and transitive binary relation $\leq$ that is $a \leq a \quad \forall a \in P$

$a \leq b, b \leq c \Rightarrow a \leq c$

A partially ordered set (poset) is a preordered set $(P, \leq)$ s.t. $a \leq b, b \leq a \Rightarrow a = b$

e.g. A preordered set $(P, \leq)$ can be regarded as a category as follows:

objects: elements of P

morphisms: $\mathrm{Hom}(a,b) = \begin{cases} \{*\} & a \leq b \\ \phi & \text{otherwise} \end{cases}$ $\quad a \xrightarrow{\leq} b$

Conversely if $\mathcal{C}$ is a small category in which every hom set is empty or a singleton, then $\mathcal{C}$ is the category associated to the preordered set $(Ob(\mathcal{C}), \leq)$

where $X \leq Y \Leftrightarrow \mathrm{Hom}_{\mathcal{C}}(X,Y) \neq \phi$

e.g. A monoid M is a category as follows:

it has exactly one object $*$

$\mathrm{Hom}(*,*) = M$ where composition is given by multiplication

Conversely, if $\mathcal{C}$ is a category of one object X, then $\mathcal{C}$ is the category associated to the monoid $\mathrm{Hom}_{\mathcal{C}}(X,X)$

Def. A morphism $f: X \to Y$ is an isomorphism if $\exists\, g: Y \to X$ s.t. $gf = id_X$, $fg = id_Y$

Such a g is unique if it exists. It is denoted by f^{-1} and called the inverse of f

Two objects X, Y are isomorphic, denoted by $X \cong Y$ if $\exists$ an isom $f: X \to Y$

Def. Let $\mathcal{C}$ be a category. The opposite category $\mathcal{C}^{op}$ is given as follows:

$Ob(\mathcal{C}^{op}) = Ob(\mathcal{C})$

$\mathrm{Hom}_{\mathcal{C}^{op}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(Y,X) \quad f^{op} \leftarrow\!\shortmid f$

composition $f^{op} g^{op} = (gf)^{op}$

identity morphisms: same as $\mathcal{C}$

$\mathcal{C}$: $X \xrightarrow{f} Y \xrightarrow{g} Z$

$\mathcal{C}^{op}$: $X \xleftarrow{f^{op}} Y \xleftarrow{g^{op}} Z$

e.g. If $\mathcal{C}$ is the category associated to a preordered set $(P, \leq)$

then $\mathcal{C}^{op}$ is ——— $(P, \leq')$

where $a \leq' b \Leftrightarrow b \leq a$

MEMO NO: _____
DATE: / /

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e.g. If $\mathcal{C}$ is the category associated to a monoid $(M,\cdot)$
then $\mathcal{C}^{op}$ is ——— $(M,*)$
where $a*b := b\cdot a$

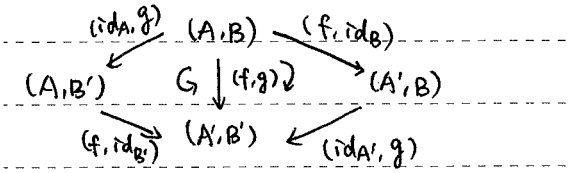
Def A subcategory $\mathcal{D}$ of $\mathcal{C}$ is a category where
$Ob(\mathcal{D})$ is a subcollection of $Ob(\mathcal{C})$
$Hom_{\mathcal{D}}(X,Y)$ is a subset of $Hom_{\mathcal{C}}(X,Y)$
s.t. the morphisms in $\mathcal{D}$ are closed under composition and include id_X for every object X of $\mathcal{D}$

A subcategory is a full subcategory if $\forall X,Y \in \mathcal{D}$ $Hom_{\mathcal{D}}(X,Y) = Hom_{\mathcal{C}}(X,Y)$

e.g. $Ab \subseteq Grp$, $CRing \subseteq Ring$
$Vect_k^{f.d.} \subseteq Vect_k$ (f.d. = finite dimension) $FinSet \subseteq Set$

Def Let $\mathcal{A}, \mathcal{B}$ be categories. The product category $\mathcal{A}\times\mathcal{B}$ has
objects (A,B) where $A\in\mathcal{A}$ $B\in\mathcal{B}$
morphisms $Hom_{\mathcal{A}\times\mathcal{B}}((A,B),(A',B')) = Hom_{\mathcal{A}}(A,A')\times Hom_{\mathcal{B}}(B,B')$
composition $(f',g')(f,g) = (f'f, g'g)$
identity morphisms $id_{(A,B)} = (id_A, id_B)$

For $f: A\to A'$ in $\mathcal{A}$ $g: B\to B'$ in $\mathcal{B}$
$(f,g) = (f; id_{B'})(id_A, g) = (id_{A'}, g)(f, id_B)$ $(A,B)\to(A',B')$



Def Let $\mathcal{A}, \mathcal{B}$ be categories. A (covariant) ~~factor~~ functor $F: \mathcal{A}\to\mathcal{B}$ is a rule which assigns
to each object A of $\mathcal{A}$, an object $F(A)$ of $\mathcal{B}$ (If $\mathcal{A}$ $\mathcal{B}$ small $F: Ob(\mathcal{A})\to Ob(\mathcal{B})$ set map)
to every morphism $f: A_1\to A_2$ in $\mathcal{A}$, an a morphism $F(f): F(A_1)\to F(A_2)$
$\Rightarrow$ $Hom_{\mathcal{A}}(A_1,A_2) \to Hom_{\mathcal{B}}(F(A_1), F(A_2))$
$f \mapsto F(f)$

s.t. $F(id_A) = id_{F(A)}$ $F(gf) = F(g)F(f)$ $\forall f: A_1 \to A_2$ $g: A_2 \to A_3$

A contravariant functor $F: \mathcal{A} \to \mathcal{B}$ is a covariant functor $\mathcal{A}^{op} \to \mathcal{B}$ (or a covariant functor $\mathcal{A} \to \mathcal{B}^{op}$)

A functor $F: \mathcal{A} \to \mathcal{B}$ is faithful if the set map $\mathrm{Hom}_{\mathcal{A}}(A_1, A_2) \to \mathrm{Hom}_{\mathcal{B}}(F(A_1), F(A_2))$ is injective

full : surjective

fully faithful : bijective

e.g. (1) $\mathcal{D}$ subcategory of $\mathcal{C}$

The inclusion functor $i: \mathcal{D} \to \mathcal{C}$ is faithful

if $\mathcal{D}$ is a full subcategory of $\mathcal{C}$ then $i: \mathcal{D} \to \mathcal{C}$ is fully faithful

(2) Forgetful functors $Grp \to Set$ $Ring \to Set$ $A\text{-Mod} \to Set$ $Top \to Set$ are all faithful

(3) Dual $D: Vect_k^{op} \to Vect_k$ $V \mapsto D(V) = V^* = \mathrm{Hom}_k(V, k)$

$D = \mathrm{Hom}(-, k)$ $(f: V \to W) \mapsto D(f) = (f^*: W^* \to V^*)$

(4) $\mathcal{C}$ category $\mathrm{Hom}_{\mathcal{C}}(-,-): \mathcal{C}^{op} \times \mathcal{C} \to Set$

objects $(X, Y) \mapsto \mathrm{Hom}_{\mathcal{C}}(X, Y)$

$f: X' \to X$, $g: Y \to Y'$: $(f, g) \downarrow$, $\varphi \downarrow$

$(X', Y') \mapsto \mathrm{Hom}_{\mathcal{C}}(X', Y')$ $g\varphi f$

Given an object X in $\mathcal{C}$ we have two functors

$h^X := \mathrm{Hom}_{\mathcal{C}}(X, -)$ $\mathcal{C} \to Set$

$h_X := \mathrm{Hom}_{\mathcal{C}}(-, X)$ $\mathcal{C}^{op} \to Set$

(5) Let A be a ring $A\text{-Mod}$ Category of A-modules

$\mathrm{Hom}_A(-,-): A\text{-Mod}^{op} \times A\text{-Mod} \to A\text{-Mod}$

$(M, N) \mapsto \mathrm{Hom}_A(M, N)$

$(f, g) \downarrow$, $\varphi \downarrow$, $g\varphi f$

$(M', N') \mapsto \mathrm{Hom}_A(M', N')$

$$-\underset{A}{\otimes}- : A\text{-Mod} \times A\text{-Mod} \to A\text{-Mod}$$

$$\begin{array}{ccc} (M,N) & \mapsto & M \underset{A}{\otimes} N \\ \downarrow (f,g) & & \downarrow T(f,g) \\ (M',N') & \mapsto & M' \underset{A}{\otimes} N' \end{array}$$

Given an A-module M we have functors

$$\mathrm{Hom}_A(M,-) : A\text{-Mod} \to A\text{-Mod}$$

$$\mathrm{Hom}_A(-,M) : A\text{-Mod}^{op} \to A\text{-Mod}$$

$$M \underset{A}{\otimes} - : A\text{-Mod} \to A\text{-Mod}$$

(6) Let $f: A \to B$ be a ring map

$$|_A : B\text{-Mod} \to A\text{-Mod} \; ; \; N \mapsto N|_A$$

$$B \underset{A}{\otimes} - : A\text{-Mod} \to B\text{-Mod} \; ; \; M \mapsto M_B$$

$$\mathrm{Hom}_A(B,-) \; A\text{-Mod} \to B\text{-Mod} \; ; \; M \mapsto \mathrm{Hom}_A(B,M)$$

Note $|_A$ is faithful

(7) Let A be a ring $\mathfrak{a} \subseteq A$ an ideal $\pi : A \to A/\mathfrak{a}$

$A/\mathfrak{a}$-Mod $\to A$-Mod is fully faithful

M, N $A/\mathfrak{a}$ modules $\mathrm{Hom}_{A/\mathfrak{a}}(M,N) = \mathrm{Hom}_A(M,N)$

Def. If $F: \mathcal{A} \to \mathcal{B}$ $G: \mathcal{B} \to \mathcal{C}$ are functors then the composite functor $GF: \mathcal{A} \to \mathcal{C}$ is defined as follows: $GF(A) = G(F(A))$ $GF(f) = G(F(f))$

If $\mathcal{C}$ is a category then the identity functor $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is the rule fixing all objects and all morphisms

$F: \mathcal{A} \to \mathcal{B}$ $G: \mathcal{B} \to \mathcal{C}$ $H: \mathcal{C} \to \mathcal{D}$ functors

$\Rightarrow (HG)F = H(GF)$ $\mathrm{id}_{\mathcal{B}} F = F = F \mathrm{id}_{\mathcal{A}}$

We can form a category Cat whose objects are small categories and whose morphisms are functors.

§2. Natural transformations

Def. Let $F, G: \mathcal{A} \to \mathcal{B}$ be functors, A natural transformation $\alpha: F \Rightarrow G$ is a rule that associates to each object A of $\mathcal{A}$ a morphism $\alpha_A: F(A) \to G(A)$ in $\mathcal{B}$ s.t. $\forall$ morphism $f: A_1 \to A_2$ in $\mathcal{A}$ the left diagram commutes

$\mathcal{A}$: A_1, A_2 ; $\mathcal{B}$: $F(A_1)$, $F(A_2)$, $G(A_1)$, $G(A_2)$

$\alpha = (\alpha_A)_{A \in \mathcal{A}}$

Say α_A is natural in A

Furthermore if α_A is an isom for every $A \in \mathcal{A}$, then α is said to be a natural isomorphism. Two functors F, G are said to be natural isomorphic if $\exists$ a natural isom $\alpha: F \Rightarrow G$

e.g. $D: Vect_k \to Vect_k^{op}$ $V \mapsto D(V) = V^*$ $f \mapsto D(f) = f^*$

$D^{op}: Vect_k^{op} \to Vect_k$

$D^2: Vect_k \to Vect_k$: $V \mapsto V^{**}$ $(f: V \to W) \mapsto (f^{**}: V^{**} \to W^{**})$

$\alpha_V: V \to V^{**}$ $v \mapsto (f \mapsto f(v))$

$\Rightarrow \alpha = (\alpha_V)_{V \in Vect_k}$ is a natural transf. from id_{Vect_k} to D^2

$V \xrightarrow{\alpha_V} V^{**}$, $\varphi \downarrow$, $\varphi^{**} \downarrow$, $W \xrightarrow{\alpha_W} W^{**}$

$\alpha_V: V \to V^{**}$ is an isom if $\dim_k V < \infty$

$\Rightarrow \alpha$ is a natural isom from $id_{Vect_k^{f.d.}}$ to $D^2_{Vect_k^{f.d.}}$

e.g. $\mathcal{C}$ category $X, Y \in \mathcal{C} \longrightarrow h^X = Hom_{\mathcal{C}}(X, -)$ $\mathcal{C} \to Set$ functor

$h^Y = Hom_{\mathcal{C}}(Y, -)$

$f: X \to Y$ a morphism in $\mathcal{C}$

induces a natural transf $\alpha(f): Hom_{\mathcal{C}}(Y, -) \Rightarrow Hom_{\mathcal{C}}(X, -)$

The value of $\alpha(f)$ at $Z \in \mathcal{C}$ is the set map

$\alpha(f)_Z: Hom_{\mathcal{C}}(Y, Z) \longrightarrow Hom_{\mathcal{C}}(X, Z)$

$\varphi \longmapsto \varphi f$

$h_X = \mathrm{Hom}_{\mathcal{C}}(-, X) : \mathcal{C}^{op} \to \mathrm{Set}$

$h_Y = \mathrm{Hom}_{\mathcal{C}}(-, Y)$

$f: X \to Y$ induces a natural transf $\beta(f): \mathrm{Hom}_{\mathcal{C}}(-, X) \Rightarrow \mathrm{Hom}_{\mathcal{C}}(-, Y)$

the value of which at $Z \in \mathcal{C}$ is given by $\beta(f)_Z : \mathrm{Hom}_{\mathcal{C}}(Z, X) \to \mathrm{Hom}_{\mathcal{C}}(Z, Y)$
$\psi \mapsto f\psi$

functor category of $\mathcal{A}, \mathcal{B}$: $[\mathcal{A}, \mathcal{B}]$

Def. Let $F: \mathcal{A} \to \mathcal{B}$ be a functor

The identity natural transf $\mathrm{id}_F : F \Rightarrow F$ is defined by $(\mathrm{id}_F)_A = \mathrm{id}_{F(A)}$ $\forall A \in \mathcal{A}$

Given functors $F, G, H : \mathcal{A} \to \mathcal{B}$ and natural transf $\alpha: F \Rightarrow G$

$\beta: G \Rightarrow H$ the composite natural transf $\beta\alpha: F \Rightarrow H$ is defined by

$(\beta\alpha)_A = \beta_A \alpha_A \quad \forall A \in \mathcal{A}$

Prop Given a small category $\mathcal{A}$ and a category $\mathcal{B}$ there is a functor category $[\mathcal{A}, \mathcal{B}]$ whose objects are functors from $\mathcal{A}$ to $\mathcal{B}$ and whose morphisms are natural transf. $\mathrm{Ob}([\mathcal{A}, \mathcal{B}]) = \{F: \mathcal{A} \to \mathcal{B}\}$

$\mathrm{Hom}_{[\mathcal{A}, \mathcal{B}]}(F, G) = \{\alpha: F \Rightarrow G\} =: \mathrm{Nat}(F, G)$

The identity morphisms in $[\mathcal{A}, \mathcal{B}]$ are identity natural transf

The composition fcn in $[\mathcal{A}, \mathcal{B}]$ is given by composition of natural transf defined above.

Ex. Check that the isoms in $[\mathcal{A}, \mathcal{B}]$ are precisely the natural isoms defined earlier. Hence, two functors $F, G: \mathcal{A} \to \mathcal{B}$ are naturally isom (in the sense of last lecture) iff they are isomorphic as objects of $[\mathcal{A}, \mathcal{B}]$

Def. A functor $F: \mathcal{A} \to \mathcal{B}$ is an isomorphism of categories if

$\exists$ a functor $G: \mathcal{B} \to \mathcal{A}$ s.t. $GF = \mathrm{id}_{\mathcal{A}}$ $FG = \mathrm{id}_{\mathcal{B}}$

Ex. Let $\mathcal{A}, \mathcal{B}$ be small categories $F: \mathcal{A} \to \mathcal{B}$ a functor Then TFAE.

(1) F is an isom of categories

(2) F is an isomorphism in Cat

(3) F is bijective on objects and hom-sets. That is $F: \mathrm{Ob}(\mathcal{A}) \to \mathrm{Ob}(\mathcal{B})$

is a bijection and $\forall A_1, A_2 \in \mathcal{A}$, $F: \mathrm{Hom}_{\mathcal{A}}(A_1, A_2) \to \mathrm{Hom}_{\mathcal{B}}(F(A_1), F(A_2))$ is a bijection.

e.g. $\mathcal{C} \cong (\mathcal{C}^{op})^{op}$

$\mathcal{A}, \mathcal{B}$ small $[\mathcal{A} \times \mathcal{B}, \mathcal{C}] \cong [\mathcal{A}, [\mathcal{B}, \mathcal{C}]]$

Def. A functor $F: \mathcal{A} \to \mathcal{B}$ is an *equivalence of categories* if $\exists$ a functor $G: \mathcal{B} \to \mathcal{A}$ s.t. $GF \cong id_{\mathcal{A}}$ $FG \cong id_{\mathcal{B}}$

e.g. $D: \mathrm{Vect}_k^{f.d.} \to \mathrm{Vect}_k^{f.d\,op}$ is an equivalence of categories since $D^2 \cong id_{\mathrm{Vect}_k^{f.d.}}$.

Def. A functor $F: \mathcal{A} \to \mathcal{B}$ is *essentially surjective* if $\forall B \in \mathcal{B}$ $\exists A \in \mathcal{A}$ s.t. $F(A) \cong B$

Ex. Let $\mathcal{A}, \mathcal{B}$ be small categories, $F: \mathcal{A} \to \mathcal{B}$ a functor. Then TFAE

(1) F is an equivalence of categories

(2) F is essentially surjective and fully faithful

Def. A *skeleton* of a category $\mathcal{C}$ is a small subcategory $\mathcal{S}$ of $\mathcal{C}$ s.t. every object of $\mathcal{C}$ is isomorphic (in $\mathcal{C}$) to a unique object in $\mathcal{S}$.

Cor. If $\mathcal{S}$ is a skeleton of $\mathcal{C}$, then the inclusion functor $i: \mathcal{S} \to \mathcal{C}$ is an equivalence of categories

e.g. $\mathrm{Vect}_k^{f.d.}$ is equivalent to the category whose objects are $\{k^n, n \geq 0\}$ and whose morphisms are $\mathrm{Hom}_k(k^n, k^m) = M_{m \times n}(k)$

Fix a ~~$n \geq 0$~~ $n \geq 1$, Vect_k^n = the category of n-dim'l k-vector spaces is equivalent to the category of one object k^n with endomorphisms $\mathrm{Hom}_k(k^n, k^n) = M_n(k)$.

§3 Adjoint functors.

Def. A pair of functors $F: \mathcal{A} \to \mathcal{B}$ and $G: \mathcal{B} \to \mathcal{A}$ is *adjoint* if there is a set bijection

$$\alpha_{A,B}: \mathrm{Hom}_{\mathcal{B}}(F(A), B) \to \mathrm{Hom}_{\mathcal{A}}(A, G(B))$$

for every $A \in \mathcal{A}$, $B \in \mathcal{B}$ s.t. $\alpha_{A,B}$ is natural in both A and B.

MEMO NO: _____
DATE: / /

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That is $\forall f: A_1 \to A_2$ $\quad \mathrm{Hom}_{\mathcal{B}}(F(A_1), B) \xrightarrow[\cong]{\alpha_{A_1,B}} \mathrm{Hom}_{\mathcal{A}}(A_1, G(B))$

$\uparrow -F(f) \quad \curvearrowright \quad \uparrow -f$

$\mathrm{Hom}_{\mathcal{B}}(F(A_2), B) \xrightarrow[\cong]{\alpha_{A_2,B}} \mathrm{Hom}_{\mathcal{A}}(A_2, G(B))$

$\forall g: B_1 \to B_2$ $\quad \mathrm{Hom}_{\mathcal{B}}(F(A), B_1) \xrightarrow[\cong]{\alpha_{A,B_1}} \mathrm{Hom}_{\mathcal{A}}(A, G(B_1))$

$\downarrow g- \quad \curvearrowright \quad \downarrow G(g)-$

$\mathrm{Hom}_{\mathcal{B}}(F(A), B_2) \xrightarrow[\cong]{\alpha_{A,B_2}} \mathrm{Hom}_{\mathcal{A}}(A, G(B_2))$

$\Leftrightarrow \forall f: A_1 \to A_2 \quad g: B_1 \to B_2$ $\quad \mathrm{Hom}_{\mathcal{B}}(F(A_2), B_1) \xrightarrow[\cong]{\alpha_{A_2,B_1}} \mathrm{Hom}_{\mathcal{A}}(A_2, G(B_1))$

$\downarrow g - F(f) \quad \curvearrowright \quad \downarrow G(g) - f$

$\mathrm{Hom}_{\mathcal{B}}(F(A_1), B_2) \xrightarrow[\cong]{\alpha_{A_1,B_2}} \mathrm{Hom}_{\mathcal{A}}(A_1, G(B_2))$

$\Leftrightarrow \alpha = (\alpha_{A,B})_{A,B} : \mathrm{Hom}_{\mathcal{B}}(F(-),-) \Rightarrow \mathrm{Hom}_{\mathcal{A}}(-, G(-))$ is a natural ~~transf~~ isomorphism

both are functors from $\mathcal{A}^{op} \times \mathcal{B}$ to Set

e.g. (1) Let M be an A-module

Then $M \otimes_A - : A\text{-Mod} \to A\text{-Mod}$ is left-adjoint to $\mathrm{Hom}_A(M,-) : A\text{-Mod} \to A\text{-Mod}$

pf. Show that the isomorphism $\mathrm{Hom}_A(M \otimes_A N, P) \cong \mathrm{Hom}_A(N, \mathrm{Hom}_A(M,P))$

is natural in N and P

(2) Let $f: A \to B$ be a ring map

Then $B \otimes_A - : A\text{-Mod} \to B\text{-Mod}$ is left-adjoint to $f_A : B\text{-Mod} \to A\text{-Mod}$

$f_A : B\text{-Mod} \to A\text{-Mod}$ - - - $\mathrm{Hom}_A(B,-) : A\text{-Mod} \to B\text{-Mod}$

(3) $\mathrm{Hom}_{A\text{-Mod}}(A^{(I)}, M) \cong \mathrm{Map}(I, M) = \mathrm{Hom}_{\mathrm{Set}}(I, F'(M))$ A ring I set

$F' : A\text{-Mod} \to \mathrm{Set}$ ~~forgetful~~ forgetful functor has a left adjoint

$F : \mathrm{Set} \to A\text{-Mod}$; $I \to A^{(I)}$, $(f: I \to J) \mapsto (F(f) : A^{(I)} \to A^{(J)}$, $e_i \mapsto e_{f(i)})$

§4. Yoneda Lemma and representable functors

$\mathcal{C}$ category $\quad x \in \mathcal{C} \quad h^x = \mathrm{Hom}_{\mathcal{C}}(X,-) : \mathcal{C} \to \mathrm{Set} \quad \in [\mathcal{C}, \mathrm{Set}]$

$h_x = \mathrm{Hom}_{\mathcal{C}}(-, X) : \mathcal{C}^{op} \to \mathrm{Set} \quad \in [\mathcal{C}^{op}, \mathrm{Set}]$

$f: X \to Y \rightsquigarrow \alpha(f) : h^Y \Rightarrow h^X \quad \beta(f) : h_X \Rightarrow h_Y$

Yoneda embeddings

$h^{\cdot}: \mathcal{C}^{op} \to [\mathcal{C}, Set]$ $\quad$ $h_{\cdot}: \mathcal{C} \to [\mathcal{C}^{op}, Set]$

$X \mapsto Hom_{\mathcal{C}}(X, -)$ $\quad$ $X \to Hom_{\mathcal{C}}(-, X)$

$f \downarrow$ $\quad \Uparrow \alpha(f)$ $\quad$ $f \downarrow$ $\quad \Downarrow \beta(f)$

$Y \mapsto Hom_{\mathcal{C}}(Y, -)$ $\quad$ $Y \to Hom_{\mathcal{C}}(-, Y)$

$Hom_{\mathcal{C}}(-,-): \mathcal{C}^{op} \times \mathcal{C} \to Set$

$\uparrow [\mathcal{C}^{op} \times \mathcal{C}, Set] \cong [\mathcal{C}^{op}, [\mathcal{C}, Set]] \cong [\mathcal{C}, [\mathcal{C}^{op}, Set]]$

Prop. For any functor $F: \mathcal{C} \to Set$ $X \in \mathcal{C}$ there is a set bijection

$Nat(h^X, F) \cong F(X)$ $\quad \alpha_X: h^X(X) \to F(X)$

given by $\alpha \mapsto \alpha_X(id_X)$

pf. The inverse is given by $u \in F(X) \mapsto \alpha(u): h^X \to F$

$\alpha(u)_Y: h^X(Y) \to F(Y)$ $\quad (\varphi \mapsto F(\varphi)(u))$

$\| \quad\quad\quad F(\varphi): F(X) \to F(Y)$

$Hom_{\mathcal{C}}(X, Y)$ $\quad\quad\quad

(1) If $h^X \simeq h^Y$, then $X \simeq Y$

(2) If $h_X \simeq h_Y$, then $X \simeq Y$

e.g. $(\bigoplus_{i\in I} M_i) \underset{A}{\otimes} N \simeq \bigoplus_{i\in I} (M_i \underset{A}{\otimes} N)$ as A-Mod

$\forall P \in A\text{-Mod}$

$$\mathrm{Hom}_A((\bigoplus_{i\in I} M_i) \underset{A}{\otimes} N, P) \overset{\text{tensor-hom adj}}{\simeq} \mathrm{Hom}_A(\bigoplus_{i\in I} M_i, \mathrm{Hom}_A(N,P))$$

$$\overset{\text{u.p of } \oplus}{\simeq} \prod_{i\in I} \mathrm{Hom}_A(M_i, \mathrm{Hom}_A(N,P))$$

$$\simeq \prod_{i\in I} \mathrm{Hom}_A(M_i \underset{A}{\otimes} N, P) \simeq \mathrm{Hom}_A(\bigoplus_{i\in I} (M_i \underset{A}{\otimes} N), P)$$

and these bijections are natural in P

$$\Rightarrow \mathrm{Hom}_A((\bigoplus_{i\in I} M_i) \underset{A}{\otimes} N, -) \cong \mathrm{Hom}_A(\bigoplus_{i\in I} (M_i \underset{A}{\otimes} N), -)$$

$$\Rightarrow (\bigoplus_{i\in I} M_i) \underset{A}{\otimes} N \cong \bigoplus_{i\in I} (M_i \underset{A}{\otimes} N)$$

Def A functor $F: \mathcal{C} \to \mathrm{Set}$ is <u>representable</u> if $\exists X \in \mathcal{C}$ s.t. $F \cong h^X$

A <u>representation</u> of a (representable) functor $F: \mathcal{C} \to \mathrm{Set}$ is a pair (X, u) with $X \in \mathcal{C}$, $u \in F(X)$ s.t. $\forall Y \in \mathcal{C}$

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \to F(Y), \quad f \mapsto F(f)(u) \text{ is a bijection}$$

X is a <u>representing object</u> for F, u is called a <u>universal elt</u>

e.g. Let M, N be A-modules.

Define a functor $\mathrm{Bil}_A(M\times N, -): A\text{-Mod} \to \mathrm{Set}$ $\quad P \mapsto \mathrm{Bil}_A(M\times N, P)$

(the set of all A-bilinear maps from $M\times N$ to P)

$$\varphi: P \to P' \mapsto \mathrm{Bil}_A(M\times N, \varphi): \mathrm{Bil}_A(M\times N, P) \to \mathrm{Bil}_A(M\times N, P')$$
$$f \mapsto \varphi f$$

Q: Is $\mathrm{Bil}_A(M\times N, -)$ representable?

$\Leftrightarrow$ Does there exist an A-module T and an A-bilinear map $g: M\times N \to T$

s.t. $\forall P \in A\text{-Mod}$ $\quad \mathrm{Hom}_A(T, P) \to \mathrm{Bil}_A(M\times N, P)$ is a bijection $\varphi \mapsto \varphi g$

✓ $(M \underset{A}{\otimes} N,\ t: M\times N \to M \underset{A}{\otimes} N)$ is a representation of $\mathrm{Bil}_A(M\times N, -)$

e.g. Let $\{M_i\}_{i\in I}$ be a family of A-modules

The functor $\prod_{i\in I} \mathrm{Hom}_A(M_i, -): A\text{-Mod} \to \mathrm{Set}$ is represented by

$(\bigoplus_{i\in I} M_i, (\tau_i)_i)$ $\tau_i : M_i \to \bigoplus_{j\in I} M_j$

Def. A contravariant functor $F: \mathcal{C} \to \mathrm{Set}$ is <u>representable</u> if $\exists X \in \mathcal{C}$ s.t. $F \cong h_X$

A <u>representation</u> of F is a pair (X, u) where $X \in \mathcal{C}$, $u \in F(X)$ s.t.

$\forall Y \in \mathcal{C}$ $\mathrm{Hom}_{\mathcal{C}}(Y, X) \to F(Y)$; $f \mapsto F(f)(u)$ is a bijection

e.g. $\prod_{i\in I} \mathrm{Hom}_A(-, M_i) : A\text{-Mod}^{op} \to \mathrm{Set}$

is represented by $(\prod_{i\in I} M_i, (\pi_i)_i)$ where $\pi_i : \prod_{j\in I} M_j \to M_i$ is the ith projection

Prop. Let (X_1, u_1) and (X_2, u_2) be two representations of a functor $F: \mathcal{C} \to \mathrm{Set}$

Let $f: X_1 \to X_2$ be the unique morphism s.t. $F(f)(u_1) = u_2$

$g: X_2 \to X_1$ $F(g)(u_2) = u_1$

Then $fg = id_{X_2}$ $gf = id_{X_1}$ in particular $X_1 \cong X_2$

Ex. Let $\mathcal{A}, \mathcal{B}$ be small categories $F: \mathcal{A} \to \mathcal{B}$ is a functor.

Then (1) F has a right adjoint iff $\forall Y \in \mathcal{B}$ the functor

$\mathrm{Hom}_{\mathcal{B}}(F(-), Y) : \mathcal{A}^{op} \to \mathrm{Set}$ is representable

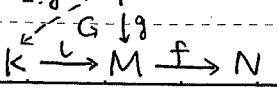
(2) If the right adjoint of F exists, then it is unique up to natural isomorphism.

Chapter 4 Functors on module categories Projective / injective / flat modules

§1. Categorical kernels and cokernels

Def. Let $f: M \to N$ be an A-~~m~~linear map. A <u>kernel</u> of f is a pair (K, ι) consisting of an A-module K and an A-~~m~~linear map $\iota: K \to M$ s.t. ① $f\iota = 0$

② $\forall$ A-~~lineg~~linear map $g: P \to M$ s.t. $fg = 0$ $\exists!$ A-linear map $g': P \to K$ s.t. $g = \iota g'$



$(\ker f, i: \ker f \to M)$ is a kernel of f

Prop Let $f: M \to N$, $L: K \to M$ be A-module homomorphisms

Then TFAE.

(1) (K, L) is a kernel of f

(2) (K, L) is a representation of the functor

$$K(f): A\text{-Mod}^{op} \to \text{Set}$$
$$P \mapsto \{\varphi \in \text{Hom}_A(P, M): f\varphi = 0\}$$

(3) L is injective and $\text{Im}L = \ker f$

Def Let $f: M \to N$ be an A-module homomorphism.

A cokernel of f is a pair (C, π) consisting of an A-module C and an A-linear map $\pi: N \to C$ s.t.

$$\begin{array}{ccccc} M & \xrightarrow{f} & N & \xrightarrow{\pi} & C \\ & & {\scriptstyle g}\downarrow & \swarrow_{\exists! g'} & \\ & & P & & \end{array}$$

① $\pi f = 0$

② $\forall$ A-linear map $g: N \to P$ s.t. $gf = 0$ $\exists!$ A-linear map $g': C \to P$ s.t. $g = g'\pi$

$(N/\text{Im}f, q)$ where $q: N \to N/\text{Im}f$ is the quotient map is a cokernel of f

Prop $f: M \to N$ $\pi: N \to C$ TFAE

(1) (C, π) is a cokernel of f

(2) (C, π) is a representation of the functor

$$\text{Cok}(f): A\text{-Mod} \to \text{Set}$$
$$P \mapsto \{\varphi \in \text{Hom}_A(N, P): \varphi f = 0\}$$

(3) π is surjective and $\ker\pi = \text{Im}f$

$\text{Coim}f = \text{coker}(\ker f)$

$\text{Im}f = \ker(\text{coker} f)$

$$\begin{array}{ccccccc} \ker f & \xrightarrow{i} & M & \xrightarrow{f} & N & \xrightarrow{q} & N/\text{Im}f = \text{coker} f \\ & & \downarrow & & \uparrow & & \\ & & \text{coim}(f) = M/\ker f & \xrightarrow{\cong} & \text{Im}f & & \end{array}$$

§2. exact sequences and snake lemma

Def A sequence of A-modules and A-module homomorphisms

$$\to M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \to \cdots$$

is exact at M_i if $\operatorname{Im} f_{i-1} = \ker f_i$

The sequence is exact if it is exact at every M_i

e.g. (1) $0 \to M \xrightarrow{f} N$ is exact $\Leftrightarrow \ker f = 0 \Leftrightarrow f$ is injective

(2) $M \xrightarrow{f} N \to 0$ is exact $\Leftrightarrow \operatorname{Im} f = N \Leftrightarrow f$ is surjective

(3) $0 \to M' \xrightarrow{f} M \xrightarrow{g} M''$ is exact

$\Leftrightarrow f$ is injective and $\operatorname{Im} f = \ker g$

$\Leftrightarrow (M', f)$ is a kernel of g

(4) $M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ is exact

$\Leftrightarrow g$ is surjective and $\operatorname{Im} f = \ker g$

$\Leftrightarrow (M'', g)$ is a cokernel of f

(5) $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ is exact

$\Leftrightarrow f$ is injective, g is surjective and $\operatorname{Im} f = \ker g$

Def A short exact sequence (ses) is an exact sequence of the form

$$0 \to M' \to M \to M'' \to 0$$

Rmk. If $N \subseteq M$ is a submodule then $0 \to N \xrightarrow{i} M \xrightarrow{q} M/N \to 0$ is a ses

Conversely, given a ses $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$, f induces an isom $\tilde{f}: M' \to \operatorname{Im} f$, g induces an isom $\tilde{g}: M/\operatorname{Im} f \to M''$

and the following diagram commutes

$$\begin{array}{ccccccc} 0 & \to & M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' \to 0 \\ & & \cong\downarrow \tilde{f} & \circlearrowright & \| \mathrm{id} & \circlearrowright & \cong\uparrow \tilde{g} \\ 0 & \to & \operatorname{Im} f & \to & M & \to & M/\operatorname{Im} f \to 0 \end{array}$$

An exact sequence $\cdots \to M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \xrightarrow{f_{i+1}} \cdots$ can be split into ses

Let $N_i = \ker f_i = \operatorname{Im} f_{i-1}$ then $0 \to N_i \xrightarrow{\iota} M_i \xrightarrow{f_i} N_{i+1} \to 0$ is a ses

Lemma If $f: M \to N$ and $g: N \to M$ are A-linear map s.t. $gf = \mathrm{id}_M$ then $N = \operatorname{Im} f \oplus \ker g$

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Prop Let $0 \to M' \xrightarrow{L} M \xrightarrow{\pi} M'' \to 0$ (*) be a ses. Then TFAE

(1) $\mathrm{Im}\, L$ is a direct summand of M

(2) $\exists$ A-linear map $\pi' : M \to M'$ s.t. $\pi' L = id_{M'}$

(3) $\exists$ A-linear map $L' : M'' \to M$ s.t. $\pi L' = id_{M''}$

When the above conditions hold, we say the ses (*) splits

pf. (1) $\Rightarrow$ (2)+(3) $M = \mathrm{Im}\, L \oplus N$

let π' be the unique A-linear map s.t. $\pi'|_{\mathrm{Im}\, L} = (\tilde{L} : M' \xrightarrow{\cong} \mathrm{Im}\, L)^{-1}$

and $\pi'|_N = 0$

Verify that $\pi|_N : N \to M''$ is an isom. Define $L' = j (\pi|_N)^{-1}$ where $j : N \hookrightarrow M$ is inclusion

Then $\pi' L = id_{M'}$ $\pi L' = id_{M''}$ $\pi' L' = 0$ $L\pi' + L'\pi = id_M$

(2) $\Rightarrow$ (1) $M = \mathrm{Im}\, L \oplus \ker \pi'$ by lemma

(3) $\Rightarrow$ (1) $M = \ker \pi \oplus \mathrm{Im}\, L'$ by lemma

Cor Let M be an A-module $N \subseteq M$ a submodule. Then TFAE

(1) $0 \to N \xrightarrow{i} M \xrightarrow{q} M/N \to 0$ splits

(2) N is a direct summand of M

(3) $\exists\, p : M \to N$ s.t. $pi = id_N$

(4) $\exists\, j : M/N \to M$ s.t. $qj = id_{M/N}$

Rmk Let $0 \to M' \underset{\pi'}{\overset{L}{\rightleftarrows}} M \underset{L'}{\overset{\pi}{\rightleftarrows}} M'' \to 0$ be a split ses

and Let π', L' be the A-linear map constructed earlier from a direct sum decomp $M = \mathrm{Im}\, L \oplus N$

Then we have the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & M' & \xrightarrow{L} & M & \xrightarrow{\pi} & M'' & \to & 0 \\ & & \| & & \downarrow \binom{\pi'}{\pi} & & \| & & \\ 0 & \to & M' & \xrightarrow[\binom{1}{0}]{} & M' \oplus M'' & \xrightarrow[(0\ 1)]{} & M'' & \to & 0 \end{array}$$

$(L\ L')\binom{\pi'}{\pi} = L\pi' + L'\pi = id_M$

$\binom{\pi'}{\pi}(L\ L') = \begin{pmatrix} \pi' L & \pi' L' \\ \pi L & \pi L' \end{pmatrix} = \begin{pmatrix} id_{M'} & \\ & id_{M''} \end{pmatrix} = id_{M' \oplus M''}$

Thm (Snake lemma) Let

$$\begin{array}{ccccccc} M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' & \to & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \to & N' & \xrightarrow[u']{} & N & \xrightarrow[v']{} & N'' \end{array}$$

be a commutative diagram of A-modules and A-module homo with the rows exact. Then there is an exact sequence of A-modules

$$\ker f' \xrightarrow{\bar{u}} \ker f \xrightarrow{\bar{v}} \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \xrightarrow{\bar{u}'} \operatorname{coker} f \xrightarrow{\bar{v}'} \operatorname{coker} f''$$

in which $\bar{u}, \bar{v}$ are induced by u, v, $\bar{u}', \bar{v}'$ are induced by u', v'

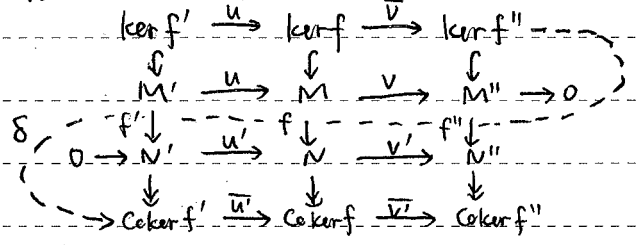
Moreover if u is injective, then so is $\bar{u}$

if v' is surjective, then so is $\bar{v}'$

pf. The commutativity of

$$\begin{array}{ccc} M' & \xrightarrow{u} & M \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ N' & \xrightarrow[u']{} & N \end{array} \quad \Rightarrow u(\ker f') \subseteq \ker f \qquad \bar{u}: \ker f' \to \ker f$$

Similarly $u'(\operatorname{Im} f') \subseteq \operatorname{Im} f$ $\quad \bar{u}': N'/\operatorname{Im} f' \to N/\operatorname{Im} f$



$\forall x'' \in \ker f''$ choose $x \in M$ s.t. $v(x) = x''$ $\qquad x \overset{v}{\mapsto} x''$

$\exists! y'$ s.t. $u'(y') = f(x)$ $\qquad y' \mapsto \bar{y}' \in \operatorname{coker} f'$ $\qquad \exists! y' \overset{u'}{\mapsto} f(x) \overset{v'}{\mapsto} 0$

Define $\delta(x'') = \bar{y}'$

Prop Let $0 \to M' \xrightarrow{f} M \xrightarrow{g} M''$ be an exact sequence of A-modules

Then for any A-module P

$0 \to \operatorname{Hom}_A(P, M') \xrightarrow{f\square} \operatorname{Hom}_A(P, M) \xrightarrow{g\square} \operatorname{Hom}_A(P, M'')$ is exact

pf. $gf = 0 \Rightarrow (g\square)(f\square) = 0 \Rightarrow \operatorname{Im}(f\square) \subseteq \ker(g\square)$

If $\varphi \in \ker(g\square)$ i.e. $g\varphi = 0$

$\exists! \varphi'$ $\quad$ (M', f) is a kernel of $g \Rightarrow \exists! \varphi' \in \operatorname{Hom}_A(P, M')$

$$\begin{array}{ccccccc} & & & & P & & \\ & & & \swarrow^{\exists! \varphi'} & \downarrow{\scriptstyle \varphi} & & \\ 0 & \to & M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' \end{array} \quad \text{s.t. } \varphi = f\varphi' = (f\square)(\varphi')$$

$\Rightarrow \ker(g_\sharp) \subseteq \mathrm{Im}(f_\sharp)$ and $f_\sharp$ is injective

Prop Let $M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ be an exact sequence of A-modules

Then for any A-module P

$0 \to \mathrm{Hom}_A(M'', P) \xrightarrow{\sharp g} \mathrm{Hom}_A(M, P) \xrightarrow{\sharp f} \mathrm{Hom}_A(M', P)$ is exact

Rmk If $S.$ is a sequence in A-Mod and $\forall P \in A\text{-Mod}$

$\mathrm{Hom}_A(P, S.)$ is ~~a se~~ exact, then ~~es~~ $S.$ is exact

(Yoneda embedding reflects exact sequences)

Prop Let $M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$ be an exact sequence of A-modules

Then for any A-module N

$$M' \underset{A}{\otimes} N \xrightarrow{T(f, id_N)} M \underset{A}{\otimes} N \xrightarrow{T(g, id_N)} M'' \underset{A}{\otimes} N \to 0$$

Pf $gf = 0 \Rightarrow T(g, id_N)\, T(f, id_N) = T(gf, id_N) = T(0, id_N) = 0$

$$M' \underset{A}{\otimes} N \xrightarrow{T(f,id)} M \underset{A}{\otimes} N \xrightarrow{\pi} M \underset{A}{\otimes} N / \mathrm{Im}\, T(f, id) \to 0$$

$$M \underset{A}{\otimes} N \xrightarrow{T(g,id)} M'' \underset{A}{\otimes} N \quad \text{(the induced map } \overline{T(g,id)} \text{ on the quotient)}$$

Goal: $\overline{T(g,id)}$ is an isom

Define $\varphi: M'' \times N \to M \underset{A}{\otimes} N / \mathrm{Im}\, T(f, id)$ $(x'', y) \mapsto \overline{x \otimes y}$ where x is any elt in M s.t. $g(x) = x''$

If $x_1, x_2 \in M$ $g(x_1) = g(x_2) = x''$ $x_1 - x_2 \in \ker g = \mathrm{Im} f$

$\Rightarrow x_1 \otimes y - x_2 \otimes y = (x_1 - x_2) \otimes y \in \mathrm{Im}\, T(f, id)$ φ is well-defined

Verify that φ is A-bilinear hence $\exists$ an A-linear map $\psi: M'' \underset{A}{\otimes} N \to M \underset{A}{\otimes} N / \mathrm{Im}\, T(f, id)$

s.t. $\psi(x'' \otimes y) = \overline{x \otimes y}$ where $g(x) = x''$

Verify $\psi\, \overline{T(g, id)} = id$ $\overline{T(g, id)}\, \psi = id \Rightarrow \overline{T(g, id)}$ is an isom

$\Rightarrow (M'' \underset{A}{\otimes} N, T(g, id))$ is a cokernel of $T(f, id)$

§3. Left exact and right exact functors

Def Let A, B be rings. A functor $F: A\text{-Mod} \to B\text{-Mod}$ is **additive** if $\forall M_1, M_2 \in A\text{-Mod}$ $F: \mathrm{Hom}_A(M_1, M_2) \to \mathrm{Hom}_B(F(M_1), F(M_2))$ (*)

is a homomorphism of ~~ab~~ abelian gps That is $F(f+g) = F(f) + F(g)$ $\forall f, g: M_1 \to M_2$

Furthermore if a ring map $f: A \to B$ is given and ~~then F is A-linear if~~ $(*)$ is an A-linear map, then we say F is A-linear

e.g. Let M be an A-module

Then $\mathrm{Hom}_A(M, -) : A\text{-Mod} \to A\text{-Mod}$ are all A-linear

$\mathrm{Hom}_A(-, M) : A\text{-Mod}^{op} \to A\text{-Mod}$

$M \otimes_A - : A\text{-Mod} \to A\text{-Mod}$

If $f: A \to B$ is a ring map, then the functors

$B \otimes_A - : A\text{-Mod} \to B\text{-Mod}$ are A-linear

$\mathrm{Hom}_A(B, -) : A\text{-Mod} \to B\text{-Mod}$

Prop. If $F: A\text{-Mod} \to B\text{-Mod}$ is an additive functor,

then F transforms zero morphisms to zero morphisms and zero module to zero module

Moreover $\forall\, F(M_1 \oplus M_2) \simeq F(M_1) \oplus F(M_2)$ as B-modules

Def Let $F: A\text{-Mod} \to B\text{-Mod}$ be an additive functor

F is left exact if F preserves kernels. That is, $\forall$ se exact sequence

$0 \to M' \xrightarrow{f} M \xrightarrow{g} M''$ of A-modules, the sequence of B-modules

$0 \to F(M') \xrightarrow{F(f)} F(M) \xrightarrow{F(g)} F(M'')$ is exact

F is right exact if F preserves cokernels. That is $\forall$ exact seq

$M' \to M \to M'' \to 0$ in A-Mod the seq $F(M') \to F(M) \to F(M'') \to 0$ in B-Mod is exact

F is exact if it is both left exact and right exact

Def Let $F: A\text{-Mod} \to B\text{-Mod}$ be a contravariant additive functor

then F is left exact if F transforms cokernels into kernels

i.e. $M' \to M \to M'' \to 0$ exact $\Rightarrow$ $0 \to F(M'') \to F(M) \to F(M')$ exact

F is right exact if F transforms kernels into cokernels

F is exact if it is both left exact and right exact

e.g. $\mathrm{Hom}_A(M, -) : A\text{-Mod} \to A\text{-Mod}$ is left exact

$\mathrm{Hom}_A(-, M) : A\text{-Mod}^{op} \to A\text{-Mod}$ is left exact

$M \otimes_A -$: A-Mod → A-Mod is right exact

Prop Let F: A-Mod → B-Mod be an exact functor. Then TFAE

(1) F is faithful i.e. $\forall M_1, M_2 \in$ A-Mod $\mathrm{Hom}_A(M_1, M_2) \to \mathrm{Hom}_B(F(M_1), F(M_2))$ is injective

(2) F reflects zero modules i.e. if $M \in$ A-Mod and $F(M) = 0$ then $M = 0$

(3) F reflects zero homomorphisms i.e. if $f: M_1 \to M_2$ is A-linear and $F(f) = 0$ then $f = 0$

(4) F reflects exact sequences i.e. if $M_1 \to M_2 \to M_3$ is a sequence in A-Mod and $F(M_1) \to F(M_2) \to F(M_3)$ is exact in B-Mod then $M_1 \to M_2 \to M_3$ is exact in A-Mod

Prop Let $f: A \to B$ be a ring map. Then

(1) $|_A$: B-Mod → A-Mod is faithfully exact. That is, a sequence of B-modules and B-module homomorphism is exact in B-Mod iff its restriction to A is exact in A-Mod

(2) $B \otimes_A -$: A-Mod → B-Mod is right exact

(3) $\mathrm{Hom}_A(B, -)$: A-Mod → B-Mod is left exact

pf. (1) given a sequence $M' \xrightarrow{f} M \xrightarrow{g} M''$ of A-modules

$\ker g$, $\mathrm{Im} f$ only depend on the ~~abl~~ abelian group structure of M', M, M''

Rmk A-Mod $\underset{G}{\overset{F}{\rightleftarrows}}$ B-Mod adjoint pair of additive functors

⇒ F is right exact and G is left exact

$(B \otimes_A -, |_A)$ $(|_A, \mathrm{Hom}_A(B, -))$ $(M \otimes_A -, \mathrm{Hom}_A(M, -))$

Def Let M be an A-module. Then

M is <u>projective</u> if $\mathrm{Hom}_A(M, -)$ is exact

<u>injective</u> if $\mathrm{Hom}_A(-, M)$ is exact

<u>flat</u> if $M \otimes_A -$ is exact

§4. projective modules

Prop An A-module P is projective iff $\forall$ $\pi: M \twoheadrightarrow N$, $f: P \to N$ $\exists \tilde{f}: P \to M$ s.t. $f = \pi\tilde{f}$

$$\begin{array}{ccccc} & & P & & \\ & \swarrow^{\tilde{f}} & \circlearrowright & \downarrow f & \\ M & \xrightarrow{\pi} & N & \to & 0 \end{array}$$

pf. Since $\mathrm{Hom}_A(P,-)$ is left exact. P projective $\Leftrightarrow$ $\mathrm{Hom}_A(P,-)$ is exact

$\Leftrightarrow$ $\mathrm{Hom}_A(P,-)$ preserves surjective homo $\Leftrightarrow \forall \pi: M \twoheadrightarrow N$ $\pi_*: \mathrm{Hom}_A(P,M) \to \mathrm{Hom}_A(P,N)$ is surjective, $\varphi \mapsto \pi\varphi$

Cor Free modules are projective.

pf. F has basis $\{e_i\}_{i\in I}$

$$\begin{array}{ccc} & & F \\ & \swarrow^{\tilde{f}} \quad \swarrow^{\pi} & \downarrow f \\ M & \xrightarrow{\quad} & N \end{array}$$

$\forall i$ choose an $x_i \in M$ s.t. $\pi(x_i) = f(e_i)$

u.p. of free modules $\Rightarrow \exists!$ A-linear map $\tilde{f}: F \to M$ s.t. $\tilde{f}(e_i) = x_i$

$\pi\tilde{f}(e_i) = \pi(x_i) = f(e_i)$ $\pi\tilde{f}$ and f agree on $\{e_i\}_{i\in I}$ $\Rightarrow \pi\tilde{f} = f$

($\pi\tilde{f}$: A-linear; $\{e_i\}_{i\in I}$: generate F)

Prop $\bigoplus_{i\in I} P_i$ is projective $\Leftrightarrow$ each P_i is projective

Thm Let P be an A-module. Then TFAE

(1) P is projective

(2) $\forall M$ $\pi: M \twoheadrightarrow N$, $f: P \to N$ $\exists \tilde{f}: P \to M$ s.t. $\pi\tilde{f} = f$

(3) Every ses $0 \to N \to M \to P \to 0$ splits

Equivalently $\forall$ $\pi: M \twoheadrightarrow P$ $\exists l: P \to M$ s.t. $\pi l = \mathrm{id}_P$

(4) P is a direct summand of a free module.

pf. (1)$\Leftrightarrow$(2) Prop (2)$\Rightarrow$(3) take $f = \mathrm{id}_P$

(3)$\Rightarrow$(2) Let

$$\begin{array}{ccc} L & \xrightarrow{P_2} & P \\ {\scriptstyle P_1}\downarrow & \circlearrowright & \downarrow f \\ M & \xrightarrow[\pi]{} & N \end{array}$$

be the "pullback" diagram where $L = \{(x,z) \in M \oplus P : \pi(x) = f(z)\}$ is a submodule of $M \oplus P$

P_1, P_2 are projections

π is surjective $\Rightarrow P_2$ is surjective Then $\exists l: P \to L$ s.t. $P_2 l = \mathrm{id}_P$

Set $\tilde{f} = P_1 l$ Then $\pi\tilde{f} = f$

(3)⇒(4) ∃ free module $\supseteq F$ and a surjection $\pi: F \twoheadrightarrow P$

$\exists \iota: P \to F$ $F = \ker\pi \oplus \mathrm{Im}\iota$ $\mathrm{Im}\iota \simeq P$

(4)⇒(3) Prop + Cor

Cor. Projective modules over a PID are free.

pf. every submodule of a free module over a PID is free.

Thm. (Quillen-Suslin) All f.g. projective modules over $k[x_1, \cdots, x_n]$ (k → field) are free

eg. $A = \mathbb{Z}[\sqrt{-5}] \subseteq \mathbb{Q}[\sqrt{-5}]$ $\mathcal{O}_1 = (2, 1+\sqrt{-5})$ is not principal, hence not free

$\pi: A^2 \twoheadrightarrow \mathcal{O}_1$; $(x, y) \mapsto 2x + (1+\sqrt{-5})y$

TFAE: (1) $\mathcal{O}_1$ is projective

(2) $0 \to \ker\pi \to A^2 \to \mathcal{O}_1 \to 0$ splits

(3) $\ker\pi$ is a direct summand of A^2

Fact. Let A be a ring. $S \in M_n(A)$ idempotent matrix ($S^2 = S$). Then the submodule P of A^n generated by the columns of S is a direct summand of A^n, hence projective.

$S^2 = S \Rightarrow f^2 = f$ where $f: A^n \to A^n$ $x \mapsto Sx$ $\Rightarrow A^n = \ker f \oplus \mathrm{Im} f$

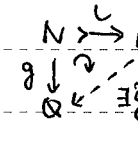
Verify $\ker\pi = \mathrm{span}\left\{ \begin{pmatrix} 1+\sqrt{-5} \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ -1+\sqrt{-5} \end{pmatrix} \right\}$, $\begin{pmatrix} 3 & 1+\sqrt{-5} \\ -1+\sqrt{-5} & -2 \end{pmatrix}$ is idempotent. $\overset{\text{H.P.}}{\Rightarrow}$

§5. Injective modules

Thm. Let Q be an A-module. Then TFAE.

(1) Q is injective

(2) $\forall\, \iota: N \overset{\text{injective}}{\rightarrowtail} M$ $g: N \to Q$ $\exists\, \tilde{g}: M \to Q$ s.t. $\tilde{g}\iota = g$



(3) Every ses $0 \to Q \to M \to N \to 0$ splits

equiv. $\forall\, \iota: Q \rightarrowtail M$ $\exists \pi: M \to Q$ s.t. $\pi\iota = \mathrm{id}_Q$

(4) (Baer's criterion) For every ideal $\mathcal{O}_1$ of A, any A-linear map $g: \mathcal{O}_1 \to Q$ can be extended to an A-linear map $\tilde{g}: A \to Q$

$\Leftrightarrow \exists\, q \in Q$ s.t. $g(a) = aq$ $\forall\, a \in \mathcal{O}_1$

pf. (1)⇔(2) Since the contravariant $\mathrm{Hom}_A(-, Q)$ is left exact

Q injective $\Leftrightarrow$ $\mathrm{Hom}_A(-, Q)$ exact

$\Leftrightarrow$ $\mathrm{Hom}_A(-, Q)$ transforms injective homo into surjective homo

$\Leftrightarrow$ $\forall\, L: N \to M$ $\mathrm{Hom}_A(M, Q) \to \mathrm{Hom}_A(N, Q)$ is surj.

(2) $\Rightarrow$ (3) take $g = \mathrm{id}_Q$

(3) $\Rightarrow$ (2) consider the "pushout" diagram

$$\begin{array}{ccc} N & \xrightarrow{L} & M \\ {\scriptstyle g}\downarrow & \circlearrowright & \downarrow{\scriptstyle i_2} \\ Q & \xrightarrow[i_1]{} & Q \oplus M / \{(g(y), -L(y)) : y \in N\} \end{array}$$

If $i_1(q) = 0$ for some $q \in Q$ then

$\exists\, y \in N$ s.t. $(q, 0) = (g(y), -L(y))$

$\Rightarrow L(y) = 0 \Rightarrow y = 0 \Rightarrow q = 0$

By (3) $\exists\, \pi: Q \oplus M / \{-\} \to Q$ s.t. $\pi i_1 = \mathrm{id}$

Set $\tilde{g} = \pi i_2$ verify $\tilde{g} L = g$

(2) $\Rightarrow$ (4) trivial

(4) $\Rightarrow$ (2) Let $L: N \rightarrowtail M$ be an injective A-linear map and $g: N \to Q$ an A-linear map

Assume wlog that N is a submodule of M and L is inclusion map

Consider the set Σ of all extensions of g to submodules of M containing N

$\Sigma = \{(K, \nu) : K$ submodule of M and $N \subseteq K \subseteq M$, $\nu: K \to Q$ A-linear, $\nu|_N = g\}$

$(N, g) \in \Sigma \Rightarrow \Sigma \neq \phi$

Order Σ as follows $(K, \nu) \leq (K', \nu')$ $(\Leftrightarrow K \subseteq K'$ and $\nu'|_K = \nu$

Given a chain $\{(K_i, \nu_i) : i \in I\}$ in Σ if we set $K = \bigcup_{i \in I} K_i$ and $\nu: K \to Q$ be the unique map s.t. $\nu|_{K_i} = \nu_i$. Then K is a submodule of M containing N ν is A-linear and (K, ν) is an upper bound of the chain in Σ

By Zorn's lemma, Σ has a maximal elt (L, μ) It remains to show $L = M$

If $L \subsetneq M$, then choose an $x \in M$ s.t. $x \notin L$ $N \hookrightarrow L \subsetneq L + Ax \subseteq M$

Define $\mathfrak{A} = \{a \in A : ax \in L\} = \mathrm{Ann}_{M/L}(\bar{x})$

$$\begin{array}{ccc} N & \hookrightarrow & L \\ & \searrow^{g} & \downarrow{\scriptstyle \mu} \\ & & Q \end{array} \qquad \mu'$$

$\mathfrak{A}$ is an ideal of A and $\mathfrak{A} \to Q$; $a \mapsto \mu(ax)$ is an A-linear map

By (4) $\exists\, q \in Q$ s.t. $\mu(ax) = aq$ $\forall\, a \in \mathfrak{A}$

Define a map $\mu': L + Ax \to Q$ $y + ax \mapsto \mu(y) + aq$

Verify that μ' is well-defined μ' is A-linear and $\mu'|_L = \mu$

$\Rightarrow (L, \mu) \lneq (L+Ax, \mu')$ contradicting the maximality of (L, μ)

Hence $L = M$

Prop. $\prod_{i\in I} Q_i$ is injective $\Leftrightarrow$ each Q_i is injective

Def. Let A be an integral domain. An A-module M is divisible if $\forall a\in A\ a\neq 0$ $x\in M$ $\exists y\in M$ s.t. $x=ay$. Equivalently $aM = M$ $\forall a\in A\ a\neq 0$

Prop. Let A be an integral domain, M is an A-module. Then

(1) If M is injective, then M is divisible

(2) If A is a PID, then M is injective $\Leftrightarrow$ M is divisible

pf. By Baer's criterion M is injective $\Leftrightarrow$ $\forall$ nonzero ideal $\mathfrak{a} \subseteq A$, the map $\mathrm{Hom}_A(A, M) \to \mathrm{Hom}_A(\mathfrak{a}, M)$ is surjective.

When $\mathfrak{a} = aA$ is a nonzero principal ideal then $aA \cong A$ as A-modules and $\mathrm{Hom}_A(aA, M) \cong M$; $g \mapsto g(a)$ Similarly $\mathrm{Hom}_A(A, M) \cong M$; $f \mapsto f(1)$

$$\begin{array}{ccc} \mathrm{Hom}_A(A, M) & \longrightarrow & \mathrm{Hom}_A(aA, M) \\ \text{IS} & & \text{IS} \\ M & \xrightarrow{a} & M \end{array} \qquad \begin{array}{ccc} (f: \square \to \square x) & \longrightarrow & f|_{aA} \\ \updownarrow & & \downarrow \\ x & \longrightarrow & ax \end{array}$$

Hence $(*)$ is surj. $\Leftrightarrow$ $a: M \to M$ is surj $\Leftrightarrow$ $aM = M$

e.g. $\mathbb{Q}$ $\mathbb{Q}/\mathbb{Z}$ divisible $\overset{\mathbb{Z}\ \text{PID}}{\Rightarrow}$ $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ injective $\mathbb{Z}$-module

$\mathbb{Z}$ not divisible $\Rightarrow$ not injective

Thm. Let A be a ring. Then the category of A-modules has enough projectives and enough injectives. That is

(1) Any A-module M is the homomorphic image of a projective A-module

(2) Any A-module M is isomorphic to a submodule of an injective A-module

pf. (1) free module is projective ✓

$A^* = \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$

$\pi: \bigoplus_{\mathrm{Hom}_A(A,M)} A \twoheadrightarrow M$ or equiv. $\bigoplus_M A \twoheadrightarrow M$ (1) ✓

$e_x \mapsto x$

$j: M \to \prod_{\mathrm{Hom}_A(M,A^*)} A^*$ It remains to show that ⓑ A^* is injective

$x \mapsto (\psi(x))_{\psi \in \mathrm{Hom}_A(M,A^*)}$ ⓐ j is injective

Def. Let A be a ring. A *projective generator* P of A-Mod is an A-module P s.t. $\mathrm{Hom}_A(P,-)$ is faithfully exact. That is P is projective and $\mathrm{Hom}_A(P,M) \neq 0$ if $M \neq 0$

An A-module Q is an injective *cogenerator* of A-Mod if $\mathrm{Hom}_A(-,Q)$ is faithfully exact. Equivalently, Q is injective and $\mathrm{Hom}_A(M,Q) \neq 0$ if $M \neq 0$

Prop. Let A be a ring P a proj generator of A-Mod, Q an inj cogenerator of A-Mod, M, an A-modamodule. Then

(1) The canonical map $\pi: \bigoplus_{\mathrm{Hom}_A(P,M)} P \to M$ is surjective

(2) The canonical map $j: M \to \prod_{\mathrm{Hom}_A(M,Q)} Q$ is injective

pf. (2) If j is not injective then $\ker j \neq 0$; hence $\mathrm{Hom}_A(\ker j, Q) \neq 0$

Choose a nonzero A-linear map $g: \ker j \to Q$ and let $i: \ker j \to M$ be inclusion

$(0 \neq) \ker j \xrightarrow{i} M \xrightarrow{j} \prod_{\mathrm{Hom}_A(M,Q)} Q$ $ji = 0 \Rightarrow \psi i = 0 \ \forall \psi \in \mathrm{Hom}_A(M,Q)$

$0 \neq g \downarrow \curvearrowright \exists \tilde{g}$ $\Rightarrow \tilde{g} i = 0 \neq g$ contradicting $g \neq 0$

Q

e.g. A as an A-module, is a projective generator $\mathrm{Hom}_A(A,-) \cong \mathrm{id}$

Prop. $\mathbb{Q}/\mathbb{Z}$ is an injective cogenerator of $\mathbb{Z}$-Mod

pf. already know that $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module

It remains to show that ~~if~~ $\mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}) \neq 0$ if $M \neq 0$

Let $M \neq 0$ be a $\mathbb{Z}$-module and let $x \in M$, $x \neq 0$

If x has infinite order ($\Leftrightarrow \mathrm{Ann}_{\mathbb{Z}}(x) = 0$) then $\mathbb{Z}x \cong \mathbb{Z}$

$\exists!$ $\mathbb{Z}$-linear map $g: \mathbb{Z}x \to \mathbb{Q}/\mathbb{Z}$ s.t. $g(x) = \overline{\frac{1}{2}}$ (or any $\bar{q}$ where $q \in \mathbb{Q}\setminus\mathbb{Z}$)

If x has order $n \geq 2$ then $\mathbb{Z}x \cong \mathbb{Z}/n\mathbb{Z}$ and $\exists!$ $\mathbb{Z}$-linear map $g: \mathbb{Z}x \to \mathbb{Q}/\mathbb{Z}$ s.t. $g(x) = \overline{\frac{1}{n}}$ In any case $g \neq 0$

Since $\mathbb{Q}/\mathbb{Z}$ is injective, g extends to a $\mathbb{Z}$-linear map $\tilde{g}: M \to \mathbb{Q}/\mathbb{Z}$

Finally $g \neq 0 \Rightarrow \tilde{g} \neq 0 \Rightarrow \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}) \neq 0$

Prop Let A be a ring. Then $A^* = \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$ is an injective cogenerator of A-Mod

pf. recall that $A\text{-Mod} \underset{\mathrm{Hom}_{\mathbb{Z}}(A,-)}{\overset{|_{\mathbb{Z}}}{\rightleftarrows}} \mathbb{Z}\text{-Mod}$ is an adjoint pair of additive functors and $|_{\mathbb{Z}}$ is exact. HW $\Rightarrow \mathrm{Hom}_{\mathbb{Z}}(A,-)$ transforms injective modules into injective modules $\Rightarrow \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$ is an injective A-module

For any A-module M, we have the adjunction isom

$$\mathrm{Hom}_{\mathbb{Z}}(M|_{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) \cong \mathrm{Hom}_A(M, \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})) = \mathrm{Hom}_A(M, A^*)$$

$M \neq 0 \Rightarrow M|_{\mathbb{Z}} \neq 0 \Rightarrow \mathrm{Hom}_{\mathbb{Z}}(M|_{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) \neq 0 \Rightarrow \mathrm{Hom}_A(M, A^*) \neq 0$

Hence A^* is an injective cogenerator of A-Mod

§6. Flat modules

Prop Let A be a ring, $\mathfrak{a}$ an ideal of A, M an A-module. If we make the canonical identification $A \otimes_A M \cong M$ then the A-linear map

$\mathfrak{a} \otimes_A M \to A \otimes_A M$ (induced by $\mathfrak{a} \hookrightarrow A$) is identified with

$$\tau: \mathfrak{a} \otimes_A M \to M; \quad \sum_{i=1}^{n} a_i \otimes x_i \mapsto \sum_{i=1}^{n} a_i x_i$$

Moreover, τ has image $\mathfrak{a}M$ and $A/\mathfrak{a} \otimes_A M \cong M/\mathfrak{a}M$ as $A/\mathfrak{a}$-module

pf. $0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0 \curvearrowright -\otimes_A M$

$$\mathfrak{a} \otimes_A M \to A \otimes_A M \to A/\mathfrak{a} \otimes_A M \to 0$$

$\searrow$ $\cong$ M $\qquad$ $\cong$ $M/\mathfrak{a}M$

e.g. $A/\mathfrak{a} \otimes_A A/\mathfrak{b} \cong A/\mathfrak{a} \,/\, \mathfrak{a}A/\mathfrak{b} = A/\mathfrak{b} \,/\, \mathfrak{a}+\mathfrak{b}/\mathfrak{b} \cong A/\mathfrak{a}+\mathfrak{b}$

Thm. Let M be an A-module, then TFAE.

(1) M is flat

(2) $\forall N \hookrightarrow N'$ $M\otimes N \to M\otimes N'$ is injective

(2a) $\forall N \hookrightarrow N'$ with N' finitely generated $M\otimes N \to M\otimes N'$ is injective

(2b) $\forall N \hookrightarrow N'$ with N, N' f.g. $M\otimes N \to M\otimes N'$ is injective

(3) $\forall$ ideal $\mathfrak{a} \subset A$ $\mathfrak{a}\otimes M \to M$ $(\cong A\otimes M)$ is inj equivalently $\mathfrak{a}\otimes M \xrightarrow{\sim} \mathfrak{a}M$

(3c) $\forall$ f.g $\mathfrak{a} \subset A$ $\mathfrak{a}\otimes M \to M$ is inj.

pf. (1) $\Leftrightarrow$ (2) $\Rightarrow$ (2a) $\Rightarrow$ (2b) ✓

$\Downarrow$ (3) $\Rightarrow$ $\Downarrow$ (3c)

Lemma Let M, N be A-modules and $v \in M\otimes N$

Then (1) $\exists$ a f.g submodule $M_1 \subseteq M$ a s f.g submod $N_1 \subseteq N$

and $u \in M_1 \otimes N_1$ s.t. $I_{(i,j)}(u) = v$ where $i: M_1 \hookrightarrow M$ $j: N_1 \hookrightarrow N$ are inclusions

(2) If $v = \sum_{i=1}^n x_i \otimes y_i = 0$ in $M\otimes N$ then $\exists$ f.g. submod $M_2 \subseteq M$ containing x_i f.g. submod $N_2 \subset N$ containing y_i s.t. $\sum_{i=1}^n x_i \otimes y_i = 0$ in $M_2 \otimes N_2$

pf. (1) Write $v = \sum_{i=1}^n x_i \otimes y_i$ let $M_1 = Ax_1 + \cdots + Ax_n$ $N_1 = Ay_1 + \cdots + Ay_n$

$u = \sum_{i=1}^n x_i \otimes y_i \in M_1 \otimes N_1$

(2) $\sum_{i=1}^n e_{(x_i, y_i)}$ is a $\overset{(*)}{A}$ finite A-linear combination of the elts of the types

$e_{(x+x', y)} - e_{(x,y)} - e_{(x',y)}$, $e_{(x, y+y')} - e_{(x,y)} - e_{(x,y')}$, $e_{(ax, y)} - a e_{(x,y)}$

$e_{(x, ay)} - a e_{(x,y)}$

Let M_2 be the submod of M generated by the x_i and the x which appears in the above expression. Define N_2 analogously.

$\Rightarrow$ $(*)$ holds in $A^{(M_2 \times N_2)}$ $\Rightarrow \sum_{i=1}^n x_i \otimes y_i = 0$ in $M_2 \otimes N_2$

(2b) $\Rightarrow$ (2) $N \subseteq N'$ Goal: $M\otimes N \to M \otimes N'$ injective

Suppose $v = \sum_{i=1}^n x_i \otimes y_i \in M\otimes N$ maps to 0 in $M\otimes N'$ Want to show $v = 0$

MEMO NO: _____
DATE: 1/4/17

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let $N_1 = \mathrm{Span}(y_1, \cdots, y_n) \subseteq N$

~~let~~ lemma (2) $\Rightarrow$ $\exists$ a f.g. submod $N_1' \subseteq N'$ containing N_1 s.t. $\sum x_i \otimes y_i = 0$ in $M \otimes N_1'$

N_1, N_1' f.g $\overset{(2b)}{\Rightarrow}$ $M \otimes N_1 \to M \otimes N_1'$ injective

$\Rightarrow u = 0 \Rightarrow v = 0$

(3) $\Rightarrow$ (2a) Assume wlog that $N \subseteq N'$ is a submodule N' f.g. $\Rightarrow$ N'/N f.g.

$\Rightarrow$ $\exists$ a chain of submodules $N = N_0 \subseteq N_1 \subseteq \cdots \subseteq N_r = N'$ N_i/N_{i-1} cyclic

Explicitly, $N'/N = A\bar{x}_1 + \cdots + A\bar{x}_r$ $N_i = N + Ax_1 + \cdots + Ax_i$

$N_i = N_{i-1} + Ax_i$ $\Rightarrow$ N_i/N_{i-1} is generated by the image of x_i

To show $M \otimes N \to M \otimes N'$ is injective it suffices to show that

$\forall\, i = 1, \cdots, r$ $M \otimes N_{i-1} \to M \otimes N_i$ is injective

We may assume that N'/N is cyclic $\Rightarrow N' = N + Ax$

let $\mathfrak{a} = (N : x) = \{a \in A : ax \in N\} = \mathrm{Ann}_{N'/N}(\bar{x})$ then $\mathfrak{a}$ is an ideal of A

and $N'/N = A\bar{x} \cong A/\mathfrak{a}$

Goal: use the injectivity of $\mathfrak{a} \otimes M \to M$ to show the injectivity of $N \otimes M \to N' \otimes M$

Check that $0 \to \mathfrak{a} \to N \oplus A \to N' \to 0$ is exact

($\Leftarrow$ $\begin{array}{ccc} \mathfrak{a} & \to & A \\ \downarrow & & \downarrow \\ N & \to & N' \end{array}$ is pullback & pushout) $a \mapsto (-ax, a)$ $(y, b) \mapsto y + bx$

$\Rightarrow \mathfrak{a} \otimes M \to (N \otimes M) \oplus M \to N' \otimes M \to 0$ exact

($\Leftarrow$ $\begin{array}{ccc} \mathfrak{a} \otimes M & \to & M \\ \downarrow & & \downarrow \\ N \otimes M & \to & N' \otimes M \end{array}$ is pushout)

Suppose $t \in N \otimes M$ maps to 0 in $N' \otimes M$ Then $(t, 0) \in (N \otimes M) \oplus M$ maps to 0 in $N' \otimes M$

$\Rightarrow \exists\, s \in \mathfrak{a} \otimes M$ s.t. $s \mapsto (t, 0)$ ~~Since $\mathfrak{a} \otimes M \to$~~

$\Rightarrow$ s maps to $\neq 0$ under the injective map $\mathfrak{a} \otimes M \to M$, hence $s = 0$

$\Rightarrow$ $N \otimes M \to N' \otimes M$ is injective

Prop. $\bigoplus_{i \in I} M_i$ is flat $\Leftrightarrow$ each M_i is flat

pf. tensor product commutes with direct sums

Prop. Projective modules (in particular free modules) are flat

pf. $A \otimes_A - \cong \mathrm{id}_{A\text{-Mod}}$ $\Rightarrow$ A flat A-module $\Rightarrow$ $A^{(I)}$ is flat

$\Rightarrow$ direct summands of $A^{(I)}$ (i.e. proj mod) are flat

Def. Let A be an integral domain, M an A-module, $x \in M$

If $\mathrm{Ann}_A(x) \neq 0$ i.e. $\exists a \in A$ $a \neq 0$ s.t. $ax = 0$ then x is called a torsion element.

Prop. Let A be an integral domain. M an A-module

(1) The set of torsion elts in M forms a submodule $\mathrm{Tor}(M)$ called the torsion submodule of M

We say that M is torsion if $\mathrm{Tor}(M) = M$; M is ~~toris~~ torsion-free if $\mathrm{Tor}(M) = 0$

(2) $\mathrm{Tor}(M/\mathrm{Tor}(M)) = 0$

pf. (1) It suffices to show that $\mathrm{Tor}(M)$ is closed under addition and scalar multiplication

If $ax = 0$ for some $a \neq 0$ then $a(bx) = 0$ for all $b \in A$

If $ax = by = 0$ for some $a \neq 0$, $b \neq 0$ then $ab(x+y) = 0$ and $ab \neq 0$

(2) omitted

Rmk. $\mathrm{Tor}(M) = \bigcup_{a \in A, a \neq 0} \ker(a)$ where $a: M \to M;\ x \mapsto ax$

M is torsion free $\Leftrightarrow \forall a \in A$ $a \neq 0$ the map $a: \underset{x \mapsto ax}{M \to M}$ is injective

e.g. In an abelian gp (i.e. $\mathbb{Z}$-module)

a torsion elt is the same thing as an elt of finite order

(1) $\mathbb{Q}/\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$ $(n \geq 2)$ are torsion $\mathbb{Z}$-modules.

(2) $\mathbb{Z}$, $\mathbb{Q}$, $\bigoplus_{i=1}^{\infty} \mathbb{Z}$, $\prod_{i=1}^{\infty} \mathbb{Z}$ are torsion-free $\mathbb{Z}$-modules.

(3) $\mathrm{Tor}(\underbrace{\mathbb{Z}^r \oplus \mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_s)}_{M}) = \underset{a_i \geq 2}{\mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_s)}$ $M/\mathrm{Tor}(M) \cong \mathbb{Z}^r$

Prop. Let A be an integral domain

(1) If $N \subseteq M$ is a submodule then $\mathrm{Tor}(N) = \mathrm{Tor}(M) \cap N$

Hence any submodule of a torsion (resp. torsion-free) module is torsion (resp. torsion-free)

(2) $\mathrm{Tor}(\bigoplus_{i \in I} M_i) = \bigoplus_{i \in I} \mathrm{Tor}(M_i)$

(3) $\mathrm{Tor}(\prod_{i\in I} M_i) \subseteq \prod_{i\in I} \mathrm{Tor}(M_i)$ and the equality does not hold in general

(4) Submodules of free modules are torsion-free (e.g. proj modules are torsion-free, every ideal of A is torsion-free). The converse is not true.

Prop. Let A be a PID, M a f.g. A-module

Then M is torsion-free $\Leftrightarrow$ M is free

pf. $\Leftarrow$ Prop(4)

$\Rightarrow M \cong A^r \oplus \underbrace{A/(a_1) \oplus \cdots \oplus A/(a_s)}_{\mathrm{Tor}(M)}$ So if $\mathrm{Tor}(M)=0$ then $M \cong A^r$ is free.

Rmk. If A is not an integral domain, the Tor defined above is not a submodule in general (e.g. $\mathrm{Tor}(A) =$ {zero-divisors of A}). The correct defn is as follows.

Def. Let A be a ring (not necessarily an integral domain), M A-module

$x \in M$ is a torsion elt if $\exists$ a non-zero-divisor a of A s.t. $ax=0$

The torsion submodule $\mathrm{Tor}(M)$ is the set of torsion elements in M

(use the fact that the product of two nzds is again a nzd to show that $\mathrm{Tor}(M)$ is a submodule of M) $\mathrm{Tor}(M/\mathrm{Tor}(M)) = 0$

Thm. (1) Flat modules are torsion-free

(2) If A is a PID, then an A-module M is flat iff it is torsion-free

pf. Let $\mathfrak{a} = aA$ be the principal ideal generated by a nzd a of A

Then $aA \cong A$ and after making the following identifications

$$aA \otimes_A M \cong A \otimes_A M \cong M$$
$$ab \otimes x \mapsto b \otimes x \mapsto bx$$

the map $\mathfrak{a} \otimes_A M \to A \otimes_A M$ is identified with $a: M \to M$, $x \mapsto ax$

$$\begin{array}{ccc} aA \otimes_A M & \to & A \otimes_A M \\ \| & & \| \\ M & \to & M \\ \uparrow & & \downarrow \\ x & & ax \end{array} \qquad \begin{array}{c} a \otimes x \\ \mapsto \\ a \otimes x \end{array}$$

M is torsion free $\Leftrightarrow$ $\forall$ nzd a of A the map $a: M \to M$, $x \mapsto ax$ is injective

$\Leftrightarrow$ $\forall$ principal ideal $\mathfrak{a}$ of A generated by a nzd a of A

the map $\mathfrak{a} \otimes M \to M$ is injective

Rmk. For modules over a PID: projective ($\Leftrightarrow$ free) $\Rightarrow$ injective ($\Leftrightarrow$ divisible) $\Rightarrow$ flat ($\Leftrightarrow$ torsion-free)

Def. Let A be a ring. An A-algebra is a ring B together with a ring map $f: A \to B$. e.g. $A \subseteq A[x]$, $A \to A/\mathfrak{a}$, $\mathbb{Z} \subset \mathbb{Q}$

Def. A ring map $f: A \to B$ is flat or B is a flat A-algebra if B is flat as an A-module.

e.g. (1) $A[x]$ is a flat A-algebra because $A[x] \cong \bigoplus_{i=0}^{\infty} A$ is free.

(2) $\mathbb{Q}$ is a flat $\mathbb{Z}$-algebra. In general if A is an integral domain with fraction field K, then K is a flat A-algebra.

Prop. Let $f: A \to B$ be a ring map. Then

(1) B is a flat A-algebra $\Leftrightarrow B \otimes_A - : A\text{-Mod} \to B\text{-Mod}$ is exact

(2) If M is a flat A-module, then $M_B = B \otimes_A M$ is a flat B-module

(3) If B is a flat A-algebra, N is a flat B-module, then N is a flat as an A-module

(4) If B is a flat A-algebra and C is a flat B-alg, then C is a flat A-algebra.

pf. (1) B is a flat A-algebra $\Leftrightarrow$ B is flat as an A-module

$\Leftrightarrow B \otimes_A -' : A\text{-Mod} \to A\text{-Mod}$ is exact $\Leftrightarrow B \otimes_A - : A\text{-Mod} \to B\text{-Mod}$ is exact

$B \otimes_A -$: $A\text{-Mod} \to B\text{-Mod}$, I_A : $B\text{-Mod} \to A\text{-Mod}$ faithfully exact

Fact. If $F: A\text{-Mod} \to B\text{-Mod}$ and $G: B\text{-Mod} \to C\text{-Mod}$ are additive functors s.t. G is faithfully exact, then F is exact $\Leftrightarrow GF$ is exact

(2) $M_B \otimes_B N = (M \otimes_A B) \otimes_B N \cong M \otimes_A (B \otimes_B N) \cong M \otimes_A N$

$$\begin{array}{ccc} B\text{-Mod} & \xrightarrow{M_B \otimes_B -} & B\text{-Mod} \\ {\scriptstyle I_A}\downarrow & & \downarrow{\scriptstyle I_A} \\ A\text{-Mod} & \xrightarrow[M \otimes_A -]{} & A\text{-Mod} \end{array} \qquad \begin{array}{ccc} N & \longmapsto & M_B \otimes_B N \\ \downarrow & & \downarrow \\ N|_A & \longrightarrow & M \otimes_A (N|_A) \end{array}$$

commutes up to a natural isomorphism of functors

e.g. $A = k[x,y]$ k field

$\mathfrak{a} = (x,y) \subseteq A$ is torsion-free A-module

but $\mathfrak{a}$ is not flat

$\mathfrak{b} = (x) \subseteq A$ $\quad A \to A/\mathfrak{b} = k[x,y]/(x) \cong k[y]$ PID

Recall $A/\mathfrak{a} \otimes_A M \cong M/\mathfrak{a}M$, $\quad \mathfrak{a} \otimes_A A/\mathfrak{b} \cong \mathfrak{a}/\mathfrak{a}\mathfrak{b} = (x,y)/(x^2,xy)$ is an $A/\mathfrak{b}$ module is $k[y]$

$\bar{x} \neq 0$ $\bar{x} \in (x,y)/(x^2,xy)$ $\quad y \neq 0$ $y \in k[y]$ $\quad \bar{x}y = 0$ in $(x,y)/(x^2,xy)$ $\quad k[y]$

Thm Let M be an A-module. Then TFAE

(1) M is flat

(2) (equational criterion for flatness) If $a_i \in A$ and $\sum_{i=1}^{r} x_i \in M$ $(1 \le i \le r)$ satisfy $\sum_{i=1}^{r} a_i x_i = 0$, then $\exists$ an integer s and $y_j \in M$ $b_{ij} \in A$ $(1 \le i \le r, 1 \le j \le s)$ s.t. $x_i = \sum_{j=1}^{s} b_{ij} y_j$ $(1 \le i \le r)$ and $\sum_{i=1}^{r} a_i b_{ij} = 0$ $(1 \le j \le s)$

(2)′ If $a_{ki} \in A$ $x_i \in M$ $(1 \le i \le r, 1 \le k \le m)$ satisfy $\sum_{i=1}^{r} a_{ki} x_i = 0$ $(1 \le k \le m)$

then $\exists$ s and $y_j \in M$, $b_{ij} \in A$ s.t. $x_i = \sum_{j=1}^{s} b_{ij} y_j$ $\forall i$. and $\sum_{i=1}^{r} a_{ki} b_{ij} = 0$ $\forall j,k$

In other words, if $\begin{pmatrix} a_{11} & \cdots & a_{1r} \\ \vdots & & \\ a_{m1} & \cdots & a_{mr} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = 0$ (I) then $\exists$ s, y_j, b_{ij} s.t.

$\begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = \begin{pmatrix} b_{11} & \cdots & b_{1s} \\ \vdots & & \\ b_{r1} & \cdots & b_{rs} \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_s \end{pmatrix}$ (II) and $\begin{pmatrix} a_{11} & \cdots & a_{1r} \\ \vdots & & \\ a_{m1} & \cdots & a_{mr} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1s} \\ \vdots & & \\ b_{r1} & \cdots & b_{rs} \end{pmatrix} = 0$ (III)

(3) For every morphism $\alpha : F \to M$ from a f.g. free module F and for every cyclic submodule K of $\ker\alpha$, there is a commutative diagram.

$K \subseteq \ker\alpha \hookrightarrow F \xrightarrow{\nu} G$ with G f.g. free s.t. $\nu(K) = 0$

$\alpha \searrow \quad \swarrow \beta$

M

(3)′ same as (3) but for any f.g. submodule K of $\ker\alpha$

pf. (1) $\Rightarrow$ (2)′ Let $\lambda : A^r \to A^m$ be the A-linear map defined by the matrix $(a_{ki}) \in M_{m\times r}(A)$

and let $K = \ker\lambda = \{(x_1, \cdots, x_r) \in A^r \mid \sum_{i=1}^{r} a_{ki} x_i = 0 \ \forall k = 1, \cdots, m\}$

Then $0 \to K \to A^r \xrightarrow{\lambda} A^m$ is exact

M flat $\Rightarrow 0 \to K\otimes M \to A^r\otimes M \xrightarrow{T(\lambda, id_M)} A^m \otimes M$ is exact

φ is M^r $\quad \psi$ is M^m

Then ψ is also defined by the matrix (a_{ki}) and φ $\varphi\left(\begin{pmatrix} b_1 \\ \vdots \\ b_r \end{pmatrix}\otimes y\right) = \begin{pmatrix} b_1 y \\ \vdots \\ b_r y \end{pmatrix}$

(I) $\Leftrightarrow \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} \in \ker\psi = \mathrm{Im}\varphi \Rightarrow \exists$ an integer s $\begin{pmatrix} b_{1j} \\ \vdots \\ b_{rj} \end{pmatrix} \in K$ $y_j \in M$ $(1\le j\le s)$

s.t. $\varphi\left(\sum_{j=1}^{s}\begin{pmatrix} b_{1j} \\ \vdots \\ b_{rj} \end{pmatrix}\otimes y_j\right) = \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} \Leftrightarrow$ (II)

(III) $\Leftrightarrow \begin{pmatrix} b_{1j} \\ \vdots \\ b_{rj} \end{pmatrix} \in K$ $j=1,\cdots,s$

(2)$'\Rightarrow$(2) set $m=1$

(2)$\Rightarrow$(1) By the ideal criterion it suffices to show that $\forall$ ideal $\mathfrak{a}\subseteq A$

$\tau: \mathfrak{a}\otimes M \to M$ is injective

Suppose $\tau\left(\sum_{i=1}^{r} a_i\otimes x_i\right) = \sum_{i=1}^{r} a_i x_i = 0$ By(2) $\exists\, s$ y_j $(1\le j\le s)$ $b_{ij}\in A$ $1\le i\le r$, $1\le j\le s$

s.t. $x_i = \sum_{j=1}^{s} b_{ij} y_j$ $\forall i$ and $\sum_{i=1}^{r} a_i b_{ij} = 0$ $\forall j$

Then $\sum_{i=1}^{r} a_i\otimes x_i = \sum_{i=1}^{r} a_i\otimes\left(\sum_{j=1}^{s} b_{ij} y_j\right) = \sum_{i=1}^{r}\sum_{j=1}^{s} a_i b_{ij}\otimes y_j = \sum_{j=1}^{s}\underbrace{\left(\sum_{i=1}^{r} a_i b_{ij}\right)}_{=0}\otimes y_j =$

$= \sum_{j=1}^{s} 0\otimes y_j = 0$

(2)$'\Leftrightarrow$(3)$'$

Set $F = A^r$ $\quad G = A^s$ α is given by $\alpha: (a_1,\cdots,a_r) \mapsto (a_1,\cdots,a_r)\begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix}$ for some $\begin{pmatrix} \eta_1 \\ \vdots \\ \eta_r \end{pmatrix} \in M^r$

Suppose K is generated by $(a_{11},\cdots,a_{1r}),\cdots,(a_{m1},\cdots,a_{mr}) \in A^r$

Then $K\subseteq \ker\alpha \Leftrightarrow (a_{k1},\cdots,a_{kr})\begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix} = 0$ $\forall\, 1\le k\le m \Leftrightarrow$ (I)

ν is given by $\nu: (a_1,\cdots,a_r) \mapsto (a_1,\cdots,a_r)\begin{pmatrix} b_{11}\cdots b_{1s} \\ \vdots \\ b_{r1}\cdots b_{rs} \end{pmatrix}$ for some $(b_{ij}) \in M_{r\times s}(A)$

β is given by $\beta: (c_1,\cdots,c_s) \mapsto (c_1,\cdots,c_s)\begin{pmatrix} y_1 \\ \vdots \\ y_s \end{pmatrix}$ for some $\begin{pmatrix} y_1 \\ \vdots \\ y_s \end{pmatrix} \in M^s$

$\beta\nu = \alpha \Leftrightarrow$ (II) $\quad \nu(K) = 0 \Leftrightarrow$ (III)

Cor. Let $(A,\mathfrak{m})$ be a local ring with residue field $k = A/\mathfrak{m}$. Let M be a flat A-module. Recall that $M/\mathfrak{m}M$ is an $A/\mathfrak{m}$-module i.e. a k-vector space. If $x_1,\cdots,x_n \in M$ and their images $\bar{x}_1,\cdots,\bar{x}_n$ in $M/\mathfrak{m}M$ are linearly independent over k, then $x_1,\cdots,x_n$ are linearly indep over A. pf. (eqn. criterion for flatness)

Rmk. Every f.g flat module over a local ring is free (later)

§7. Modules of finite presentation

Lemma. Let A be a ring. M an A-module. Then $\exists$ an exact sequence

$$A^{(\Omega)} \to A^{(\Lambda)} \to M \to 0 \quad (*)$$

Pf. Pick a set of generators $\{x_\lambda\}_{\lambda\in\Lambda}$ for M

$\Rightarrow$ a surjective morphism $\pi: A^{(\Lambda)} \to M$; $e_\lambda \mapsto x_\lambda$

$\ker\pi = \{(a_\lambda)_\lambda \in A^{(\Lambda)} : \sum_{\lambda\in\Lambda} a_\lambda x_\lambda = 0\}$ is the submodule of linear relations among the x_λ

Pick a set of generators $\{\eta_w\}_{w\in\Omega}$ for $\ker\pi$

$\Rightarrow$ surjective morphism $\sigma: A^{(\Omega)} \to \ker\pi$ $e_w \mapsto \eta_w$

Put together we get an exact sequence

$$A^{(\Omega)} \xrightarrow{j\sigma} A^{(\Lambda)} \xrightarrow{\pi} M \to 0$$

($\sigma: A^{(\Omega)} \to \ker\pi$, $j: \ker\pi \to A^{(\Lambda)}$)

Def. $(*)$ is called a free presentation of M; $(*)$ is called a finite free presentation of M if both Ω and Λ are finite sets.

An A-module M is finitely presented if it admits a finite free presentation

e.g. M f.g $\Leftrightarrow$ $\exists$ an exact sequence $A^n \to M \to 0$

M f.p $\Rightarrow$ M f.g

If $\mathfrak{a}$ is a non-f.g. ideal of a ring A, then $A/\mathfrak{a}$ is not a f.p. A-module (cyclic generated by one elt)

Prop. Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of A-modules

Then (1) M_2 f.g. $\Rightarrow$ M_3 f.g

(2) M_1, M_3 f.g $\Rightarrow$ M_2 f.g

(3) M_3 f.p., M_2 f.g $\Rightarrow$ M_1 f.g

(4) M_2 f.p., M_1 f.g $\Rightarrow$ M_3 f.p

(5) M_1, M_3 f.p $\Rightarrow$ M_2 f.p

pf (1) ✓ (2) $0 \to A^m \xrightarrow{i} A^m \oplus A^n \xrightarrow{q} A^n \to 0$

$\pi_1 \downarrow$ $\qquad\qquad \downarrow \pi_2$

$0 \to M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0$

A^n free $\Rightarrow$ projective $\exists\, \pi_2'$ s.t. $g\pi_2' = \pi_2$

$\pi_2' \swarrow$ $A^n \downarrow \pi_2$

$M_2 \xrightarrow{g} M_3$

$\pi(x,y) = f\pi_1(x) + \pi_2'(y)$ $\quad x \in A^m$ $\quad y \in A^n$ then

$A^m \xrightarrow{i} A^m \oplus A^n$

$\pi_1 \downarrow$ $\circlearrowright$ $\downarrow \pi$

$M_1 \xrightarrow{f} M_2$

Snake lemma $\Rightarrow$ $\operatorname{coker}\pi_1 \to \operatorname{coker}\pi \to \operatorname{coker}\pi_2$ is exact

$\qquad\qquad\quad \| \qquad\qquad\quad \|$

$\qquad\qquad\quad 0 \qquad\qquad\quad 0$

$\Rightarrow \operatorname{coker}\pi = 0 \Rightarrow \pi$ is surjective

(3) Choose a finite free presentation $A^m \to A^n \to M_3 \to 0$

$A^m \xrightarrow{v} A^n \xrightarrow{\pi} M_3 \to 0$ $\quad A^n$ projective $\Rightarrow \pi' : A^n \to M_2$ s.t. $g\pi' = \pi$

$\pi'' \downarrow \circlearrowright$ $\downarrow \pi' \circlearrowright$ $\|$

$0 \to M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0$

$g\pi' v = \pi v = 0 \Rightarrow^{\text{u.p of kernel}} \exists\, \pi'' : A^m \to M_1$ s.t. $\pi' v = f\pi''$

Snake lemma $\Rightarrow \ker id_M = 0 \to \operatorname{coker}\pi'' \to \operatorname{coker}\pi' \to 0 = \operatorname{coker} id_M$ is exact

$\Rightarrow \operatorname{coker}\pi'' \cong \operatorname{coker}\pi'$

M_2 f.g. $\Rightarrow \operatorname{coker}\pi'$ f.g. $\Rightarrow \operatorname{coker}\pi''$ f.g.

$0 \to \underset{\text{f.g.}}{\operatorname{Im}\pi''} \to M_1 \to \underset{\text{f.g.}}{\operatorname{coker}\pi''} \to 0$ exact $\overset{(2)}{\Rightarrow} M_1$ f.g.

(5). Snake lemma $\Rightarrow 0 \to \ker\pi_1 \to \ker\pi \to \ker\pi_2 \to 0$ is exact

$\overset{(3)}{\Rightarrow} M_1, M_3$ f.p. $\quad \ker\pi_1, \ker\pi_2$ f.g. $\overset{(2)}{\Rightarrow} \ker\pi$ f.g. $\Rightarrow M_2$ f.p.

Prop Let M_i $(i \in I)$ N be A-modules If N is f.p then the natural morphism $\alpha_N : (\prod_{i\in I} M_i) \otimes_A N \to \prod_{i\in I}(M_i \otimes_A N)$ is an isom

pf Choose a finite free presentation $A^m \to A^n \to N \to 0$

$(\prod_{i\in I} M_i) \otimes A^m \to (\prod_{i\in I} M_i) \otimes A^n \to (\prod_{i\in I} M_i) \otimes N \to 0$ exact

$M_i \otimes A^m \to M_i \otimes A^n \to M_i \otimes N \to 0$ exact

$\Rightarrow \prod_{i\in I}(M_i \otimes A^m) \to \prod_{i\in I}(M_i \otimes A^n) \to \prod_{i\in I}(M_i \otimes N) \to 0$ exact

$(\prod_{i\in I} M_i) \otimes A^m \longrightarrow (\prod_{i\in I} M_i) \otimes A^n \longrightarrow (\prod_{i\in I} M_i) \otimes N \to 0$

$\cong \downarrow \alpha_{A^m}$ $\circlearrowright$ $\cong \downarrow \alpha_{A^n}$ $\circlearrowright$ $\downarrow \alpha_N$

$\prod_{i\in I}(M_i \otimes A^m) \longrightarrow \prod_{i\in I}(M_i \otimes A^n) \longrightarrow \prod_{i\in I}(M_i \otimes N) \to 0$

$$\left(\prod_{i\in I} M_i\right)\otimes_A A^n \cong \left(\prod_{i\in I} M_i\right)^n \cong \prod_{i\in I} M_i^n \cong \prod_{i\in I}(M_i\otimes_A A^n)$$

$\Rightarrow \alpha_{A^m}, \alpha_{A^n}$ are isoms $\Rightarrow \alpha_N$ is an isom

Thm. Let A be a ring, M an A-module. Then TFAE

(1) M is f.g. and projective

(2) M is f.p and flat

(3) M is the direct summand of a f.g. free module

pf. (1) $\Leftrightarrow$ (3) exercise

(3) $\Rightarrow$ (2) Suppose $A^n = M\oplus K \Rightarrow M$ projective $\Rightarrow M$ flat

$A^n \xrightarrow{\pi_K} K \hookrightarrow A^n \xrightarrow{\pi_M} M \to 0$ is a finite free presentation

(2) $\Rightarrow$ (3) M f.g. $\Rightarrow$ surjective morphism $\pi: A^n \to M$

M f.p $\overset{\text{Prop(3)}}{\Rightarrow}$ $\ker\pi$ f.g

$\overset{\text{Thm(3)'}}{\Rightarrow}$ $\exists$ $\ker\pi \hookrightarrow A^n \xrightarrow{\nu} A^m$ s.t. $\nu(\ker\pi)=0 \Rightarrow \exists\alpha$ s.t. $\nu=\alpha\pi$

$\pi \searrow M \swarrow \beta$ $\quad \pi=\beta\nu$

$\Rightarrow \pi=\beta\alpha\pi$

π surjective $\Rightarrow \beta\alpha = id_M \Rightarrow A^m = \mathrm{Im}\,\alpha \oplus \ker\beta \cong M\oplus\ker\beta$

Chapter 5 Noetherian rings and modules

§1 Noetherian modules and Noetherian rings

Lemma Let M be an A-module. Then TFAE

(1) Every ascending chain of submodules in M is stationary (a.c.c. ascending chain condition)

That is if $M_1\subseteq M_2\subseteq\cdots$ is a sequence of submodules in M then $\exists n$ s.t. $M_n = M_{n+1}$

(2) Every non-empty set of submodules of M has a maximal elt

(3) Every submodule of M is f.g.

pf. (1) $\Rightarrow$ (2) If a nonempty set of submodules $\{M_i\}_{i\in I}$ has no maximal elt

then $\exists$ a sequence of indices $i_1, i_2, \cdots$ in I s.t. $M_{i_1}\subsetneq M_{i_2}\subsetneq\cdots$ contradicting (1)

(2) ⇒ (1) $\{M_i\}_{i\geq 1}$ has a maximal ett say M_n then $M_n = M_{n+1} = \cdots$

(1) ⇒ (3) If a submod $N \subseteq M$ is not f.g. then $\exists\, y_1, y_2, \cdots \in N$ s.t. $(y_1) \subsetneq (y_1, y_2) \subsetneq \cdots$

(3) ⇒ (1) $\bigcup_{i\geq 1} M_i$ is a submod of M hence is f.g. Suppose $\bigcup_{i\geq 1} M_i = \langle x_1, \cdots, x_k \rangle$

Then for each $i = 1, \cdots, k$ $\exists\, n_i$ s.t. $x_i \in M_{n_i}$ Let $n = \max n_i$ then $\bigcup_{i\geq 1} M_i \subseteq M_n$

Def. A module satisfying the above conditions is called a Noetherian module

Prop. Let M be an A-module, $N \subseteq M$ a submod

Then M is Noet ⇔ both N and M/N are Noet

Equivalently, if $0 \to M' \to M \to M'' \to 0$ is a ses of A-modules

then M is Noet ⇔ M' and M'' are Noet

pf. ⇒ $L_1/N \subseteq L_2/N \subseteq \cdots$ where $L_1 \subseteq L_2 \subseteq \cdots \subseteq M$ $L_n = L_{n+1} = \cdots \Rightarrow L_n/N = L_{n+1}/N = \cdots$

⇐ Let $L_1 \subseteq L_2 \subseteq \cdots$ be an a.c.c. of submod in M

Then $L_1 \cap N \subseteq L_2 \cap N \subseteq \cdots$ is an a.c.c. of submod in N

$L_1 + N/N \subseteq L_2 + N/N \subseteq \cdots$ M/N

By assumption $\exists\, n$ s.t. $L_n \cap N = L_{n+1} \cap N = \cdots$ $L_n + N = L_{n+1} + N = \cdots$

$\Rightarrow L_n = L_{n+1} = \cdots$

Cor. If M_i $(1 \leq i \leq n)$ are Noet A-modules then $\bigoplus_{i=1}^{n} M_i$ is Noet

pf. $0 \to M_n \to \bigoplus_{i=1}^{n} M_i \to \bigoplus_{i=1}^{n-1} M_i \to 0$ induction on n

Def. A ring A is Noetherian if A is Noet as an A-module, that is if any one of the following equiv conditions holds (1). ideals of A satisfy a.c.c.

(2) every non-empty set of ideals has a maximal ett (3) every ideal of A is f.g.

e.g. PIDs (include fields, $\mathbb{Z}$, $k[x]$ ···) are Noetherian rings

HBT A Noet ⇒ $A[x_1, \cdots, x_n]$ Noet e.g. $\mathbb{Z}[x_1, \cdots, x_n]$ $k[x_1, \cdots, x_n]$ Noet

$k[x_1, x_2, \cdots] = \bigcup_{n\geq 1} k[x_1, \cdots, x_n]$ is not Noet

Note $k[x_1, x_2, \cdots]$ is a UFD hence satisfies a.c.c.p (principal)

$(x) \subsetneq (x^{\frac{1}{2}}) \subsetneq (x^{\frac{1}{4}}) \subsetneq \cdots$

$\mathbb{Z}[x, x^{\frac{1}{2}}, x^{\frac{1}{4}}, \cdots] = \bigcup_{n\geq 1} \mathbb{Z}[x^{\frac{1}{2^n}}]$ is not Noet and not UFD

$\mathcal{C}(\mathbb{R}) = \{\text{cont fcns on } \mathbb{R}\}$ is not Noet $\mathcal{O}_n = \{f \in \mathcal{C}(\mathbb{R}) : f|_{[n,\infty)} = 0\}$ $\mathcal{O}_1 \subsetneq \mathcal{O}_2 \subsetneq \mathcal{O}_3 \subsetneq \cdots$

⇔ $\bigcup_{n\geq 1} \mathcal{O}_n$ is not f.g.

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Prop. If A is a Noetherian ring and $\mathfrak{a}$ is an ideal of A then $A/\mathfrak{a}$ is a Noet ring.

Rmk. A ~~submodule~~ subring of a Noet ring need not be Noet

e.g. Let A be a non-Noetherian domain and K be its fraction field

then K is noetherian but its subring A is not

Prop. Let A be a Noet ring M an A-module. Then TFAE

(1) M is f.g.

(2) M is Noetherian

(3) M is f.p.

Pf. (2)⇒(1), (3)⇒(1) trivial.

(1)⇒(2)(3) M f.g ⇒ $\exists\ \pi: A^n \to M$ surj. ⇒ $0 \to K \to A^n \to M \to 0$ ses ($K = \ker\pi$)

A Noet ring ⇔ A is a Noet A-module ⇒ $A^n = \bigoplus_{i=1}^{n} A$ is Noet

Prop⇒ K, M Noet. ⇒ K f.g. ⇒ M f.p. ($0 \to K \to A^m \to A^n \to M \to 0$)

Hilbert's Basis Theorem If A is a Noet ring, then $A[x]$ is a Noet ring.

Pf. Let $\mathfrak{b}$ be an ideal of $A[x]$. For each $n \geq 0$ define $\mathfrak{a}_n = \{a \in A : \exists\, ax^n + \text{l.o.t} \in \mathfrak{b}\}$

Then $\mathfrak{a}_n$ is an ideal of A and $\mathfrak{a}_0 \subseteq \mathfrak{a}_1 \subseteq \cdots$

A is Noet ⇒ $\exists\, n$ s.t. $\mathfrak{a}_n = \mathfrak{a}_{n+1} = \cdots$ and $\mathfrak{a}_0, \cdots \mathfrak{a}_n$ are f.g.

For each $i = 0, \cdots, n$ choose a finite set of generators for $\mathfrak{a}_i$:

$\mathfrak{a}_i = (a_{i1}, a_{i2}, \cdots a_{ik_i})$ and for each $j = 1, \cdots k$ choose a polynomial $f_{ij}(x)$

in $\mathfrak{b}$ s.t. $f_{ij} = a_{ij}x^i + \text{l.o.t}$

Claim. $\mathfrak{b}$ is generated by $\{f_{ij}(x) : 0 \leq i \leq n,\ 1 \leq j \leq k_i\}$ (finite set)

Suppose $f(x) \in \mathfrak{b}$ and $\deg f(x) \overset{=}{\leq} d$

If $d \leq n$ then $\exists\, c_1, \cdots, c_{k_d}$ s.t. $f(x) - \sum_{j=1}^{k_d} c_j f_{dj}(x)$ has degree $< d$

If $d > n$ then $\exists\, c_1, \cdots, c_{k_n}$ s.t. $f(x) - \sum_{j=1}^{k_n} c_j f_{nj}(x) \cdot x^{d-n}$ has degree $< d$

We conclude by induction on d.

Exercise If A is a Noetherian ring, then $A[[x]]$ is a Noetherian ring

Hint $\mathfrak{b} \subseteq A[[x]]$, $n \geq 0$ Let $\mathfrak{a}_n = \{a \in A : \exists\, ax^n + \text{h.o.t} \in \mathfrak{b}\}$ $\mathfrak{a}_0 \subseteq \mathfrak{a}_1 \subseteq \cdots \mathfrak{a}_n \subseteq \cdots$

Prop In a Noet ring A, every ideal $\mathfrak{a}$ contains a power of its radical $\sqrt{\mathfrak{a}}$

That is $\exists n \geq 1$ s.t. $(\sqrt{\mathfrak{a}})^n \subseteq \mathfrak{a}$

In particular the nilradical of a Noet ring is nilpotent.

That is if A is a Noet ring, then $\exists n \geq 1$ s.t. $\mathrm{nil}(A)^n = 0$

pf $(\sqrt{\mathfrak{a}})^n \subseteq \mathfrak{a} \iff (\sqrt{\mathfrak{a}/\mathfrak{a}})^n = 0 \iff (\mathrm{nil}(A/\mathfrak{a}))^n = 0$ A Noet $\Rightarrow A/\mathfrak{a}$ Noet

PS1 6(2): a f.g. ideal consisting of nilpotent elts is nilpotent

Prop In a Noet ring A, every ideal $\mathfrak{a}$ has finitely many minimal prime ideals over $\mathfrak{a}$.

In particular, a Noet ring has finitely many minimal prime ideals.

Hint. Σ = the set of (proper) ideals which has infinitely many minimal prime ideals over it.

Exercise. Over a Noet ring, direct sums of injective modules is injective.

and direct product of flat modules is flat

§2. Finitely generated algebras and finite algebra

Let A be a ring. Recall that an **A-algebra** is a ring B together with a ring map $f: A \to B$ e.g. $A[x_1, \cdots, x_n]$ $A/\mathfrak{a}$ are A-algebras.

An A-algebra is a ring and an A-module at the same time.

$$ab := f(a)b \quad a \in A \quad b \in B$$

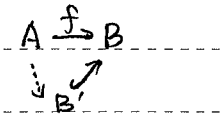
If f is injective then B contains a subring $f(A)$ or $A \cdot 1$ which is isomorphic to A. In general, B only contains a subring which is isomorphic to a quotient of A

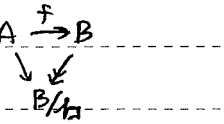
If k is a field, then any nonzero k-algebra contains a subring $k \cdot 1$ isomorphic to k

Every ring A is a $\mathbb{Z}$-algebra via the unique ring map $\mathbb{Z} \to A$

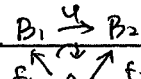
Def Let B be an A-algebra with structure map $f: A \to B$

Then an **A-subalgebra** of B is a subring B' of B s.t. $f(A) \subseteq B'$





Let B_1, B_2 be A-algebras. An **A-algebra homomorphism** $\varphi: B_1 \to B_2$ is a ring map s.t. $B_1 \xrightarrow{\varphi} B_2$ commutes. $\Rightarrow \mathrm{Im}\varphi$ is an A-subalg of B_2



MEMO NO: _____
DATE: 1 4 / 29

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Rmk. We may construct a category A-alg whose objects are A-algebras and whose morphisms are A-algebra homomorphisms.

Def. Let B be an A-algebra. $g(x_1,\cdots,x_n)=\sum a_{i_1\cdots i_n}x_1^{i_1}\cdots x_n^{i_n}\in A[x_1,\cdots,x_n]$ $b_1,\cdots,b_n\in B$ Then we define $g(b_1,\cdots,b_n)=\sum a_{i_1\cdots i_n}b_1^{i_1}\cdots b_n^{i_n}\in B$

Prop Let B be an A-algebra $b_1,\cdots,b_n\in B$

Then (1) $A[b_1,\cdots,b_n]=\{g(b_1,\cdots,b_n)=g(x_1,\cdots,x_n)\in A[x_1,\cdots,x_n]\}$ is an A-subalgebra of B, In fact, it is the smallest A subalg of B containing $b_1,\cdots,b_n$, called by A-subalgebra generated by $b_1,\cdots,b_n$

(2) $\exists!$ A-algebra homomorphism $\varphi: A[x_1,\cdots,x_n]\to B$ s.t. $\varphi(x_i)=b_i$ $(1\le i\le n)$

Moreover $\mathrm{Im}\varphi=A[b_1,\cdots,b_n]$ $\left(\begin{smallmatrix}\nwarrow & \nearrow\\ & A\end{smallmatrix}\right)$

pf. (2) $\varphi(g(x_1,\cdots,x_n))=g(b_1,\cdots,b_n)$

Rmk. $\mathrm{Hom}_{A\text{-alg}}(A[x_1,\cdots,x_n],B)\simeq B^n$ $\varphi\mapsto(\varphi(x_i))_{1\le i\le n}$

Def. Let $f:A\to B$ be a ring map. We say that f is of finite type, or that B is a finite type A-algebra or a finitely generated A-algebra if $\exists\, b_1,\cdots,b_n\in B$ s.t. $B=A[b_1,\cdots,b_n]$.

We say that f is finite or that B is a finite A-algebra if B is f.g as an A-module. That is, $\exists\, b_1,\cdots,b_n$ s.t. $B=Ab_1+\cdots+Ab_n$

Lemma. B is a f.g. A-algebra $\Leftrightarrow B\simeq A[x_1,\cdots,x_n]/I$

pf. $\Rightarrow$ $B=A[b_1,\cdots,b_n]$ $\varphi: A[x_1,\cdots,x_n]\twoheadrightarrow B$ $x_i\mapsto b_i$ $B\simeq A[x_1,\cdots,x_n]/\ker\varphi$

$\Leftarrow$ $A[x_1,\cdots,x_n]/I=A[\bar{x}_1,\cdots,\bar{x}_n]$

Prop If A is a Noetherian ring, then every f.g. A-algebra is Noetherian.

In particular, if k is a field, then every f.g. k-algebra is Noet.

Eg. $A[x]$ is a f.g A-algebra but not a finite A-algebra because $A[x]$ is a free A-module with basis $1,x,x^2,\cdots$

$A[x]/(f(x))$ $f(x)=x^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$ is a free A-module with basis $\bar{1},\bar{x},\cdots,\overline{x^{n-1}}$

$\forall g(x)$ $g(x)=f(x)q(x)+r(x)$ $\deg r<n$

Prop. Let A be an integral domain with fraction field K

(1) K is a finite A-algebra $\Leftrightarrow K=A$

(2) K is a f.g. A-algebra $\Leftrightarrow K=A[\frac{1}{u}]$ where $u\in A$ $u\neq 0$

(3) If A is a UFD, K is a f.g. A-algebra $\Leftrightarrow$ A has finitely many prime elts.

e.g. $\mathbb{Q}$ is not a f.g. $\mathbb{Z}$-algebra

$\mathbb{Z}\subseteq\mathbb{Z}_{(2)}\subseteq\mathbb{Q}$ $\mathbb{Z}_{(2)}=\{\frac{m}{n}: 2\nmid n\}$ but $\mathbb{Q}$ is a f.g. $\mathbb{Z}_{(2)}$-algebra $\mathbb{Q}=\mathbb{Z}_{(2)}[\frac{1}{2}]$

Prop Let $f:A\to B$, $g:B\to C$ be ring maps.

(1) A finite ring map is of finite type

(2) If f,g are of finite type, then so is gf

(3) If f,g are finite, then so is gf.

(4) If gf is finite type, then g is of finite type

(5) If gf is finite, then g is finite.

pf. (1)✓(2) $B=A[b_1,\cdots,b_m]$ $C=B[c_1,\cdots,c_n]\Rightarrow C=A[g(b_1),\cdots,g(b_m),c_1,\cdots,c_n]$

(3) $B=Ab_1+\cdots+Ab_m$ $C=Bc_1+\cdots+Bc_n\Rightarrow C=\sum_{i=1}^{m}\sum_{j=1}^{n}Ag(b_i)c_j$

(4) $C=A[c_1,\cdots,c_n]\Rightarrow C=B[c_1,\cdots,c_n]$ (5) $C=Ac_1+\cdots+Ac_n\Rightarrow C=Bc_1+\cdots+Bc_n$

Rmk $k\subset k[x,xy,xy^2,\cdots]\subseteq k[x,y]$ (not a f.g. k-algebra)

e.g. If K/k is a field extension then K is a k-algebra

Given two field extensions K/k and L/k a k-algebra homo from K to L is the same thing as k-embedding of K into L

K is a finite k-algebra $\Leftrightarrow$ K/k is a finite field extension i.e. $[K:k]<\infty$

Prop (Artin-Tate) Let $A\subseteq B\subseteq C$ be ring extensions.

Suppose i) A is Noetherian ii) C is a f.g. A-algebra iii) C is a finite B-algebra

Then B is a f.g. A-algebra

pf. It suffices to construct a subring B_0 s.t. $A\subseteq B_0\subseteq B\subseteq C$ B_0 is a f.g A-algebra C is a finite B_0-algebra. If such B_0 exists, then B_0 is Noetherian

Since B is a submodule of the f.g. B_0-module C. B is a f.g B_0-module

MEMO NO: _____
DATE: / /

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i.e. B is a finite B_0-algebra $\Rightarrow$ B is a f.g. B_0-algebra (B_0 f.g A-alg) $\Rightarrow$ B is a f.g. A-alg

$C = A[x_1, \cdots, x_m]$ $x_i \in C$ $C = By_1 + \cdots + By_n$ $y_j \in C$

Then $\exists\, b_{ij}$ $(1 \le i \le m,\ 1 \le j \le n)$, b_{jkl} $(1 \le j,k,l \le n) \in B$ s.t.

$$x_i = \sum_{j=1}^{n} b_{ij} y_j \qquad y_j y_k = \sum_{l=1}^{n} b_{jkl} y_l$$

$\Rightarrow C = B_0 y_1 + \cdots + B_0 y_n$ where $B_0 = A[b_{ij}, b_{jkl}]$

Thm (Zariski's lemma) Let k be a field and A be a f.g. k-algebra. If A is a field, then A is a finite k-algebra (i.e. A is a finite algebraic extension of k)

Proof Let k be a field, $x_1, \cdots, x_r$ $(r \ge 1)$ be indeterminates.

Then (1) $k[x_1, \cdots, x_r]$ has infinitely many prime elts (i.e. irreducible poly)

(2) $k(x_1, \cdots, x_r)$ is not a f.g. $k[x_1, \cdots, x_r]$-algebra

Pf. If $k[x_1, \cdots, x_r]$ has finitely many prime elts $p_1, \cdots, p_n$ then since each p_i is non-constant, $f = p_1 \cdots p_n + 1$ is a non-constant polynomial and $p_i \nmid f$ $(1 \le i \le n)$. This is a contradiction.

Def. Let K/k be a field extension and $a_1, \cdots, a_n \in K$.

If there is nonzero polynomial $f(x_1, \cdots, x_n) \in k[x_1, \cdots, x_n]$ s.t. $f(a_1, \cdots a_n) = 0$ then we say that $a_1, \cdots, a_n$ are <u>algebraically independent over k</u>.

Equivalently, the unique k-algebra homomorphism $\varphi: k[x_1, \cdots, x_n] \to K$, $x_i \mapsto a_i$ is injective. ($f(x_1, \cdots, x_n) \mapsto f(a_1, \cdots, a_n)$)

In the case φ extends to an k-embedding $k(x_1, \cdots, x_n) \hookrightarrow K$ whose image is the subfield $k(a_1, \cdots a_n)$ generated by $a_1, \cdots a_n$ over k.

Proof Let K/k be a f.g. field extension i.e. $K = k(a_1, \cdots a_n)$.

Then after possibly relabeling the generators a_i, there exists an $r \ge 0$ s.t. $a_1, \cdots, a_r$ are algebraically independent over k and each a_j $(r+1 \le j \le n)$ is algebraic over $k(a_1, \cdots, a_r)$.

$$k \subseteq \underbrace{k(a_1, \cdots, a_r)}_{\text{is } k(x_1, \cdots, x_r)} \subseteq K = k(a_1, \cdots, a_r)(a_{r+1}, \cdots, a_n)$$

($k(a_1, \cdots, a_r)$: purely transcendental; $\subseteq K$: algebraic)

Pf. Let $\{a_1, \cdots, a_r\}$ be a maximal algebraically independent subset of $\{a_1, \cdots, a_n\}$

Since $a_1,\cdots,a_r,a_j$ $(r+1\le j\le n)$ are algebraically dependent $\exists$ a nonzero polynomial $g_j(x_1,\cdots,x_r,y)\in k[x_1,\cdots,x_r,y]$ s.t. $g_j(a_1,\cdots,a_r,a_j)=0$ $\deg_y g_j\ge 1$

$g_j = g_{j,m}(x_1,\cdots,x_r)y^m+\cdots+g_{j,1}(x_1,\cdots,x_r)y+g_{j,0}(x_1,\cdots,x_r)$ where $m\ge 1$ $g_{j,m}\neq 0$

$y^m+\frac{g_{j,m-1}(a_1,\cdots,a_r)}{g_{j,m}(a_1,\cdots,a_r)}y^{m-1}+\cdots+\frac{g_{j,0}(a_1,\cdots,a_r)}{g_{j,m}(a_1,\cdots,a_r)}$ is a monic polynomial of y with coeff in $k(a_1,\cdots,a_r)$ which annihilate a_j $\Rightarrow$ a_j algebraic over $k(a_1,\cdots,a_r)$

Proof of Zariski's lemma

Let $A=k[a_1,\cdots,a_n]$ be a f.g. k-algebra and a field. By prop, we may assume $a_1,\cdots,a_r$ are alg independent over k and each a_j $(j=r+1,\cdots,n)$ is algebraic over $k(a_1,\cdots,a_r)$ $\quad k\subseteq k[a_1,\cdots,a_r]\subseteq k(a_1,\cdots,a_r)\subseteq A=k[a_1,\cdots,a_r][a_{r+1},\cdots,a_n]$
$=k(a_1,\cdots,a_r)[a_{r+1},\cdots,a_n]$

Note (1) $k[x_1,\cdots,x_r]$ is Noetherian

(2) A is a f.g. $k[x_1,\cdots,x_r]$-algebra

(3) A is a finite algebraic extension of $k(a_1,\cdots,a_r)$

Artin-Tate $\Rightarrow$ $k(a_1,\cdots,a_r)$ is a f.g. $k[a_1,\cdots,a_r]$-alg $\Rightarrow r=0$

$\Rightarrow$ A is a finite extension of k

Chapter 6 Zariski topology and Hilbert's Nullstellensatz

Def. Fix a ground field k. The affine n-space over k is $\mathbb{A}^n_k=\{(a_1,\cdots,a_n): a_i\in k, 1\le i\le n\}$

Each polynomial $f(x_1,\cdots,x_n)\in k[x_1,\cdots,x_n]$ defines a polynomial function $f:\mathbb{A}^n_k\to k$;
$p=(a_1,\cdots,a_n)\mapsto f(p)=f(a_1,\cdots,a_n)$.

The zeros of f is defined to be $Z(f)=\{p\in\mathbb{A}^n_k : f(p)=0\}$

More generally, given any subset $T\subseteq k[x_1,\cdots,x_n]$ the zero of set of T is
$Z(T)=\bigcap_{f\in T}Z(f)=\{p\in\mathbb{A}^n_k : f(p)=0\ \forall f\in T\}$.

Lemma. $Z(T)=Z((T))$ where (T) is the ideal generated by T

Pf. $T\subseteq(T)\Rightarrow Z((T))\subseteq Z(T)$. Conversely, each poly $h\in(T)$ $h=\sum_{i=1}^m f_i g_i$ $f_i\in T$ $g_i\in k[x_1,\cdots,x_n]$
$\forall p\in Z(T)$ $f_i(p)=0\Rightarrow h(p)=0\Rightarrow Z(T)\subseteq Z((T))$

Rmk. Since $k[x_1,\cdots,x_n]$ is Noetherian, for any $T\subseteq k[x_1,\cdots,x_n]$, there exists a finite subset $T_0\subseteq k[x_1,\cdots,x_n]$

MEMO NO: _____
DATE: / /

清華大學 Tsinghua University

s.t. $(T)=(T_0)$, In fact T_0 can be chosen to be a subset of T

Thus $Z(T)=Z((T))=Z((T_0))=Z(T_0)$

Notation $f_1,\cdots f_s \in k[x_1,\cdots x_n]$ $Z(f_1,\cdots f_s)=Z(\{f_1,\cdots f_s\})=Z((f_1,\cdots f_s))=\bigcap_{i=1}^{s} Z(f_i)$

Def. A subset $X\subseteq \mathbb{A}_k^n$ is an algebraic set if $\exists T\subseteq k[x_1,\cdots x_n]$ s.t. $X=Z(T)$

equivalently $X=Z(\mathcal{O})$ for some ideal $\mathcal{O}\subseteq k[x_1,\cdots x_n]$ or $T=Z(f_1,\cdots f_s)$ for some $f_1,\cdots f_s\in k[x_1,\cdots x_n]$

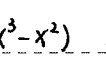
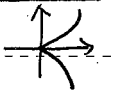
e.g. (1) $Z(0)=\mathbb{A}_k^n$ (2) $Z(k[x_1,\cdots x_n])=Z(1)=\phi$

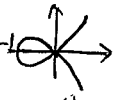
(3) $Z(x_1-a_1,\cdots,x_n-a_n)=\{(a_1,\cdots a_n)\}$ $a_1,\cdots a_n\in k$

(4) Algebraic sets in $\mathbb{A}_k^1=k$ are ϕ, k and finite subset of k ($k[x]$ is a PID)

(5) Let $f(x_1,\cdots x_n)\in k[x_1,\cdots x_n]\setminus k$ then $Z(f)$ is a hypersurface

$f(x,y)\in k[x,y]\setminus k$ $Z(f)$ is an affine plane curve

$k=\mathbb{R}$ parabola $Z(y-x^2)$  cuspidal curve  $Z(y^2-x^3)$

nodal curve $Z(y^2-x^3-x^2)$ 

Prop. The algebraic sets in $\mathbb{A}_k^n$ form the closed subsets of a topology on $\mathbb{A}_k^n$.

That is (1) $\phi, \mathbb{A}_k^n$ are algebraic sets (2) the union of two alg sets is an alg set

(3) the intersection of an arbitrary family of alg sets is an alg set

pf. (1)(3) ✓ (2) $Z(S)\cup Z(T)=Z(ST)$ $ST=\{fg: f\in S, g\in T\}$

This is called the Zariski topology on $\mathbb{A}_k^n$

Rmk (1) $Z(\mathcal{O})\cup Z(f)=Z(\mathcal{O}f)=Z(\mathcal{O}\cap(f))$ (2) $Z(f_1,\cdots f_r)\cup Z(g_1,\cdots g_s)=Z(f_ig_j: \substack{1\le i\le r\\ 1\le j\le s})$

(3) $Z(f)\cup Z(g)=Z(fg)$, (4) $Z(f_1)\cup\cdots\cup Z(f_s)=Z(f_1\cdots f_s)$, (5) $Z(f_1)\cap\cdots\cap Z(f_s)=Z(f_1,\cdots f_s)$

(6) $\bigcap_{i\in I} Z(\mathcal{O}_i)=Z(\sum_{i\in I}\mathcal{O}_i)$

For $f\in k[x_1,\cdots x_n]$ $D(f)=\mathbb{A}_k^n\setminus Z(f)=\{p\in\mathbb{A}_k^n : f(p)\neq 0\}$

$\mathbb{A}_k^n\setminus Z(T)=\mathbb{A}_k^n\setminus(\bigcap_{f\in T}Z(f))=\bigcup_{f\in T}D(f)$ the collection $\{D(f): f\in k[x_1,\cdots x_n]\}$ is a basis of the Zariski topology

Ex. $\mathbb{A}_k^n$ is compact in the Zariski topology

e.g. Any finite subset of $\mathbb{A}_k^n$ is Zariski-closed

The Zariski topology on $\mathbb{A}_k^n=k$ is the cofinite topology

That is $X \subseteq k$ is Zariski-open $\Leftrightarrow X=\phi$ or X^c is finite

$k[x_1,\cdots,x_n]$ $\qquad$ $\mathbb{A}_k^n$

$T \longrightarrow Z(T)=\{p\in \mathbb{A}_k^n : f(p)=0\ \forall f\in T\}$

$I(X)=\{f\in k[x_1,\cdots,x_n]: \longleftarrow X$

$f(p)=0,\ \forall p\in X\}$ $\qquad$ X: A subset of $\mathbb{A}_k^n$

Fact. $I(X)$ is a radical ideal of $k[x_1,\cdots,x_n]$, called the ideal of X

Prop. Let $T\subseteq k[x_1,\cdots,x_n]$ and $X\subseteq \mathbb{A}_k^n$. Then

(1) $T\subseteq I(Z(T))$, $X\subseteq Z(I(X))$

(2) $T_1\subseteq T_2 \Rightarrow Z(T_1)\supseteq Z(T_2)$ $\quad X_1\subseteq X_2 \Rightarrow I(X_1)\supseteq I(X_2)$

((1),(2) $\Rightarrow$) (3) $Z(T)=ZIZ(T)$ $\quad I(X)=IZI(X)$

(4) There is an inclusion-reversing bijection between the set

$\{\text{radical ideals}\}\supseteq\{I(X): X\subseteq \mathbb{A}_k^n\}=\{\mathfrak{a}\subseteq k[x_1,\cdots,x_n]: \mathfrak{a}=IZ(\mathfrak{a})\}$ and the set

$\{Z(T): T\subseteq k[x_1,\cdots,x_n]\}=\{X\subseteq \mathbb{A}_k^n : X=ZI(X)\}$ given by $\mathfrak{a}\mapsto Z(\mathfrak{a})$, $I(X)\leftarrow\!\shortmid X$

$\qquad$ $\|$

$\qquad$ {algebraic sets in $\mathbb{A}_k^n$}

Lemma. $\forall$ ideal $\mathfrak{a}\subseteq k[x_1,\cdots,x_n]$, one has $\sqrt{\mathfrak{a}}\subseteq IZ(\mathfrak{a})$

pf. $\mathfrak{a}\subseteq IZ(\mathfrak{a})$ $\quad IZ(\mathfrak{a})$ is a radical ideal

Rmk. The containment $\sqrt{\mathfrak{a}}\subseteq IZ(\mathfrak{a})$ can be strict if k is not alg closed

Consider the case $\mathfrak{a}=0$ $\quad \sqrt{0}=0$ $\quad Z(0)=\mathbb{A}_k^n$ $\quad IZ(0)=I(\mathbb{A}_k^n)$

Ex. If k is an infinite field, then $0=I(\mathbb{A}_k^n)$ (by induction on n)

$k=\mathbb{F}_q$, every elt in $\mathbb{F}_q$ is a root of the nonzero poly x^q-x

If $\mathfrak{a}\subsetneq k[x_1,\cdots,x_n]$ is a proper ideal s.t. $Z(\mathfrak{a})=\phi$ then $\sqrt{\mathfrak{a}}\subsetneq k[x_1,\cdots,x_n]$

$IZ(\mathfrak{a})=I(\phi)=k[x_1,\cdots,x_n] \Rightarrow \sqrt{\mathfrak{a}}\subsetneq IZ(\mathfrak{a})$

k not alg-closed $\Rightarrow \exists$ an irred. poly $p(x)\in k[x]$ deg $p\geq 2 \Rightarrow \mathfrak{a}=(p(x))$ is a maximal ideal in $k[x]$, yet $Z(p(x))=\phi$

e.g. $k=\mathbb{R}$ $\quad p(x)=x^2+1$

MEMO NO: _____
DATE: / 5 / 8

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Hilbert's Nullstellensatz Let k be an alg closed field and let $\mathcal{I} \subseteq k[x_1, \cdots, x_n] = A$ be an ideal. Then $I(Z(\mathcal{I})) = \sqrt{\mathcal{I}}$

That is, if a poly $f \in A$ vanishes at all points of $Z(\mathcal{I})$, then $\exists\, r \geq 1$ s.t. $f^r \in \mathcal{I}$

Coro. Let k be an alg closed field. Then there is an inclusion-reversing one-to-one correspondence between radical ideals of $k[x_1, \cdots, x_n]$ and alg sets in $\mathbb{A}^n_k$ given by $\mathcal{I} \mapsto Z(\mathcal{I})$, $I(X) \hookleftarrow X$

Thm (Weak Nullstellensatz) Let k be an alg closed field

(1) Every maximal ideal of $k[x_1, \cdots, x_n]$ has the form $(x_1 - a_1, \cdots, x_n - a_n)$ for some $(a_1, \cdots, a_n) \in \mathbb{A}^n_k$ ($\mathbb{A}^n_k = A$)

(2) If $\mathcal{I} \subsetneq k[x_1, \cdots, x_n]$ is a proper ideal then $Z(\mathcal{I}) \neq \phi$

Pf. (1) $p = (a_1, \cdots, a_n) \in \mathbb{A}^n_k$ denote $m_p = (x_1 - a_1, \cdots, x_n - a_n)$ m_p is the kernel of $k[x_1, \cdots, x_n] \to k$ $f(x_1, \cdots, x_n) \mapsto f(a_1, \cdots, a_n)$ surjective

($\forall f \in A\ \exists\, q_1, \cdots, q_n \in A$ s.t. $f = (x_1 - a_1) q_1 + \cdots + (x_n - a_n) q_n + f(a_1, \cdots, a_n)$)

That is $m_p = I(\{p\})$

$\Rightarrow k[x_1, \cdots, x_n]/m_p \simeq k \Rightarrow m_p$ is a maximal ideal

Let m be a maximal ideal in $k[x_1, \cdots, x_n]$. Then $k[x_1, \cdots, x_n]/m$ is a field and a f.g. k-algebra Zariski's lemma $\Rightarrow k \subseteq k[x_1, \cdots, x_n]/m$ is a finite alg extension k alg closed $\Rightarrow k \xrightarrow{\sim} k[x_1, \cdots, x_n]/m$ $\forall i = 1, 2, \cdots, n\ \exists\, a_i \in k$ s.t. $\overline{x_i} = \overline{a_i}$

$\Rightarrow m_p = (x_1 - a_1, \cdots, x_n - a_n) \subseteq m \Rightarrow m = m_p$ since m_p is maximal.

(2) $\mathcal{I}$ proper $\Rightarrow \exists\, m_p$ s.t. $\mathcal{I} \subseteq m_p \Rightarrow \{p\} = Z(m_p) \subseteq Z(\mathcal{I}) \neq \phi$

Thm (Strong Nullstellensatz) Let k be an alg closed field, $n \geq 1$ $\mathcal{I} \subseteq k[x_1, \cdots, x_n]$ is an ideal. Then $IZ(\mathcal{I}) = \sqrt{\mathcal{I}}$

Pf. (Rabinowitsch's trick) Suppose $f \in k[x_1, \cdots, x_n]$ satisfies $f(p) = 0$ $\forall p \in Z(\mathcal{I})$

Goal: $\exists\, r \geq 1$ s.t. $f^r \in \mathcal{I}$

Let $\mathcal{I}' = (\mathcal{I}, x_0 f - 1) \subseteq k[x_0, x_1, \cdots, x_n]$. Then $(a_0, p) \in Z(\mathcal{I}') \Leftrightarrow p \in Z(\mathcal{I})$ and $a_0 f(p) - 1 = 0$

$\Leftrightarrow p \in Z(\mathcal{I}) \cap D(f)$ and $a_0 = \frac{1}{f(p)}$

By assumption $Z(\mathcal{O}_1) \cap D(f) = \{p \in Z(\mathcal{O}_1) : f(p) \neq 0\} = \phi \Rightarrow Z(\mathcal{O}_1') = \phi$

Weak Nullstellensatz $\Rightarrow \mathcal{O}_1' = (1)$ i.e. $1 \in \mathcal{O}_1'$

$\Rightarrow \exists f_1, \cdots f_m \in \mathcal{O}_1$ $g_0, g_1, \cdots, g_m \in k[x_0, x_1, \cdots, x_n]$ s.t. $\sum_{i=1}^{m} f_i(x_1, \cdots x_n) g_i(x_0, x_1, \cdots x_n)$ *

Setting $x_0 = \frac{1}{f(x_1, \cdots, x_n)}$ yields an identity in $k(x_1, \cdots, x_n)$ $+ (x_0 f(x_1, \cdots, x_n) - 1) g_0(x_0, x_1, \cdots, x_n) = 1$

$\sum_{i=1}^{m} f_i(x_1, \cdots x_n) g_i\left(\frac{1}{f(x_1, \cdots x_n)}, x_1, \cdots, x_n\right) = 1$

$\Rightarrow \exists r \geq 1$ large enough $h_i \in k[x_1, \cdots x_n]$ $1 \leq i \leq m$ s.t. $\sum_{i=1}^{m} f_i \frac{h_i}{f^r} = 1 \Rightarrow f^r = \sum_{i=1}^{m} f_i h_i \in \mathcal{O}_1$

Def Let k be an alg closed field $X \subseteq \mathbb{A}_k^n$ algebraic set

The coordinate ring of X is $A(X) = k[x_1, \cdots, x_n] / I(X)$

e.g. $A(\mathbb{A}_k^n) = k[x_1, \cdots, x_n]$

$X = Z(y^2 - x^3) \subseteq \mathbb{A}_k^2$ $A(X) = k[x,y]/(y^2 - x^3)$ ($(y^2 - x^3)$ is a prime ideal)

Rmk. $\sqrt{I(X)} = I(X) \Rightarrow \mathrm{nil}(A(X)) = 0$ i.e. $A(X)$ is reduced.

$$\begin{array}{ccc} \mathbb{A}_k^n & \xrightarrow{1:1} & \mathrm{Spec}_m\, k[x_1, \cdots, x_n] \\ \uparrow & & \uparrow \\ p & \longmapsto & m_p \\ \uparrow & & \uparrow \\ \text{algebraic set } X & \xrightarrow{1:1} & \mathrm{Spec}_m A(X) = \mathrm{Spec}_m k[x_1, \cdots, x_n]/I(X) \\ \| & & \\ Z I(X) & & \end{array}$$

$p \in X = Z I(X) \Leftrightarrow f(p) = 0 \ \forall f \in I(X) \Leftrightarrow f \in m_p \ \forall f \in I(X) \Leftrightarrow I(X) \subseteq m_p$

$p \in Z(\mathcal{O}_1) \Leftrightarrow \mathcal{O}_1 \subseteq m_p$

If we transport the Zariski topology on $\mathbb{A}_k^n$ to $\mathrm{Spec}_m\, k[x_1, \cdots, x_n]$ then a closed subset in $\mathrm{Spec}_m\, k[x_1, \cdots, x_n]$ has the form $\{m \in \mathrm{Spec}_m k[x_1, \cdots x_n] : \mathcal{O}_1 \subseteq m\}$

where $\mathcal{O}_1$ is an ideal of $k[x_1, \cdots, x_n]$

Def. Let A be a ring Define $\mathrm{Spec} A = \{\text{prime ideals of } A\}$

$\cup$

$\mathrm{Spec}_m A = \{\text{maximal ideals of } A\}$

Prop. $\mathrm{Spec} A$ can be made into a topological space by defining the closed subsets to be $\mathcal{V}(\mathcal{O}_1) = \{p \in \mathrm{Spec} A : \mathcal{O}_1 \subseteq p\}$ where $\mathcal{O}_1$ is any ideal of A. This is called the Zariski topology on $\mathrm{Spec} A$. This topology has a basis $\{D(a) : a \in A\}$ where

MEMO NO: _____
DATE: / /

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$D(a) = \{\mathfrak{p} \in \operatorname{Spec} A : a \notin \mathfrak{p}\}$

pf. $V(A) = \emptyset$ $V(0) = \operatorname{Spec} A$ $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b})$

$\bigcap_{i \in I} V(\mathfrak{a}_i) = V(\sum_{i \in I} \mathfrak{a}_i)$ $D(a) = \operatorname{Spec} A \setminus V((a))$ $\Rightarrow D(a)$ is open

$\operatorname{Spec} A \setminus V(\mathfrak{a}) = \{\mathfrak{p} : \mathfrak{a} \notin \mathfrak{p}\} = \{\mathfrak{p} : \exists\, a \in \mathfrak{a} \text{ s.t. } a \notin \mathfrak{p}\}$

$= \bigcup_{a \in \mathfrak{a}} D(a)$

e.g. Let k be a field

(1) $\operatorname{Spec} k$ is a singleton

(2) $\operatorname{Spec} k[x] = \{0\} \cup \{(p(x)) : p(x) \text{ irreducible}\} = \{0\} \cup \operatorname{Spec}_m k[x]$ $\overset{0}{\uparrow}$ ——— k

k alg closed $\operatorname{Spec}_m k[x] = \{(x-a) : a \in k\} \overset{1:1}{\cong} k$

(3) $\operatorname{Spec} k[x,y] = \{0\} \cup \{(p(x,y)) : p(x,y) \text{ irred.}\} \cup \{\mathfrak{m} : \mathfrak{m} \text{ maximal}\}$

k alg closed $\operatorname{Spec}_m k[x,y] = \{(x-a, y-b) : (a,b) \in k^2\} \cong \mathbb{A}^2_k$

e.g. $\operatorname{Spec} \mathbb{Z} = \{0\} \cup \{p\mathbb{Z} : p \text{ prime}\}$

$\mathbb{Z}[x]$ has three kinds of prime ideals

(1) 0 (2) (p), $(f(x))$ where p is prime, $f(x)$ primitive over $\mathbb{Z}$, irred over $\mathbb{Q}$

(3) $(p, f(x))$ where p is prime, $f(x)$ is monic and irred mod p

Prop Let $f: A \to B$ be a ring map. Then $f^* \operatorname{Spec} B \to \operatorname{Spec} A$

is a continuous map of topological spaces. $\mathfrak{q} \mapsto \mathfrak{q}^c = f^{-1}(\mathfrak{q}) = \mathfrak{q} \cap A$

Ex (1) If $\mathfrak{a}$ is an ideal of A, then $\pi^*: \operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$ induced by the quotient map $\pi: A \to A/\mathfrak{a}$ is a homeomorphism of $\operatorname{Spec} A/\mathfrak{a}$ onto the Zariski closed subset $V(\mathfrak{a})$ of $\operatorname{Spec} A$

(2) k field $f: k[t] \to k[t]$; $t \mapsto t^2$ $\rightsquigarrow f^*: \operatorname{Spec} k[t] \to \operatorname{Spec} k[t]$

$a \mapsto a^2$ $\mathbb{A}^1_k \to \mathbb{A}^1_k$

(3) $f: k[x,y]/(y^2 - x^3) \to k[t]$ $\bar{x} \mapsto t^2$, $\bar{y} \mapsto t^3$

$\rightsquigarrow f^*: \operatorname{Spec} k[t] \to \operatorname{Spec} k[x,y]/(y^2-x^3)$

$\mathbb{A}^1_k \to Z(y^2 - x^3)$

$a \mapsto (a^2, a^3)$

MEMO NO: _____
DATE: / 5 / 13

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Chapter 7 Localization

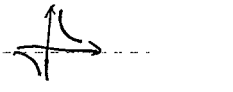
Recall Rabinowitsch's trick:

Let k be alg closed. $f \in k[x_1, \cdots, x_n]$

$D(f) = \mathbb{A}^n_k \setminus Z(f) = \{P \in \mathbb{A}^n_k : f(P) \neq 0\} \subseteq \mathbb{A}^n_k$ is open

$\downarrow$ 1:1 $\quad \downarrow P \mapsto (\frac{1}{f(P)}, P)$

$Z(x_0 f - 1) = \{(a_0, P) \in \mathbb{A}^{n+1}_k : a_0 f(P) = 1\} \subseteq \mathbb{A}^{n+1}_k$ is closed

e.g. —o— punctured line $D(x) = \mathbb{A}^1_k \setminus Z(X) \xrightarrow{1:1} Z(XY-1) \in \mathbb{A}^2_k$ 

$x \mapsto (x, \frac{1}{x})$

the coordinate ring of $Z(x_0 f - 1)$ is $k[x_0, \cdots, x_n]/(x_0 f(x_1, \cdots, x_n) - 1) = k[x_1, \cdots, x_n][\frac{1}{f(x_1, \cdots, x_n)}]$

Algebra	Geometry
$k[x_1, \cdots, x_n] \rightsquigarrow k[x_1, \cdots, x_n][\frac{1}{f}]$	$\mathbb{A}^n_k \rightsquigarrow \mathbb{A}^n_k \setminus Z(f) = D(f) \simeq Z(x_0 f - 1)$

Def. Let A be a ring. A subset $S \subseteq A$ is <u>multiplicative</u> if $1 \in S$ and $s, s' \in S \Rightarrow ss' \in S$

e.g. (1) $\{1\}$ (2) $A^\times = \{\text{units of } A\}$ (3) For $f \in A$ $S = \{f^n : n \geq 0\} = \{1, f, f^2, \cdots\}$

(4) $\{\text{non-zero-divisors of } A\}$ (5) $A \setminus P$ where $P \in \operatorname{Spec} A$

Prop Let A be a ring, $S \subseteq A$ be a multiplicative subset

Define a relation $\sim$ on $A \times S$ as follows:

$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S$ s.t. $u(at - bs) = 0$. Then

(1) $\sim$ is an equivalence relation

(2) Denote the equivalence class of (a, s) by $\frac{a}{s}$ and denote the set of equiv classes by $S^{-1}A$. Then $S^{-1}A$ is a ring with addition and mult defined as follows: $\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}$, $\frac{a}{s} \cdot \frac{b}{t} = \frac{ab}{st}$

The zero elt of $S^{-1}A$ is $\frac{0}{1}$, the identity elt is $\frac{1}{1}$

Def. $S^{-1}A$ is called the <u>localization of A with respect to S</u>

e.g. A integral domain $S = A \setminus \{0\}$ then $S^{-1}A = \operatorname{Frac}(A)$ $A \hookrightarrow \operatorname{Frac}(A)$

Prop Let A be a ring, $S \subseteq A$ a mult subset.

(1) There is a ring map $\varphi: A \to S^{-1}A$, $a \mapsto \frac{a}{1}$. Moreover, $\forall s \in S$ $\varphi(s)$ is a unit in $S^{-1}A$

(2) φ has the ~~universit~~ universal property that $\forall$ ring map $f: A \to B$ s.t. $\forall s \in S$ $f(s)$ is a unit in B, $\exists!$ ring map $g: S^{-1}A \to B$ s.t. $f = g\varphi$

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & S^{-1}A \\ {\scriptstyle f}\downarrow & \swarrow_{\exists! g} & \\ B & & \end{array}$$

(3) $\ker\varphi = \{a \in A : \exists s \in S \text{ s.t. } as = 0\} = \bigcup_{s \in S} (0:s)$

(4) $S^{-1}A = 0 \Leftrightarrow 0 \in S$

pf (1) $\forall s \in S$ $\varphi(s) = \frac{s}{1}$ $\frac{s}{1} \cdot \frac{1}{s} = \frac{s}{s} = \frac{1}{1} \Rightarrow \frac{s}{1}$ is invertible in $S^{-1}A$

(2) $g(\frac{a}{s}) = f(a)f(s)^{-1}$

(3) $a \in \ker\varphi \Leftrightarrow \frac{a}{1} = 0 = \frac{0}{1} \Leftrightarrow \exists s \in S$ s.t. $as = 0$

(4) $S^{-1}A = 0 \Leftrightarrow \frac{1}{1} = \frac{0}{1} \Leftrightarrow 1 \in \ker\varphi \Leftrightarrow \exists s \in S$ s.t. $s = 0$ i.e. $0 \in S$

Notation For $f \in A$ $S = \{f^i : i \geq 0\}$, we denote $S^{-1}A$ by A_f or $A[\frac{1}{f}]$

Note $A_f = 0 \Leftrightarrow f \in \mathrm{nil}(A)$

For $P \in \mathrm{Spec}(A)$ $S = A \setminus P$, we denote $S^{-1}A$ by A_P, called the localization of A at P $A_P \neq 0$ since $0 \in P$

e.g. Let A be an integral domain. $S \subseteq A - \{0\}$ multiplicative Then $A \to S^{-1}A$ is injective $K = \mathrm{Frac}(A)$

$$\begin{array}{ccc} A & \hookrightarrow & S^{-1}A \\ \downarrow & \swarrow_{\exists!} & \\ K & & \end{array}$$

the induced map $S^{-1}A \to K$ is also injective

$\Rightarrow A \subseteq S^{-1}A \subseteq K$ $\Rightarrow S^{-1}A$ is an integral domain and its fraction field is K

e.g. $A = \mathbb{Z}$ $\mathbb{Z}[\frac{1}{2}] = \{\frac{m}{2^r} : r \geq 0, m \in \mathbb{Z}\} \hookrightarrow \mathbb{Q}$ $\mathbb{Z}_{(2)} = \{\frac{m}{n} : m, n \in \mathbb{Z}, 2 \nmid n\} \hookrightarrow \mathbb{Q}$

Both $\mathbb{Z}[\frac{1}{2}]$ and $\mathbb{Z}_{(2)}$ are PIDs

The prime elts in $\mathbb{Z}[\frac{1}{2}]$ are $3, 5, 7, \cdots$ in $\mathbb{Z}_{(2)}$: 2

Ex. A PID (r. UFD) $S \subseteq A \setminus 0$ multiplicative then $S^{-1}A$ is a PID (r. UFD). And the prime elts in $S^{-1}A$ are the prime elts in A not dividing any of the elts in S.

e.g. $f \in A$ $A[x]/(xf-1) \cong A_f$

A ring $S \subseteq A^\times$ multiplicative $\Rightarrow$ the localization map $A \to S^{-1}A$ is an isom

Def. Let A be a ring $S = \{\text{non-zero-divisors of } A\}$. Then $S^{-1}A$ is called the total quotient ring of A. $A \to S^{-1}A$ is injective

e.g. $A = k[x,y]/(x) \cong k[y]$ total quotient ring : $\mathrm{Frac}(A) \cong k(y)$

$A = k[x,y]/(xy)$ $\mathrm{zdiv}(A) = (x)/(xy) \cup (y)/(xy)$ total quotient ring : $k(x) \times k(y)$

$A = \mathbb{Z}/6\mathbb{Z}$ $S = \{\bar{2}^i : i \geq 0\} = \{\bar{1}, \bar{2}, \bar{4}\}$ $\varphi : A \to S^{-1}A$ $\ker\varphi = \{\bar{0}, \bar{3}\} = 3\mathbb{Z}/6\mathbb{Z}$

$\mathbb{Z}/6\mathbb{Z} \xrightarrow{\varphi} (\mathbb{Z}/6\mathbb{Z})_{\bar{2}}$ π sends $\bar{1}, \bar{2}, \bar{4}$ to $\bar{1}, \bar{2}, \bar{1}$ are units in $\mathbb{Z}/3\mathbb{Z}$

$\pi \downarrow \quad \nearrow \simeq$

$\mathbb{Z}/3\mathbb{Z} \simeq \mathbb{Z}/6\mathbb{Z}/3\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/6\mathbb{Z}/\ker\varphi$

Ex. A ring, $S \subseteq A$ mult. $\varphi : A \to S^{-1}A$ $\mathfrak{a}$ an ideal of A $\mathfrak{a} \subseteq \ker\varphi$ $\bar{A} = A/\mathfrak{a}$

$\bar{S} = \{\bar{s} : s \in S\} \subseteq \bar{A}$. Then $\bar{S}^{-1}\bar{A} \simeq S^{-1}A$

$\mathbb{Z}[\frac{1}{4}] = \mathbb{Z}[\frac{1}{2}]$ $\mathbb{Z}[\frac{1}{6}] = \mathbb{Z}[\frac{1}{2}, \frac{1}{3}]$

§2. ideals of $S^{-1}A$

Prop. Let A be a ring $S \subseteq A$ mult. $\varphi : A \to S^{-1}A$ localization map

$\mathfrak{a}$ an ideal of A, $\mathfrak{b}$ an ideal of $S^{-1}A$

Then (i) $\mathfrak{a}^e = \{\frac{a}{s} : a \in \mathfrak{a}, s \in S\}$

$\mathfrak{a}^e = (1) \Leftrightarrow \mathfrak{a} \cap S \neq \emptyset$; $\mathfrak{a}^e \subsetneq (1) \Leftrightarrow \mathfrak{a} \cap S = \emptyset$

$\mathfrak{a}^{ec} = \{a \in A : \exists s \in S \text{ s.t. } sa \in \mathfrak{a}\} \overset{= \bigcup_{s \in S}(\mathfrak{a} : s)}{\supseteq} \mathfrak{a}$ $\ker\varphi = (0)^{ec}$

$\mathfrak{a}^{ec} = \mathfrak{a} \Leftrightarrow$ no elt of S is a zero-divisor in $A/\mathfrak{a}$

pf. $\mathfrak{a}^e = (1) \Leftrightarrow \frac{1}{1} \in \mathfrak{a}^e \Leftrightarrow \exists a \in \mathfrak{a}, s \in S$ s.t. $\frac{a}{s} = \frac{1}{1} \Rightarrow \exists s' \in S$ s.t. $s'a = s's \in \mathfrak{a} \cap S$

conversely if $\mathfrak{a} \cap S \neq \emptyset$ say $x \in \mathfrak{a} \cap S$ then $\frac{1}{1} = \frac{x}{x} \in \mathfrak{a}^e \Rightarrow \mathfrak{a}^e = (1)$

$\mathfrak{a}^{ec} = \{x \in A \mid \frac{x}{1} \in \mathfrak{a}^e\} = \{x \in A \mid \exists a \in \mathfrak{a}, s \in S \text{ s.t. } \frac{x}{1} = \frac{a}{s}\}$

$= \{x \in A \mid \exists s \in S \text{ s.t. } sx \in \mathfrak{a}\} = \bigcup_{s \in S}(\mathfrak{a} : s)$

$\mathfrak{a}^{ec} = \mathfrak{a} \Leftrightarrow \forall s \in S \ (\mathfrak{a} : s) = \mathfrak{a} \Leftrightarrow (\bar{0} : \bar{s}) = \bar{0}$ in $A/\mathfrak{a}$ $\forall s \in S$

i.e. $\bar{s}$ is not a zero-divisor in $A/\mathfrak{a}$

NB. For $a\in A$ $s\in S$ $\frac{a}{s}\in \mathfrak{a}^e \Leftrightarrow \frac{a}{1}\in \mathfrak{a}^e \Leftrightarrow a\in \mathfrak{a}^{ec} \Leftrightarrow \exists t\in S$ s.t. $at\in \mathfrak{a}$
$\Downarrow$
$a\in \mathfrak{a}$

(2) $\mathfrak{b}^{ce}=\mathfrak{b}$ i.e. every ideal in $S^{-1}A$ is an extended ideal

(1)(2): $e:\{\text{ideals of } A\}\to\{\text{ideals of } S^{-1}A\}$ is surjective

$c:\{\text{ideals of } S^{-1}A\}\to\{\text{ideals of } A\}$ is injective and c is a "section" of e in the sense that $ec=id$

Among the ideals of A whose extension is $\mathfrak{b}$, $\mathfrak{b}^c$ is the largest one ($\mathfrak{a}^e\subseteq\mathfrak{b} \Rightarrow \mathfrak{a}\subseteq\mathfrak{a}^{ec}=\mathfrak{b}^c$, $\mathfrak{b}^{ce}=\mathfrak{b}$)

(3) There is an inclusion-preserving one-to-one correspondence between the prime ideals of A which don't meet S and the prime ideals of $S^{-1}A$

$$\operatorname{Spec}A\supseteq\{\mathfrak{p}\in\operatorname{Spec}A:\mathfrak{p}\cap S=\phi\}\xrightarrow{1:1}\operatorname{Spec}S^{-1}A$$
$$\mathfrak{p}\mapsto\mathfrak{p}^e$$
$$\mathfrak{q}^c\longleftarrow\mathfrak{q}$$

In fact, the map $\varphi^*:\operatorname{Spec}S^{-1}A\to\operatorname{Spec}\operatorname{Spec}A$ ~~induced~~ induced by the localization map $\varphi:A\to S^{-1}A$ gives a homeomorphism of $\operatorname{Spec}S^{-1}A$ onto the subset $\{\mathfrak{p}\in\operatorname{Spec}A:\mathfrak{p}\cap S=\phi\}$ of $\operatorname{Spec}A$.

$\Sigma=\{\mathfrak{a}:\mathfrak{a}\text{ ideals of }A,\ \mathfrak{a}\cap S=\phi\}$

$$\begin{array}{ccc}\operatorname{Spec}A\cap\Sigma & \xrightarrow{1:1} & \operatorname{Spec}S^{-1}A\\ \uparrow & & \uparrow\\ \{\text{maximal elts in }\Sigma\} & \xrightarrow{1:1} & \operatorname{Spec}_m S^{-1}A\end{array}$$

Recall (ZL): $\forall\,\mathfrak{a}\in\Sigma\ \exists\,\mathfrak{p}\in\Sigma$ prime $\mathfrak{a}\subseteq\mathfrak{p}$

prove again: $\mathfrak{a}\cap S=\phi \Leftrightarrow \mathfrak{a}^e\subsetneq(1)$ in $S^{-1}A \Rightarrow \exists\,\pi\in\operatorname{Spec}_m S^{-1}A$ s.t. $\mathfrak{a}^e\subseteq\pi$
$\Rightarrow \mathfrak{a}\subseteq\mathfrak{a}^{ec}\subseteq\pi^c$ π^c prime ideal in A

$\operatorname{nil}(A)\supseteq\bigcap_{\operatorname{Spec}A}\mathfrak{p}$ prove again: $f\notin\operatorname{nil}(A)\Rightarrow A_f\neq 0\Rightarrow A_f$ has a maximal ideal π
$\Rightarrow \pi^c$ prime ideal in A and $\pi^c\cap\{f^i:i\geq 0\}=\phi \Rightarrow f\notin\pi^c$

In Rabinowitsch's trick: $A[x]=(\mathfrak{a},xf-1) \Leftrightarrow A[x]/(\mathfrak{a},xf-1)=0$
$\Leftrightarrow A[x]/(xf-1)/(\mathfrak{a},xf-1)/(xf-1)=0 \Leftrightarrow A_f/\mathfrak{a}A_f=0 \Leftrightarrow A_f=\mathfrak{a}A_f$

$\Leftrightarrow \mathfrak{a} \cap \{f^i : i \geq 0\} \neq \emptyset \Leftrightarrow f \in \sqrt{\mathfrak{a}}$

§3. Module of fractions

Prop. Let A be a ring. $S \subseteq A$ a mult subset, M an A-module.

Define a relation $\sim$ on $M \times S$ as follows:

$$(x,s) \sim (y,t) \Leftrightarrow \exists u \in S \text{ s.t. } u(tx - sy) = 0$$

Then (1) $\sim$ is an equivalence relation. Denote the equiv class of (x,s) by $\frac{x}{s}$ and the set of equiv classes by $S^{-1}M$. Then $S^{-1}M$ is an $S^{-1}A$-module via

$$\frac{x}{s} + \frac{y}{t} = \frac{tx+sy}{st} \qquad \frac{a}{s} \cdot \frac{x}{t} = \frac{ax}{st}$$

In particular, $S^{-1}M$ is an A-module via the loc map $A \to S^{-1}A$

(2) The map $\psi : M \to S^{-1}M,\ x \mapsto \frac{x}{1}$ is an A-module homo

and $\forall S^{-1}A$-module N and $\forall A$-mod homo $f : M \to N$

$$\begin{array}{ccc} M & \xrightarrow{\psi} & S^{-1}M \quad \exists!\ S^{-1}A\text{-mod homo } g \text{ s.t. } f = g\psi \\ {\scriptstyle f}\downarrow & \swarrow_{\exists! g} & \\ N & & \end{array}$$

In fact, $g(\frac{x}{s}) = \frac{1}{s} f(x)$

(3) If $f : M \to N$ is an A-mod homo, then

$S^{-1}f : S^{-1}M \to S^{-1}N,\ \frac{x}{s} \mapsto \frac{f(x)}{s}$ is a $S^{-1}A$-module homo

In fact, there is a functor $S^{-1} : A\text{-Mod} \to S^{-1}A\text{-Mod}$

$M \mapsto S^{-1}M$

$f \mapsto S^{-1}f$

Rmk. (2) is to say $\forall N \in S^{-1}A\text{-Mod}$ ($M \in A\text{-Mod}$) $\mathrm{Hom}_{S^{-1}A}(S^{-1}M, N) \overset{\gamma_{M,N}}{\cong} \mathrm{Hom}_A(M, N|_A)$

$\gamma_{M,N}$ is natural in M and N i.e. S^{-1} is a left adjoint of $|_A$

We know $S^{-1}A \underset{A}{\otimes} - : A\text{-Mod} \to S^{-1}A\text{-Mod}$ is also a left adjoint of $|_A$.

$\Rightarrow S^{-1}A \underset{A}{\otimes} - \cong S^{-1}$ as functors.

Prop There is an isom of $S^{-1}A$-modules $\alpha_M : S^{-1}A \underset{A}{\otimes} M \to S^{-1}M,\ \frac{a}{s} \otimes x \mapsto \frac{ax}{s}$

Moreover α_M is natural in M. Hence $\alpha = (\alpha_M)_{M \in A\text{-Mod}}$ defines a natural isom from $S^{-1}A \underset{A}{\otimes} -$ to S^{-1}

Rmk. The unit of the adjunction (S^{-1}, l_A, α) is $\eta : id_{A\text{-Mod}} \Rightarrow (l_A)(S^{-1})$

$\eta_M = \psi_M : M \to S^{-1}M$, $x \mapsto \frac{x}{1}$ $M \in A\text{-Mod}$

the counit $\varepsilon : S^{-1}(l_A) \Rightarrow id_{S^{-1}A\text{-Mod}}$

Prop. Let N be an $S^{-1}A$-module. Then

$$\varepsilon_N : S^{-1}(N|_A) \to N, \ \frac{y}{s} \mapsto \frac{1}{s}y \quad y \in N \ s \in S$$

is an isom. of $S^{-1}A$-modules. In particular, $S^{-1}A \otimes_A N \simeq S^{-1}N \simeq N$ as $S^{-1}A$-modules

pf. If $\varepsilon_N(\frac{y}{s}) = \frac{1}{s}y = 0$, then $y = \frac{s}{1} \cdot (\frac{1}{s}y) = \frac{s}{1} \cdot 0 = 0$ in N (injective ∪, $\Downarrow \frac{y}{s} = 0$)

Also $y = \varepsilon_N(\frac{y}{1})$ surjective ∪

e.g. Let A be an integral domain with fraction field K

Then $K \otimes_A K \simeq K$ In particular $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \simeq \mathbb{Q}$

$A \to K = (A-0)^{-1}A$ $K \otimes_A K \simeq (A-0)^{-1}K \simeq K$

Prop. Let A be a ring, $S \subseteq A$ mult subset

(1) If M is an A-module s.t. $\forall s \in S$ $h_s : M \to M$, $x \mapsto sx$ is bijective then M is an $S^{-1}A$-module via $\frac{a}{s}x = h_s^{-1}(ax)$ and the A-mod structure on M obtained from this $S^{-1}A$-mod. str by restrictions of scalars coincides with the initial one.

(2) If N is an $S^{-1}A$-module, then as an A-module, $\forall s \in S$ $h_s : N \to N$ $y \mapsto sy$ is bijective and the $S^{-1}A$-module structure on N derived from this A-module structure by the process described in (1) coincides with the one given initially

If M, M' are two A-modules s.t. $\forall s \in S$ $h_s : M \to M$ and $h_s' : M' \to M'$ are bijective, then $f : M \to M'$ is an A-mod homo iff it is an $S^{-1}A$-mod homo.

The functor $l_A : S^{-1}A\text{-Mod} \to A\text{-Mod}$ induces an isomorphism from $S^{-1}A\text{-Mod}$ to the full subcategory of A-Mod comprised of those for which $h_s : M \to M$ are bijective for all $s \in S$

Prop Let A be a ring, $S \subseteq A$ mult

(1) If M, N are A-modules, then $\exists$ an isom of $S^{-1}A$-modules

$$S^{-1}M \underset{S^{-1}A}{\otimes} S^{-1}N \to S^{-1}(M \underset{A}{\otimes} N) \quad \frac{x}{s} \otimes \frac{y}{t} \mapsto \frac{x \otimes y}{st}$$

(localization commutes with tensor prod.)

(2) If M, N are $S^{-1}A$-modules, then $\exists$ an isom of $S^{-1}A$-modules

$$M \underset{A}{\otimes} N \to M \underset{S^{-1}A}{\otimes} N \quad x \otimes y \mapsto x \otimes y$$

Recall $M, N \in A\text{-Mod}$ $\quad M/\mathfrak{a}M \underset{A/\mathfrak{a}}{\otimes} N/\mathfrak{a}N \simeq (M \underset{A}{\otimes} N)/\mathfrak{a}(M \underset{A}{\otimes} N)$

$M, N \in A/\mathfrak{a}\text{-Mod}$ $\quad M \underset{A}{\otimes} N \simeq M \underset{A/\mathfrak{a}}{\otimes} N$

pf (1) Extension of scalars commutes with tensor product

Let $f: A \to B$ be a ring map $\quad M, N \in A\text{-Mod}$

Then $M_B \underset{B}{\otimes} N_B \simeq (M \underset{A}{\otimes} N)_B$ $\quad$ Set $B = S^{-1}A$ ✓ $(A/\mathfrak{a})$

(2) $M \underset{A}{\otimes} N \simeq M \underset{A}{\otimes} (S^{-1}A \underset{S^{-1}A}{\otimes} N) \simeq (M \underset{A}{\otimes} S^{-1}A) \underset{S^{-1}A}{\otimes} N \simeq S^{-1}M \underset{S^{-1}A}{\otimes} N \simeq M \underset{S^{-1}A}{\otimes} N$

§4. localization is exact

Prop. $S^{-1}: A\text{-Mod} \to S^{-1}A\text{-Mod}$ is exact.

Equivalently $S^{-1}A$ is a flat A-module (A-algebra)

pf. $S^{-1} \simeq S^{-1}A \underset{A}{\otimes} -$ $\Rightarrow$ S^{-1} is right exact

It suffices to show that if $f: M \to N$ is an injective A-module homo

then $S^{-1}f: S^{-1}M \to S^{-1}N$ is an injective $S^{-1}A$-module homo

$\forall$ If $(S^{-1}f)(\frac{x}{s}) = \frac{f(x)}{s} = 0$ then $\exists\, t \in S$ s.t. $tf(x) = 0$ in N i.e. $f(tx) = 0$

f injective $\Rightarrow$ $tx = 0 \Rightarrow \frac{x}{s} = \frac{tx}{ts} = 0 \Rightarrow S^{-1}f$ is injective.

e.g. A integral domain $K = \mathrm{Frac}(A)$. Then K is a flat A-algebra

Prop. Let A be a ring $S \subseteq A$ mult. M an A-module.

(1) If N is an A-submodule of M with inclusion map $i: N \hookrightarrow M$,

then $S^{-1}i: S^{-1}N \to S^{-1}M$ is injective which allows us to regard

$S^{-1}N$ as an $S^{-1}A$-submodule of $S^{-1}M$. Moreover $S^{-1}M/S^{-1}N \simeq S^{-1}(M/N)$

(2) If M, N_1, N_2 are A-submodules of M, then $S^{-1}(N_1+N_2) = S^{-1}N_1 + S^{-1}N_2$

$S^{-1}(N_1 \cap N_2) = S^{-1}N_1 \cap S^{-1}N_2$

(3) If $\mathfrak{a}$ is an ideal of A then $S^{-1}\mathfrak{a}$ is an ideal of $S^{-1}A$ and $S^{-1}\mathfrak{a} = \mathfrak{a}^e$

(4) If $\mathfrak{a}_1, \mathfrak{a}_2$ are ideals of A then $S^{-1}(\mathfrak{a}_1+\mathfrak{a}_2) = S^{-1}\mathfrak{a}_1 + S^{-1}\mathfrak{a}_2$

$S^{-1}(\mathfrak{a}_1 \cap \mathfrak{a}_2) = S^{-1}\mathfrak{a}_1 \cap S^{-1}\mathfrak{a}_2$ $\quad S^{-1}(\mathfrak{a}_1\mathfrak{a}_2) = (S^{-1}\mathfrak{a}_1)(S^{-1}\mathfrak{a}_2)$

Prop Let A be a ring, $S \subseteq A$ multiplicative subset, $\mathfrak{a}$ an ideal of A. Then $S^{-1}A/S^{-1}\mathfrak{a} \simeq \bar{S}^{-1}(A/\mathfrak{a})$ as rings where $\bar{S}$ is the image of S under $A \to A/\mathfrak{a}$

($S^{-1}A/S^{-1}\mathfrak{a}$ is $S^{-1}(A/\mathfrak{a})$)

Special cases (1) $S = \{f^i : i \geq 0\}$ $A_f/\mathfrak{a}A_f \simeq (A/\mathfrak{a})_{\bar{f}}$ it's zero iff $f \in \sqrt{\mathfrak{a}}$

(2) $\mathfrak{p} \in \operatorname{Spec} A$ $\mathfrak{a} \subseteq \mathfrak{p}$ $A_{\mathfrak{p}}/\mathfrak{a}A_{\mathfrak{p}} \simeq (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}}$

Rmk $\operatorname{Spec} S^{-1}A = \{S^{-1}\mathfrak{p} \ (\mathfrak{p}^e) : \mathfrak{p} \in \operatorname{Spec} A, \mathfrak{p} \cap S = \emptyset\}$

($A_{\mathfrak{p}}$ is local)

$\mathfrak{p} \in \operatorname{Spec} A$ $\quad S = A - \mathfrak{p}$

$\operatorname{Spec} A_{\mathfrak{p}} = \{\mathfrak{q}A_{\mathfrak{p}} : \mathfrak{q} \subseteq \mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} A\}$

$\Rightarrow A_{\mathfrak{p}}$ is a local ring with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$

(3) $\mathfrak{p} \in \operatorname{Spec} A$ $\quad A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \simeq \operatorname{Frac}(A/\mathfrak{p}) =: k(\mathfrak{p})$

(4) $\mathfrak{m} \in \operatorname{Spec m} A$ $\quad A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}} \simeq A/\mathfrak{m} =: k(\mathfrak{m})$

e.g. $A = \mathbb{Z}$ p, q prime $\quad (\mathbb{Z}/p\mathbb{Z})_{(q)} \cong \mathbb{Z}_{(q)}/p\mathbb{Z}_{(q)}$

$p = q$ $\quad \mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{Z}/p\mathbb{Z}$ by (4)

$p \neq q$ $\quad p\mathbb{Z} \not\subseteq q\mathbb{Z}$ $\quad \mathbb{Z}_{(q)}/p\mathbb{Z}_{(q)} = 0$ $\quad (\mathfrak{a} \cap S \neq \emptyset \Rightarrow \mathfrak{a}S^{-1}A = (1) = S^{-1}A)$

In general $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} A$ $\mathfrak{p} \not\subseteq \mathfrak{q}$ then $A_{\mathfrak{q}}/\mathfrak{p}A_{\mathfrak{q}} = 0$

e.g. $A = k[x,y]/(xy)$ $\quad \mathfrak{p} = (x)/(xy) = (\bar{x})$ $\quad$ By (2) $A_{\mathfrak{p}} \cong k[x,y]_{(x)}/(xy)$

y is invertible in $k[x,y]_{(x)} \Rightarrow k[x,y]_{(x)}/(xy) = k[x,y]_{(x)}/(x) \underset{\text{By (3)}}{\cong} \operatorname{Frac}\left(k[x,y]/(x)\right)$

$\simeq \operatorname{Frac}(k[y]) = k(y)$

Let A be a ring $\mathfrak{p} \in \operatorname{Spec} A$

(1) $A \to A/\mathfrak{p}$ induces $\operatorname{Spec} A/\mathfrak{p} \to \operatorname{Spec} A$ continuous injective $\quad \nu(\mathfrak{p})$

which gives a homeomorphism of $\operatorname{Spec} A/\mathfrak{p}$ onto the subset $\{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{p} \subseteq \mathfrak{q}\}$ of $\operatorname{Spec} A$

(2) $A \to A_{\mathfrak{p}}$ induces $\operatorname{Spec} A_{\mathfrak{p}} \to \operatorname{Spec} A$ cont inj. homeomorphism of $\operatorname{Spec} A_{\mathfrak{p}}$ onto $\{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{q} \subseteq \mathfrak{p}\}$ of $\operatorname{Spec} A$

$$A \to A_{\mathfrak{p}} \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \quad (\text{II } S)$$
$$A \to A/\mathfrak{p} \to \operatorname{Frac}(A/\mathfrak{p}) = K(\mathfrak{p})$$

$\rightsquigarrow$ $A \to K(\mathfrak{p})$ induces $\operatorname{Spec} K(\mathfrak{p}) \to \operatorname{Spec} A$ which sends the unique point of $\operatorname{Spec} K(\mathfrak{p})$ to $\mathfrak{p} \in \operatorname{Spec} A$

Prop. Let $f: A \to B$ be a ring map $\mathfrak{p} \in \operatorname{Spec} A$. Then TFAE

(1) $\exists\, \mathfrak{q} \in \operatorname{Spec} B$ s.t. $\mathfrak{p} = \mathfrak{q}^c$

(2) $\mathfrak{p} = \mathfrak{p}^{ec}$ (i.e. $\mathfrak{p}$ is a contracted ideal.)

(3) $\mathfrak{p}B \cap f(A-\mathfrak{p}) = \emptyset$ $(\mathfrak{p}B = \mathfrak{p}^e)$

(4) $B \otimes_A K(\mathfrak{p}) \neq 0$ $\left(B \otimes_A K(\mathfrak{p}) \simeq B \otimes_A A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \simeq B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}\right)$

pf. Let $\mathfrak{b}$ be an ideal of B ($\mathfrak{p}B = \mathfrak{p}^e$; $f(A-\mathfrak{p})^{-1}B / f(A-\mathfrak{p})^{-1}(\mathfrak{p}B)$)

Then (i) $\mathfrak{p} \subseteq \mathfrak{b}^c \Leftrightarrow (f(\mathfrak{p})) = \mathfrak{p}^e \subseteq \mathfrak{b}$

(ii) $\mathfrak{p} \supseteq \mathfrak{b}^e \Leftrightarrow f(A-\mathfrak{p}) \cap \mathfrak{b} = \emptyset$

(iii) $\mathfrak{p} = \mathfrak{b}^c \Leftrightarrow \mathfrak{p}B \subseteq \mathfrak{b}$ and $\mathfrak{b} \cap f(A-\mathfrak{p}) = \emptyset$

Set $\mathfrak{b} = \mathfrak{p}^e = \mathfrak{p}B$ then $\mathfrak{p} = \mathfrak{p}^{ec} \Leftrightarrow \mathfrak{p}B \cap f(A-\mathfrak{p}) = \emptyset$ hence (2) $\Leftrightarrow$ (3)

(4) $\Leftrightarrow$ $f(A-\mathfrak{p})^{-1}(\mathfrak{p}B) \subsetneq f(A-\mathfrak{p})^{-1}(B) \Leftrightarrow f(A-\mathfrak{p}) \cap \mathfrak{p}B = \emptyset \Leftrightarrow$ (3)

(1) $\Rightarrow$ (2) ✓

(3) $\Rightarrow$ (1) by HW $\exists\, \mathfrak{q} \in \operatorname{Spec} B$ s.t. $\mathfrak{p}B \subseteq \mathfrak{q}$ and $\mathfrak{q} \cap f(A-\mathfrak{p}) = \emptyset$

$\overset{\text{(iii)}}{\Rightarrow}$ $\mathfrak{p} = \mathfrak{q}^c$

Rmk. One can show that there is a set bijection

$$(f^*)^{-1}(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q}^c = \mathfrak{p}\} \overset{1:1}{\longleftrightarrow} \operatorname{Spec} B \otimes_A K(\mathfrak{p})$$

§6 local properties

Notation Let A be a ring, AM an A-module.

(1) For $S = \{f^i, i \geq 0\}$ $f \in A$ we denote $S^{-1}M$ by $M_f \simeq M \otimes_A A_f$

(2) For $\mathfrak{p} \in \operatorname{Spec} A$ $S = A - \mathfrak{p}$ we denote $S^{-1}M$ by $M_{\mathfrak{p}} \simeq M \otimes_A A_{\mathfrak{p}}$

Def. A property P of a ring A (or of an A-module M) is said to be a local property if A (or M) has property $P \Leftrightarrow A_\mathfrak{p}$ (or $M_\mathfrak{p}$) has property P for every $\mathfrak{p} \in \operatorname{Spec} A$

Rmk. Think of $a \in A$ as a function on $\operatorname{Spec} A$

Think of $\frac{a}{1} \in A_\mathfrak{p}$ as a germ of a at $\mathfrak{p} \in \operatorname{Spec} A$

$A_\mathfrak{p}$ simpler than A

e.g. $A = k[x]$ k alg closed field $\operatorname{Spec} k[x] = \{0\} \cup \{(x-c) : c \in k\} = \{0\} \cup k$

$\forall c \in k$ $k[x]_{(x-c)}$ is a discrete valuation ring (DVR) i.e. a PID with a unique prime ett $k[x]_{(x-c)} = \{\frac{f(x)}{g(x)} : f, g \in k[x],\ x-c \nmid g(x)\}$

$(x-c)^{\nu} \frac{f(x)}{g(x)}$ $x-c \nmid f, g$ $\frac{f}{g}$ is a unit

$\forall f(x) \in k[x]$ $\frac{f(x)}{1} \in k[x]_{(x-c)}$ is determined by the order of vanishing of $f(x)$ at $x=c$ (up to units in $k[x]_{(x-c)}$)

Prop. Let M be an A-module and let $x \in M$. Then TFAE

(1) $x = 0$ in M

(2) $\frac{x}{1} = 0$ in $M_\mathfrak{p}$ $\forall \mathfrak{p} \in \operatorname{Spec} A$

(3) $\frac{x}{1} = 0$ in $M_\mathfrak{m}$ $\forall \mathfrak{m} \in \operatorname{Spec_m} A$

pf. (1)⇒(2)⇒(3) ✓

(3)⇒(1) Let $\mathfrak{a} = \operatorname{Ann}(x)$ then $x = 0 \Leftrightarrow \mathfrak{a} = (1)$

$\frac{x}{1} = 0$ in $M_\mathfrak{m} \Leftrightarrow \exists s \notin \mathfrak{m}$ s.t. $sx = 0 \Rightarrow \mathfrak{a} \not\subseteq \mathfrak{m}$ $\forall \mathfrak{m} \in \operatorname{Spec_m} A$

$\Rightarrow \mathfrak{a}$ is not proper i.e. $\mathfrak{a} = (1) = A \Rightarrow x = 0$

equiv. $j : M \to \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_\mathfrak{p}$ $x \mapsto (\frac{x}{1})$ is injective

Prop. Let M be an A-module.

(1) $M = 0$

(2) $M_\mathfrak{p} = 0$ $\forall \mathfrak{p} \in \operatorname{Spec} A$

(3) $M_\mathfrak{m} = 0$ $\forall \mathfrak{m} \in \operatorname{Spec_m} A$

pf. (1)⇒(2)⇒(3) ✓ (3)⇒(1) $\forall x \in M$ $\frac{x}{1} = 0$ in $M_\mathfrak{m} \overset{\forall x \in M}{\Rightarrow} x = 0 \Rightarrow M = 0$

MEMO NO: _____
DATE: / /

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Prop Let $f: M \to N$ be an A-module homo. Then TFAE.

(1) f is injective (resp. surjective, bijective)

(2) $f_{\mathfrak{p}}: M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ — — $\forall \mathfrak{p} \in \operatorname{Spec} A$

(3) $f_{\mathfrak{m}}: M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ — — $\forall \mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A$

Pf. $0 \to \ker f \to M \xrightarrow{f} N$ exact

$\Rightarrow 0 \to (\ker f)_{\mathfrak{p}} \to M_{\mathfrak{p}} \xrightarrow{f_{\mathfrak{p}}} N_{\mathfrak{p}}$ exact

$\Rightarrow (\ker f)_{\mathfrak{p}} = \ker(f_{\mathfrak{p}})$ by Prop before ✓

Prop Let M be an A-module. Then TFAE.

(1) M is a flat A-module.

(2) $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module $\forall \mathfrak{p} \in \operatorname{Spec} A$

(3) $M_{\mathfrak{m}}$ is a flat $A_{\mathfrak{m}}$-module $\forall \mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A$

Pf. (1)$\Rightarrow$(2) Fact M flat A module B is an A-algebra $M_B = B \otimes_A M$ is a flat B-module

✓ (2)$\Rightarrow$(3) ✓

(3)$\Rightarrow$(1) Let $N \hookrightarrow N'$ be an injective A-module homo

Goal $M \otimes_A N \to M \otimes_A N'$ injective

$\Leftrightarrow (M \otimes_A N)_{\mathfrak{m}} \to (M \otimes_A N')_{\mathfrak{m}}$ inj $\forall \mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A$

$\Leftrightarrow M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} N_{\mathfrak{m}} \to M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} N'_{\mathfrak{m}}$ inj $\forall \mathfrak{m}$

$N \overset{\text{inj}}{\hookrightarrow} N' \Rightarrow N_{\mathfrak{m}} \hookrightarrow N'_{\mathfrak{m}}$ inj, $M_{\mathfrak{m}}$ flat over $A_{\mathfrak{m}}$

$\Rightarrow M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} N_{\mathfrak{m}} \hookrightarrow M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} N'_{\mathfrak{m}}$ inj.

Prop. Let A be an integral domain with fraction field K

Then $A = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}} = \bigcap_{\mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A} A_{\mathfrak{m}}$

where A as well as $A_{\mathfrak{p}}$ for $\mathfrak{p} \in \operatorname{Spec} A$ are viewed as subrings of K

Pf. $A \subseteq \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}} \subseteq \bigcap_{\mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A} A_{\mathfrak{m}}$ obvious suffices to show $\bigcap_{\mathfrak{m} \in \operatorname{Spec}_{\mathfrak{m}} A} A_{\mathfrak{m}} \subseteq A$

Let $x = \frac{a}{b} \in K$ $a, b \in A$ $b \neq 0$

Define the ideal of denominators of x to be $\mathfrak{a} = \{c \in A : cx \in A\}$

$= \{c \in A : ca \in (b)\} = ((b) : (a))$ Note that $x \in A \Leftrightarrow \mathfrak{a} = (1)$

If $x \in \bigcap_{m \in \operatorname{Spec_m} A} A_m$ Then $\forall\, m \in \operatorname{Spec_m} A_A \ \exists\, a' \in A$ s.t. $x = \frac{a'}{s}$

$\Rightarrow sx \in A \Rightarrow s \in (0) \Rightarrow (0) \nsubseteq m \quad \forall m \in \operatorname{Spec_m} A \Rightarrow (0) = (1)$ ✓

$\Rightarrow x \in A$

Prop Let A be a ring $S \subseteq A$ multiplicative. M an A-module.

(1) If M is a f.g A-module then $S^{-1}M$ is a f.g $S^{-1}A$-module.

(2) f.p. f.p

(3) Noetherian Noetherian

(4) If A is a Noetherian ring then $S^{-1}A$ is a Noetherian ring.

Rmk Think of $a \in A$ as a fcn on $\operatorname{Spec} A$

e.g. $A = k[x_1, \cdots, x_n]$ $\bar{k} = k$ $\operatorname{Spec} A \supseteq \operatorname{Spec_m} A \simeq \mathbb{A}^n_k$

$\forall f \in A$ $f : \mathbb{A}^n_k \to k$, $p \mapsto f(p)$ $p = (a_1, \cdots, a_n) \in \mathbb{A}^n_k \rightsquigarrow m_p \in \operatorname{Spec_m} A$

$A/m_p = k[x_1, \cdots, x_n]/(x_1 - a_1, \cdots, x_n - a_n) \simeq k$

$f(p)$ is the image of f in $k[x_1, \cdots, x_n]/m_p \simeq k$

The image of a at $m \in \operatorname{Spec_m} A$ is $\bar{a} \in A/m$

The image of a at $p \in \operatorname{Spec} A$ is $\frac{\bar{a}}{1} \in \operatorname{Frac}(A/p) = K(p)$

Chapter 8

§1 Cayley-Hamilton Theorem and Nakayama's Lemma

Thm1 Let A be a ring. $\Phi = (a_{ij}) \in M_n(A)$ $P_\Phi(T) = \det(TI_n - \Phi) \in A[T]$ be the characteristic polynomial of Φ. Then $P_\Phi(\Phi) = 0$ in $M_n(A)$

Thm2 Let M be an A-module generated by $m_1, \cdots, m_n$ and let $\varphi : M \to M$ be an A-linear endomorphism of M. If $\varphi(m_i) = \sum_{j=1}^{n} a_{ij} m_j$ $(1 \le i \le n)$ with $a_{ij} \in A$ then $P_\Phi(\varphi) = 0$ in $\operatorname{End}_A(M)$ where $\Phi = (a_{ij}) \in M_n(A)$

Rmk Let A be a ring. M an A-module.

Let $\operatorname{End}_A(M)$ be the (possibly noncommutative) ring of A-linear endomorphisms of M.

MEMO NO: _____
DATE: / /

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Recall that $End_A(M)$ is an A-algebra. More specifically, there is a ring homomorphism $\rho: A \to End_A(M)$ $a \mapsto (\rho_a: x \mapsto ax)$ and $Im\,\rho \subseteq Z(End_A(M))$.

Via ρ, $End_A(M)$ acquires the structure of A an A-module.

Def Given $f(T) = \sum_{i=0}^{m} a_i T^i \in A[T]$, $\varphi \in End_A(M)$ define $f(\varphi) = \sum_{i=0}^{m} a_i \varphi^i$

Thm1 $\Rightarrow$ Thm2:

$$\varphi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} \varphi(m_1) \\ \vdots \\ \varphi(m_n) \end{pmatrix} = \Phi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}, \quad \varphi^2\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \varphi\left(\Phi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}\right) = \Phi\left(\varphi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}\right) = \Phi^2\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}$$

$\Rightarrow \forall f(T) \in A[T]$ $f(\varphi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = f(\Phi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}$. Thm1 $\Rightarrow P_\Phi(\Phi) = 0 \Rightarrow P_\Phi(\varphi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = P_\Phi(\Phi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = 0$

i.e. $\forall (1 \le i \le n)$ $P_\Phi(\varphi)(m_i) = 0 \Rightarrow P_\Phi(\varphi) = 0$

Thm2 $\Rightarrow$ Thm1:

Let $\varphi: A^n \to A^n$ be the unique A-linear map endo s.t. $\varphi(e_i) = \sum_{j=1}^{n} a_{ij} e_j$ $(1 \le i \le n)$

where $e_1, \cdots, e_n$ are the standard basis of A^n.

$\forall f(T) \in A[T]$ $f(\varphi)\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = f(\Phi)\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$ Thm2 $\Rightarrow P_\Phi(\varphi) = 0 \Rightarrow P_\Phi(\Phi)\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = 0$

$\Rightarrow P_\Phi(\Phi) = 0$ since $e_1, \cdots, e_n$ are linearly independent over A.

Pf of Thm1

Note that Thm1 holds if A is an integral domain. Indeed, we may view Φ as a matrix with coefficients in $K = Frac(A)$. Classical Cayley-Hamilton thm $\Rightarrow P_\Phi(\Phi) = 0$

Let X_{ij} $(1 \le i, j \le n)$ be n^2 indeterminates. Let $\Psi = (X_{ij})$ be a matrix with coeff in $\mathbb{Z}[X_{ij}]$, then $P_\Psi(T) = \sum_{\nu=0}^{n} f_\nu(X_{ij}) T^\nu$ is a polynomial of T with coeff in $\mathbb{Z}[X_{ij}]$

$P_\Psi(\Psi) = 0 \rightsquigarrow$ gives n^2 identities $g_{pq}(X_{ij}) = 0$ $(1 \le p, q \le n)$ in $\mathbb{Z}[X_{ij}]$

Let $\sigma: \mathbb{Z}[X_{ij}] \to A$ be the unique ring homo s.t. $\sigma(X_{ij}) = a_{ij}$

Then the image of $P_\Psi(T) \in \mathbb{Z}[X_{ij}][T]$ in $A[T]$ under σ is precisely $P_\Phi(T)$ and the images of $g_{pq}(X_{ij}) = 0$ in A under σ says precisely $P_\Phi(\Phi) = 0$.

Pf of Thm2

Lemma 1 Let M be an A-module. $\varphi \in End_A(M)$ Then M is an $A[T]$-module via $f(T)\,m = f(\varphi)\,m$ $\forall f(T) \in A[T]$ $m \in M$

Lemma 2 Let M be an A-module. Suppose $m_1, \cdots, m_n \in M$ and they satisfy n

A-linear equations $\sum_{i=1}^{n} a_{ij} m_j = 0$ $(1 \le i \le n)$ Then $\Delta m_j = 0$ $\forall j = 1, 2, \cdots, n$

where $\Delta = \det(a_{ij}) \in A$

pf of lemma 2 Let $\Phi = (a_{ij})$ $\Phi \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = 0$ $\Phi^* \Phi \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \Phi^* \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

$\Phi^* \Phi = \begin{pmatrix} \Delta & & \\ & \ddots & \\ & & \Delta \end{pmatrix} \Rightarrow \Delta m_1, \cdots, \Delta m_n = 0$

We regard M as an $A[T]$-module by letting T act as φ (Lemma 1)

Hypothesis implies $Tm_i = \sum_{j=1}^{n} a_{ij} m_j \Rightarrow \sum_{j=1}^{n} (T\delta_{ij} - a_{ij}) m_j = 0$ $1 \le i \le n$

Lemma 2 $\Rightarrow \det(T\delta_{ij} - a_{ij}) m_j = P_\Phi(T) m_j = 0$ $j = 1, \cdots, n$

$\Rightarrow P_\Phi(\varphi)(m_j) = 0$ $\forall j$ $\Rightarrow P_\Phi(\varphi) = 0$

Corollary 1 Let A be a ring with ideal $\mathfrak{a}$. Let M be an A-module that can be generated by n elts. Let $\varphi \in \mathrm{End}_A(M)$ be s.t. $\varphi(M) \subseteq \mathfrak{a}M$

Then $\exists$ a monic polynomial $f(X) = X^n + a_1 X^{n-1} + \cdots + a_{n-1} X + a_n \in A[x]$ with $a_i \in \mathfrak{a}^i$ for each i s.t. $f(\varphi) = 0$ in $\mathrm{End}_A(M)$

Pf: Suppose $M = Am_1 + \cdots + Am_n \Rightarrow \mathfrak{a}M = \mathfrak{a}m_1 + \cdots + \mathfrak{a}m_n$

$\Rightarrow \varphi(m_i) = \sum_{j=1}^{n} a_{ij} m_j$ $(1 \le i \le n)$ with $a_{ij} \in \mathfrak{a}$ Let $f(x)$ be the characteristic polynomial of $\Phi = (a_{ij})$ $f(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ Thm 2 $\Rightarrow f(\varphi) = 0$

Note that Φ has coeff in $\mathfrak{a}$ and $a_i = (-1)^i \sum$ principal i by i minors of Φ

Hence $a_i \in \mathfrak{a}^i$

Corollary 2 Let M be a f.g A-module and let $\mathfrak{a}$ be an ideal of A

If $M = \mathfrak{a}M$, then $\exists a \in \mathfrak{a}$ s.t. $(1+a)M = 0$

If we have further $\mathfrak{a} \subseteq \mathrm{rad}(A)$ then $M = 0$

pf Take $\varphi = \mathrm{id}_M$ in corollary 1 $\Rightarrow \exists a_i \in \mathfrak{a}^i$ $(1 \le i \le n)$ s.t. $\mathrm{id} + a_1 * \mathrm{id} + \cdots + a_n \mathrm{id} = 0$ in $\mathrm{End}_A(M)$ i.e. $(1 + a_1 + \cdots + a_n)\mathrm{id} = 0$ let $a = a_1 + \cdots + a_n \in \mathfrak{a}$ then $(1+a)M = 0$

$a \in \mathfrak{a} \subseteq \mathrm{rad}(A) \Rightarrow 1 + a$ is a unit in $A \Rightarrow M = 0$

Corollary 3 Let M be a f.g A-module and let $\varphi \in \mathrm{End}_A(M)$

(1) If φ is surjective, then φ is an isomorphism.

(2) Let M be a free A-module of rank n. Then every set of n elts in M

that generates M is a free basis of M.

pf. (1) View M as an $A[x]$-module where x acts by φ

φ surjective $\Rightarrow \varphi(M)=M \Rightarrow M=xM=\mathcal{O}(M)$ where $\mathcal{O}=(x)\subseteq A[x]$

M f.g over $A \Rightarrow M$ f.g over $A[x]$ Corollary 2 $\Rightarrow \exists f(x)\in A[x]$ s.t. $(1+xf(x))M=0$

$\Leftrightarrow (id+\varphi f(\varphi))M=0 \Rightarrow id+\varphi f(\varphi)=0$ in $\mathrm{End}_A(M)$

$\varphi(-f(\varphi))=(-f(\varphi))\varphi=id_M \Rightarrow \varphi$ is an isom.

(2) Let $u_1,\cdots,u_n$ be an A-basis of M. $v_1,\cdots,v_n$ be a generating set of M

Then the unique A-linear map $\varphi: M\to M$ $u_i\mapsto v_i$ is surjective

By (1), φ is an isom, ~~hence~~ hence $v_1,\cdots,v_n$ is a basis of M.

Nakayama's Lemma ~~Let~~

Let $(A,\mathfrak{m})$ be a local ring with residue field $k=A/\mathfrak{m}$. Let M be a f.g A-module ($M_k=M\otimes_A k=M\otimes_A A/\mathfrak{m}\simeq M/\mathfrak{m}M$ is a k vector space)

(1) If $M=\mathfrak{m}M$ then $M=0$. Equivalently $M_k=0\Rightarrow M=0$

(2) If N is a submodule of M s.t. $M=\mathfrak{m}M+N$ then $M=N$

pf (1) use corollary 2

(2) $M=\mathfrak{m}M+N \Leftrightarrow M/N=\mathfrak{m}(M/N) \overset{(1)}{\Rightarrow} M/N=0 \Leftrightarrow M=N$

(3) Let $x_1,\cdots,x_n\in M$ Then $\{x_1,\cdots,x_n\}$ generates M over $A \Leftrightarrow \{\bar{x}_1,\cdots,\bar{x}_n\}$ generates M_k over k

In particular, $\{x_1,\cdots,x_n\}$ is a minimal generating set for M

$\Leftrightarrow \{\bar{x}_1,\cdots,\bar{x}_n\}$ is a k-basis for M_k

Hence the minimal generating sets of M have the same cardinality i.e. $\dim_k M_k$

pf. (3) $\Rightarrow$ triv $\Leftarrow$: $\{\bar{x}_1,\cdots,\bar{x}_n\}$ generates $M_k \Leftrightarrow M/\mathfrak{m}M=A/\mathfrak{m}\bar{x}_1+\cdots+A/\mathfrak{m}\bar{x}_n$

$=(Ax_1+\cdots+Ax_n+\mathfrak{m}M)/\mathfrak{m}M$

$\Leftrightarrow M=\underbrace{Ax_1+\cdots+Ax_n}_{N}+\mathfrak{m}M$

$\overset{(2)}{\Leftrightarrow} M=N=Ax_1+\cdots+Ax_n$

Rmk (1) Could fail if M is not f.g

Let $(A,\mathfrak{m})$ be a local domain that is not a field ($\Rightarrow \mathfrak{m}\neq 0$)

Let $K=\mathrm{Frac}(A)$ hw $\Rightarrow K$ is not a f.g A-module.

Choose any $x \in \mathfrak{m}$, $x \neq 0$. Then x is invertible in $k \Rightarrow k = xk = \mathfrak{m}k$

However $k \neq 0$

Rmk. Under the hypothesis of Nakayama's lemma, let $\{x_1, \cdots, x_n\}$ be a minimal generating set for M. Let $\pi: A^n \twoheadrightarrow M : e_i \mapsto x_i$ be the surjective morphism

Then $\pi_k: k^n \to M_k$; $e_i \mapsto \bar{x}_i$ is an isom of k-vector space.

Cor. A f.g. flat module over a local ring is free.

pf. Let $(A, \mathfrak{m})$ be a local ring. M f.g flat A-module

Choose a minimal generating set $x_1, \cdots, x_n$ for M. Then $\bar{x}_1, \cdots, \bar{x}_n$ is a k-basis for $M_k \Rightarrow \bar{x}_1, \cdots, \bar{x}_n$ linearly indep over k

$\overset{\text{hw}}{\Rightarrow}$ flat $x_1, \cdots, x_n$ linearly indep over $A \Rightarrow M$ is a free A-module with basis $x_1, \cdots, x_n$

Cor. A f.g projective module over a local ring is free

pf 1. use previous corollary

pf 2. Let $(A, \mathfrak{m})$ be a local ring, M f.g projective A-module.

Choose a minimal generating set $\{x_1, \cdots, x_n\}$ for M

Then $\pi: A^n \to M$; $e_i \mapsto x_i$ is surjective and $\pi_k: k^n \to M_k$ is an isom

Let $K = \ker \pi$; then $0 \to K \to A^n \to M \to 0$ is exact and splits since M is proj.

$\Rightarrow 0 \to K_k \to k^n \overset{\cong}{\to} M_k \to 0$ ($\to$ split) $\Rightarrow K_k = 0$

$A^n \cong K \oplus M \Rightarrow K$ f.g $\overset{K_k = 0 \text{ Nakayama}}{\Rightarrow} K = 0 \Rightarrow M$ is free

Rmk. "f.g" can be removed i.e. A proj module over a local ring is free

§2. Integral extensions, Integral closure, Normal domains

Ref. If A is a subring of B, then we say $A \subseteq B$ is a ring extension

Let $A \subseteq B$ be rings, and $b \in B$. The elt b is said to be integral over A if $\exists$ a monic poly $f(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n \in A[x]$ s.t. $f(b) = b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$ in B

If every elt in B is integral over A, then we say B is integral over A, or that B is an integral extension of A

e.g. Every elt a in A is integral over A

L/K field extension $\alpha \in L$ algebraic over $K \Rightarrow K(\alpha) = K[\alpha]$

$\pi: K[x] \to K[\alpha]$; $x \mapsto \alpha$ $\ker \pi$ is an ideal of $K[x]$ (PID), is generated by a unique monic poly in $K[x]$. This poly is the minimal poly of α over K

$A \subseteq B$ ring extension $b \in B$ integral over A

$\pi: A[x] \to A[b]$; $x \mapsto b$ $\ker \pi$ contains a monic poly but it is not in general principal

Prop Let $A \subseteq B$ be rings and $b \in B$. Then TFAE.

(1) b is integral over A

(2) $A[b]$ is a finite A-algebra. i.e. $A[b]$ is a f.g A-module.

(3) $A[b]$ is contained in a subring C of B s.t. C is f.g as an A-module

pf (1)$\Rightarrow$(2) If $b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$ with $a_i \in A$

then $A[b]$ as an A-module, is generated by $1, b, b^2, \cdots, b^{n-1}$

(2)$\Rightarrow$(3) Set $C = A[b]$

(3)$\Rightarrow$(1) Suppose $A \subseteq A[b] \subseteq C \subseteq B$ where C is a f.g A-module

Then $\varphi: C \to C$; $c \mapsto bc$ is an A-linear endo of C

C f.g over $A \overset{C-H}{\Rightarrow} \exists$ a monic poly $f(x) \in A[x]$ s.t. $f(\varphi) = 0$ in $\mathrm{End}_A(C)$

It is easy to see that $f(\varphi)$ is given by $c \mapsto f(b)c$

$f(\varphi) = 0 \Rightarrow f(b)c = 0 \ \forall c \in C \overset{c=1}{\Rightarrow} f(b) = 0$

Prop2 Let $A \subseteq B$ be rings and let $b_1, \cdots, b_n$ be elts of B, each integral over A.

Then $A[b_1, \cdots, b_n]$ is a finite A-algebra

pf $\forall i = 1, \cdots, n$ $\exists f_i(x) \in A[x]$ monic s.t. $f_i(b_i) = 0$ Suppose $\deg f_i = \nu_i$

Then $A[b_1, \cdots, b_n]$, as an A-module, is generated by the finite set $\{b_1^{m_1} \cdots b_n^{m_n} : 0 \le m_i < \nu_i\}_{i=1,2,\cdots,n}$

Prop3 Let $A \subseteq B$ be rings. Then the set $C = \{b \in B : b \text{ integral over } A\}$ is a subring of B containing A.

pf. $A \subseteq C$ triv. It suffices to show that if $b_1, b_2 \in C$ then $b_1 \pm b_2$, $b_1 b_2 \in C$

Prop2 $\Rightarrow A[b_1, b_2]$ is a finite A-algebra.

$A[b_1-b_2] \subseteq A[b_1+b_2] \subseteq A[b_1,b_2]$ $\overset{(2)\Rightarrow(1)\ \text{Prop 1}}{\Longrightarrow}$ $b_1 \pm b_2$, $b_1 b_2$ are integral over A

Def. We call C the integral closure of A in B

If $C=A$, then we say A is integrally closed in B

Prop 4 Let $A \subseteq B \subseteq C$ be rings. ~~If $A \subseteq B$~~ s.t. B is integral over A and C is integral over B. Then C is integral over A.

pf. Let $c \in C$ ~~Let~~ By assumption, there is an eqn. $c^n + b_1 c^{n-1} + \cdots + b_n = 0$ with $b_i \in B$

Prop 2 $\overset{B'=}{\Rightarrow} A[b_1, \cdots, b_n]$ is a finite A-algebra

$A \subseteq B' \subseteq B \subseteq C$ (c integral over B') $\overset{\text{Prop 1}}{\Rightarrow} B'[c]$ is a finite B'-alg

$\Rightarrow B'[c]$ is a finite A-algebra $\overset{\text{Prop 1}}{\Rightarrow} c$ is integral over A.

Cor. Let $A \subseteq B$ be rings and let C be the integral closure of A in B

Then C is integrally closed ~~or~~ in B

pf. use Prop 4.

Def Let $f: A \to B$ be a ring map

We say f is integral or that B is an integral A-algebra

if B is integral over $f(A)$

Prop 5 Let $f: A \to B$ be an integral ring map

(1) If $\mathfrak{a}$ is an ideal of A $\mathfrak{b}$ is an ideal of B and $f(\mathfrak{a}) \subseteq \mathfrak{b}$

then $\bar{f}: A/\mathfrak{a} \to B/\mathfrak{b}$ is integral. In particular, if $\mathfrak{a} = \mathfrak{b}^c$

then $A/\mathfrak{a} \subseteq B/\mathfrak{b}$ is an integral extension

(2) If S is a multiplicative subset of A, then $S^{-1}f: S^{-1}A \to S^{-1}B \cong f(S)^{-1}B$

is integral. In particular, if $A \subseteq B$ is an integral extension, then $S^{-1}A \subseteq S^{-1}B$

is an integral extension

pf. (1) $\forall b \in B$, $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ in B $\Rightarrow \bar{b}^n + \bar{a}_1 \bar{b}^{n-1} + \cdots + \bar{a}_n = 0$ in $B/\mathfrak{b}$ with $\bar{a}_i \in A/\mathfrak{a}$

(2) $\forall b \in B$, $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ in B $\overset{\forall s \in S}{\Rightarrow} \left(\frac{b}{s}\right)^n + \frac{a_1}{s}\left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = 0$ in $S^{-1}B$

with $\frac{a_i}{s^i} \in S^{-1}A$

MEMO NO:
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Prop 6 (finite type + integral = finite) Let $f: A \to B$ be a ring map.

Then f is finite $\Leftrightarrow$ f is finite type and integral

Rmk A field extension L/K is finite $\Leftrightarrow$ L is a f.g. and algebraic extension of k.

pf WLOG assume f is injective (f is finite / of finite type / integral $\Leftrightarrow$ $f(A) \hookrightarrow B$ is finite / of finite type / integral)

($\Rightarrow$) finite $\Rightarrow$ finite type (known) Prop 1 $\Rightarrow$ integral

($\Leftarrow$) $B = A[b_1, \cdots, b_n]$ each b_i integral over $A \overset{\text{Prop 2}}{\Rightarrow}$ B finite A-algebra.

Def Let A be an integral domain and $K = \mathrm{Frac}(A)$

Then the integral closure of A in K is simply called the integral closure of A or the normalization of A. If A is integrally closed in K, then we say A is an integrally closed domain or normal domain

Prop UFDs are integrally closed domains.

pf Let A be a UFD $k = \mathrm{Frac}(A)$ Let $\frac{a}{b} \in k$ ($a, b \in A$, $a, b \neq 0$)

If $\frac{a}{b}$ is integral over A then there is an eqn $(\frac{a}{b})^n + c_1(\frac{a}{b})^{n-1} + \cdots + c_n = 0$ with $c_i \in A$

Multiplying by $b^n \Rightarrow a^n + c_1 a^{n-1} b + \cdots + c_n b^n = 0$ in A

Since A is a UFD, we may assume $\gcd(a,b) = 1$ $b \mid a^n \Rightarrow b$ is a unit $\Rightarrow \frac{a}{b} \in A$

e.g. $\mathbb{Z}$ $\mathbb{Z}[i]$ $\mathbb{Z}[x_1, \cdots, x_n]$, $k[x_1, \cdots, x_n]$ are normal domains.

$k[x,y]/(y^2 - x^3)$, $k[x,y]/(y^2 - x^3 - x^2)$ are not integrally closed

$\bar{k} = k$ $X \subseteq \mathbb{A}^n_k$ algebraic variety ($\Leftrightarrow$ $A(X) = k[x_1, \cdots, x_n]/I(X)$ is an integral domain)

$A(X)$ integrally closed $\Rightarrow$ singular locus of X has codim ≥ 2

Prop Let A be an integral domain. Then the integral closure of A is an integrally closed domain pf. $A \subseteq \bar{A} \subseteq K$

$A \subseteq \bar{A} \rightsquigarrow \mathrm{Spec}\bar{A} \to \mathrm{Spec}A$: normalization

e.g. Let k be a field, $A = k[x,y]/(y^2 - x^3)$ $y^2 - x^3$ irred poly $\Rightarrow A$ is an integral domain

Let $K = \mathrm{Frac}(A)$ $u = \bar{x} \neq 0$ $v = \bar{y} \neq 0$ $\in A$ $A = k[u,v]$ $v^2 = u^3$

Let $t=\frac{v}{u}\in K$ $\quad t^2=(\frac{v}{u})^2=u$ in K $\quad v=tu=t^3$ in K

$A=k[t^2,t^3]\subseteq K \quad K=k(t)$

Claim. t is transcendental over k. $\Rightarrow \dim_k A<\infty$

Pf. If not $\dim_k K=\dim_k k(t)<\infty$. On the other hand, $A=k[x][y]/(y^2-x^3)$

$\simeq k[x]\oplus k[x]\bar{y}$ as $k[x]$-modules $\Rightarrow \dim_k A=\infty$ absurd!

$t^2\in A\Rightarrow t$ is integral over A (integral extension)

not integrally closed: $A=k[t^2,t^3]\subsetneq k[t]\subseteq k(t)=K \qquad A\subseteq k[t]\subseteq \overline{A}$

$\overline{A}$ integral over $A\Rightarrow \overline{A}$ integral over $k[t]$

$k[t]$ integrally closed in $k(t)$ $\Rightarrow \overline{A}=k[t]$

$\mathrm{Spec}\, k[t]\to \mathrm{Spec}\, k[t^2,t^3] \quad t\mapsto(t^2,t^3)$ 

e.g. $K/\mathbb{Q}$ f.d. finite field extension $\quad \mathbb{Q}\subseteq K$ (∪ $\mathbb{Z}$)

$\mathcal{O}_K$ = integral closure of $\mathbb{Z}$ in K

"ring of integers"

Can show that $K=\mathrm{Frac}(\mathcal{O}_K)$ $\mathcal{O}_K$ integrally closed in $K\Rightarrow \mathcal{O}_K$ is a normal domain

$K=\mathbb{Q}(\sqrt{d})$ d square free integer $\pm p_1\cdots p_r$

$$\mathcal{O}_K=\begin{cases}\mathbb{Z}[\sqrt{d}] & d\not\equiv 1 \pmod 4\\ \mathbb{Z}[\frac{\sqrt{d}+1}{2}] & d\equiv 1\pmod 4\end{cases}$$

$d=-3$ $\quad \mathbb{Z}[\sqrt{-3}]\subsetneq \mathbb{Z}[\frac{\sqrt{-3}+1}{2}]$ (normalization)

$\mathbb{Z}[\sqrt{-3}]$: not UFD $\qquad \mathbb{Z}[\frac{\sqrt{-3}+1}{2}]$: ED

Prop. Being integrally closed is a local property for integral domains.

§3. Cohen-Seidenberg Theorems (aka going up and going down theorems)

Let $f: A\to B$ be an integral ring map.

Study properties of the continuous $f^*: \mathrm{Spec}\, B\to \mathrm{Spec}\, A$, $\mathfrak{q}\mapsto \mathfrak{q}^c=\mathfrak{q}\cap A$

Lemma Let $A\subseteq B$ be an integral extension of integral domains.

Then A is a field $\Leftrightarrow$ B is a field.

pf. $\Rightarrow$ Let $b\in B$ $b\neq 0$ and let $b^n+a_1b^{n-1}+\cdots+a_n=0$ $(a_i\in A)$ be an eqn of integral dependence for b of smallest possible degree

Then $a_n \neq 0$ (otherwise, since since B is an integral domain and $b \neq 0$ it follows that $b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1} = 0$) $\Rightarrow b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) a_n^{-1} = -1$

$\Rightarrow b$ is invertible in B

$\Leftarrow$ Let $a \in A$ $a \neq 0$ then $\frac{1}{a} \in B \Rightarrow \frac{1}{a}$ is integral over A

$\Rightarrow (\frac{1}{a})^n + a_1(\frac{1}{a})^{n-1} + \cdots + a_n = 0 \Rightarrow 1 + a_1 a + \cdots + a_n a^n = 0$

i.e. $1 + a(a_1 + \cdots + a_n a^{n-1}) = 0$ a is invertible in A

Prop1. (Maximality) Let $f: A \to B$ be an integral ring map

$q \in \operatorname{Spec} B$ $p = q^c$ Then p is maximal $\Leftrightarrow q$ is maximal

Pf. $\Rightarrow$ $A/p \subseteq B/q$ integral extension use Lemma ✓

Prop2. (Incomparability) Let $f: A \to B$ be an integral ring map

Let $q_1, q_2 \in \operatorname{Spec} B$ s.t. $q_1^c = q_2^c = p$ and $q_1 \subseteq q_2$ then $q_1 = q_2$

Pf. $f: A \to B$ integral $\Rightarrow f_p: A_p \to B_p$ integral

$i=1,2$ $q_i^c = p \Rightarrow q_i \cap f(A-p) = \phi$ (IIS $f(A-p)^{-1}B$)

$\Rightarrow q_1 B_p, q_2 B_p \in \operatorname{Spec} B_p$

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_p & \xrightarrow[f_p]{} & B_p \end{array} \qquad \begin{array}{ccc} \operatorname{Spec} A & \longleftarrow & \operatorname{Spec} B \\ \uparrow & & \uparrow \\ \operatorname{Spec} A_p & \longleftarrow & \operatorname{Spec} B_p \end{array} \qquad \begin{array}{ccc} p & \longleftarrow & q_i \\ \uparrow & & \uparrow \\ q_i B_p \cap A_p & \longleftarrow & q_i B_p \end{array}$$

implies $q_i B_p \cap A_p = pA_p$

Since f_p is integral

pA_p unique maximal ideal in A_p Maximality $\Rightarrow q_i B_p$ is maximal in B_p

$q_1 \subseteq q_2 \Rightarrow q_1 B_p \subseteq q_2 B_p \Rightarrow q_1 B_p = q_2 B_p \Rightarrow q_1 = q_2$

Def Let $f: A \to B$ be a ring map. If $q \in \operatorname{Spec} B$ and $q^c = p$ then we say q lies over p

$$\begin{array}{ccc} B & & \operatorname{Spec} B & & q \\ \uparrow & & \downarrow & & \downarrow \\ A & & \operatorname{Spec} A & & p \end{array}$$

Prop3 (Lying over) Let $A \subseteq B$ be an integral extension

Then $\forall p \in \operatorname{Spec} A$ $\exists q \in \operatorname{Spec} B$ s.t. $q^c = p$

That is $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.

pf. $A \subseteq B$ integral extension $\Rightarrow A_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ integral extension

$A_{\mathfrak{p}} \neq 0 \Rightarrow B_{\mathfrak{p}} \neq 0$ Let $\mathfrak{n}$ be a maximal ideal of $B_{\mathfrak{p}}$

Then Maximality $\Rightarrow \mathfrak{n} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$

$$\begin{array}{ccccc} A & \hookrightarrow & B & \mathfrak{p} & \leftarrow & \mathfrak{n} \cap B \\ \downarrow & & \downarrow & \uparrow & & \uparrow \\ A_{\mathfrak{p}} & \hookrightarrow & B_{\mathfrak{p}} & \mathfrak{p}A_{\mathfrak{p}} & \leftarrow & \mathfrak{n} \end{array}$$

Let $\mathfrak{q} = \mathfrak{n} \cap B$ Then $\mathfrak{q} \in \operatorname{Spec} B$ and $\mathfrak{q}$ lies over $\mathfrak{p}$ by the left diagram.

Rmk. Let $f: A \to B$ be an integral ring map

$$A \twoheadrightarrow A/\ker f \hookrightarrow B$$

($A/\ker f \hookrightarrow B$: integral extension)

$$\operatorname{Spec} A \hookleftarrow \operatorname{Spec} A/\ker f \twoheadleftarrow \operatorname{Spec} B$$

($\operatorname{Spec} A/\ker f \twoheadleftarrow \operatorname{Spec} B$: Lying-over)

element in Spec A which contains $\ker f$ $\leftarrow V(\ker f) \leftarrow$ homeomorphism

Conclusion: $f^*(\operatorname{Spec} B) = V(\ker f)$

Def. Let $f: A \to B$ be a ring map ($\operatorname{Spec} A$)

(1) We say f satisfies going up if given $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{q}_1 \in \operatorname{Spec} B$ lying over $\mathfrak{p}_1$ there exists $\mathfrak{q}_2 \in \operatorname{Spec} B$ s.t. $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and $\mathfrak{q}_2$ lies over $\mathfrak{p}_2$

$$\begin{array}{ccc} \mathfrak{q}_1 & \subseteq & \exists \mathfrak{q}_2 \\ | & & | \\ \mathfrak{p}_1 & \subseteq & \mathfrak{p}_2 \end{array}$$

(2) going down

$$\begin{array}{ccc} \exists \mathfrak{q}_1 & \subseteq & \mathfrak{q}_2 \\ | & & | \\ \mathfrak{p}_1 & \subseteq & \mathfrak{p}_2 \end{array} \qquad \mathfrak{q}_2 \mid \mathfrak{p}_2 \quad \exists\, \mathfrak{q}_1 \subseteq \mathfrak{q}_2 \text{ s.t. } \mathfrak{q}_1 \mid \mathfrak{p}_1$$

Thm 1. An integral ring map satisfies going up

Thm 2. Let $A \subseteq B$ be integral domains, A integral closed, B integral over A

Then the inclusion map $i: A \hookrightarrow B$ satisfies going down

($f: A \to B$ satisfies going up $\Leftrightarrow f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed mapping)

MEMO NO: _____
DATE: / 6 / 3

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Prop Let $f: A \to B$ be a ring map. Then TFAE

(1) f satisfies going up

(2) For every $\mathfrak{q} \in \operatorname{Spec} B$ and $\mathfrak{p} = \mathfrak{q} \cap A$, the map $\operatorname{Spec} B/\mathfrak{q} \to \operatorname{Spec} A/\mathfrak{p}$ is surjective

$$\begin{array}{ccc} \operatorname{Spec} B/\mathfrak{q} & \longrightarrow & \operatorname{Spec} A/\mathfrak{p} \\ \downarrow & & \downarrow \\ \operatorname{Spec} B & \longrightarrow & \operatorname{Spec} A \end{array}$$

$\{\mathfrak{q}' : \mathfrak{q} \subseteq \mathfrak{q}'\} = V(\mathfrak{q}) \longrightarrow V(\mathfrak{p}) = \{\mathfrak{p}' : \mathfrak{p} \subseteq \mathfrak{p}'\}$

(3) For every ideal $\mathfrak{b}$ of B and every prime ideal $\mathfrak{p}$ of A s.t. $\mathfrak{b} \cap A \subseteq \mathfrak{p}$ there exists $\mathfrak{q} \in \operatorname{Spec} B$ s.t. $\mathfrak{b} \subseteq \mathfrak{q}$ and $\mathfrak{q}^c = \mathfrak{p}$ ($\mathfrak{q}^c = \mathfrak{q} \cap A$)

(4) For every ideal $\mathfrak{b}$ of B and $\mathfrak{a} = \mathfrak{b} \cap A$ the map $\operatorname{Spec} B/\mathfrak{b} \to \operatorname{Spec} A/\mathfrak{a}$ is surjective

(5) $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed mapping

Pf. (1)$\Leftrightarrow$(2) (3)$\Leftrightarrow$(4)

(1) $\Rightarrow$ (3) **Lemma** Let $f: A \to B$ be a ring map. $\mathfrak{b}$ an ideal of B. $\mathfrak{a} = \mathfrak{b} \cap A$. Then for every minimal prime ideal $\mathfrak{p}$ over $\mathfrak{a}$, there exists a prime ideal $\mathfrak{q}$ of B s.t. $\mathfrak{b} \subseteq \mathfrak{q}$ and $\mathfrak{q}^c = \mathfrak{p}$

There exists a minimal prime ideal $\mathfrak{p}_0$ over $\mathfrak{b} \cap A$ s.t. $\mathfrak{b} \cap A \subseteq \mathfrak{p}_0 \subseteq \mathfrak{p}$

Lemma $\Rightarrow \exists\, \mathfrak{q}_0 \in \operatorname{Spec} B$ s.t. $\mathfrak{b} \subseteq \mathfrak{q}_0$ and $\mathfrak{q}_0^c = \mathfrak{p}_0$. Apply going up to $\mathfrak{q}_0$ over $\mathfrak{p}_0 \subseteq \mathfrak{p}$

$\exists\, \mathfrak{q} \in \operatorname{Spec} B$ s.t. $\mathfrak{q}_0 \subseteq \mathfrak{q}$ and $\mathfrak{q}^c = \mathfrak{p}$. $\mathfrak{b} \subseteq \mathfrak{q}_0$ and $\mathfrak{q}_0 \subseteq \mathfrak{q} \Rightarrow \mathfrak{b} \subseteq \mathfrak{q}$

(3) $\Rightarrow$ (5) Let $V(\mathfrak{b})$ be a closed subset of $\operatorname{Spec} B$. $f^*(V(\mathfrak{b})) \subseteq V(\mathfrak{a})$ ($\mathfrak{a} = \mathfrak{b}^c$)

(3) implies $f^*(V(\mathfrak{b})) = V(\mathfrak{a})$ is closed in $\operatorname{Spec} A \Rightarrow f^*$ is closed

(5) $\Rightarrow$ (1) **Lemma** Let A be a ring, $\mathfrak{p} \in \operatorname{Spec} A$. Then the Zariski closure of $\mathfrak{p}$ in $\operatorname{Spec} A$ is $V(\mathfrak{p})$

Let $\mathfrak{q} \in \operatorname{Spec} B$, $\mathfrak{p} = \mathfrak{q}^c$, then $f^*(V(\mathfrak{q})) \subseteq V(\mathfrak{p})$, $\mathfrak{p} \in V(\mathfrak{p})$, $f^*(V(\mathfrak{q}))$

(5) $\Rightarrow f^*(V(\mathfrak{q}))$ is closed in $\operatorname{Spec} A$. Lemma $\Rightarrow f^*(V(\mathfrak{q})) = V(\mathfrak{p})$

Pf of Thm1. Let $f: A \to B$ be an integral ring map, $\mathfrak{b}$ an ideal of B, $\mathfrak{a} = \mathfrak{b} \cap A$
$\Rightarrow A/\mathfrak{a} \subseteq B/\mathfrak{b}$ is an integral extension $\overset{\text{Lying over}}{\Longrightarrow}$ $\operatorname{Spec} B/\mathfrak{b} \to \operatorname{Spec} A/\mathfrak{a}$ is surjective
Prop $\Rightarrow$ f satisfies going up

Prop. Let $f: A \to B$ be a ring map. Then TFAE.
(1) f satisfies going down
(2) For every $\mathfrak{q} \in \operatorname{Spec} B$ and $\mathfrak{p} = \mathfrak{q}^c$, the map $\operatorname{Spec} B_{\mathfrak{q}} \to \operatorname{Spec} A_{\mathfrak{p}}$ is surj.

$$\begin{array}{ccc} \operatorname{Spec} B_{\mathfrak{q}} & \longrightarrow & \operatorname{Spec} A_{\mathfrak{p}} \\ \text{IS} & & \text{IS} \\ \{\mathfrak{q}' : \mathfrak{q}' \subseteq \mathfrak{q}\} & \to & \{\mathfrak{p}' : \mathfrak{p}' \subseteq \mathfrak{p}\} \end{array}$$

(3) For every $\mathfrak{p} \in \operatorname{Spec} A$ and minimal prime ideal $\mathfrak{q}_0$ over $\mathfrak{p}B$ (implies $\mathfrak{p}B \subsetneq B$) we have $\mathfrak{q}_0^c = \mathfrak{p}$

eg. Let A be a ring, S multiplicative subset of A
Then the localization map $\varphi: A \to S^{-1}A$ satisfies going down
φ^* induces an inclusion-preserving bijection from $\operatorname{Spec} S^{-1}A$ to the subset $\Sigma' = \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \cap S = \emptyset\}$ of $\operatorname{Spec} A$ and if $\mathfrak{p} \in \Sigma'$ and $\mathfrak{p}' \subseteq \mathfrak{p}$ then $\mathfrak{p}' \in \Sigma'$

Prop. A flat ring map satisfies going down

~~Pf of thm2~~

Def. Let $A \subseteq B$ be rings, $\mathfrak{a}$ an ideal of A. An elt $b \in B$ is said to be integral over $\mathfrak{a}$ if there is an eqn $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ with $a_i \in \mathfrak{a}$
The integral closure of $\mathfrak{a}$ in B is the set of all elts in B which are integral over $\mathfrak{a}$.

Rmk. $a \in A$ a integral over $\mathfrak{a} \iff a \in \sqrt{\mathfrak{a}}$ In particular, 1 is integral over $\mathfrak{a}$ $\iff \mathfrak{a} = (1)$

Lemma. Let $A \subseteq B$ be rings, $\mathfrak{a}$ an ideal of A. Let C be the integral closure of A in B. Then the integral closure of $\mathfrak{a}$ in B is $\sqrt{\mathfrak{a}C}$
and therefore is closed under addition and multiplication
pf. Suppose $b \in B$ is integral over $\mathfrak{a} \Rightarrow \exists$ an eqn $b^n + \underbrace{a_1 b^{n-1} + \cdots + a_n}_{\in \mathfrak{a}C} = 0$ with $a_i \in \mathfrak{a}$
$\Rightarrow b$ is integral over $A \Rightarrow b \in C \Rightarrow b^n \in \mathfrak{a}C \Rightarrow b \in \sqrt{\mathfrak{a}C}$

MEMO NO: _____
DATE: / /

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Conversely, if $b \in \sqrt{\mathfrak{a}C}$ then $b^k = \sum_{i=1}^{m} a_i c_i$ for some $k \geq 1$ $a_i \in \mathfrak{a}$, $c_i \in C$

Since each c_i is integral in A $C' := A[c_1, \cdots, c_m]$ is a f.g. A-module and $b^k C' \subseteq \mathfrak{a} C'$ Cayley–Hamilton Thm $\Rightarrow \exists f(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ with $a_i \in \mathfrak{a}^i \subseteq \mathfrak{a}$ s.t. $f(\varphi) = 0$ where $\varphi : C' \to C'$ is multiplication by b^k in $\mathrm{End}_A(C')$

$f(\varphi)(1) = 0 \Rightarrow f(b^k) = 0 \Rightarrow b^k$ is integral over $\mathfrak{a} \Rightarrow b$ is integral over $\mathfrak{a}$

Lemma. Let A be an integrally closed domain and $K = \mathrm{Frac}(A)$

Let L/K be a field extension. Let $\alpha \in L$ be integral over an ideal $\mathfrak{a}$ of A then α is algebraic over K and the minimal polynomial over K has the form $x^n + a_1 x^{n-1} + \cdots + a_n$ with $a_i \in \sqrt{\mathfrak{a}}$

$$\begin{array}{ccc} K & \hookrightarrow & L \\ \cup & & \cup \\ A & \supset & \mathfrak{a} \ni \alpha \end{array}$$

pf. Let E be an extension field of L s.t. $p(x) = x^n + \cdots + a_n$ splits in E

e.g. take E to be the splitting field of $p(x)$ over L

α integral over $\mathfrak{a} \Rightarrow \exists$ a monic poly $f \in A[x]$ whose non-leading coefficients lie in $\mathfrak{a}$ s.t. $f(\alpha) = 0$ $f \in A[x] \subseteq K[x]$ $p(x) = \min(\alpha, K)$

$\Rightarrow p(x) \mid f(x)$ in $K[x] \Rightarrow \exists$ a monic poly $q(x) \in K[x]$ s.t. $f(x) = p(x) q(x)$ in $K[x]$

$p(x) = (x - \alpha)(x - \alpha_2) \cdots (x - \alpha_n)$ $\alpha_2, \cdots, \alpha_n \in E \Rightarrow f(\alpha_2) = \cdots = f(\alpha_n) = 0$

$\Rightarrow \alpha_i$ is integral over $\mathfrak{a}$ $\forall i \Rightarrow$ the non-leading coefficients of $p(x)$: $a_1, \cdots, a_n$ are integral over $\mathfrak{a}$

On the other hand $a_1, \cdots, a_n \in K$ (elementary symmetric polys in $\alpha_1, \alpha_2, \cdots, \alpha_n$)

A is integrally closed in K $\Rightarrow$ (Lemma) $a_1, \cdots, a_n \in \sqrt{\mathfrak{a}} \subseteq A$

pf of Thm 2.

Let $\mathfrak{p}' \subseteq \mathfrak{p}$ be prime ideals in A, $\mathfrak{q}$ a prime ideal in B lying over $\mathfrak{p}$

Goal: Find a $\mathfrak{q}' \in \mathrm{Spec}\, B$ s.t. $\mathfrak{q}' \subseteq \mathfrak{q}$ and $\mathfrak{q}'^c = \mathfrak{p}'$

$\Leftrightarrow \mathfrak{q}' \cap (B - \mathfrak{q}) = \phi \Leftrightarrow \mathfrak{p}' B \subseteq \mathfrak{q}'$ and $\mathfrak{q}' \cap \complement(A - \mathfrak{p}') = \phi$

Fact. $\mathfrak{q}' \cap (B - \mathfrak{q}) = \phi$ $\mathfrak{q}' \cap \complement(A - \mathfrak{p}') = \phi \Leftrightarrow \mathfrak{q}' \cap (B - \mathfrak{q})(A - \mathfrak{p}') = \phi$

It suffices to show that $\mathfrak{p}' B \cap (A - \mathfrak{p}')(B - \mathfrak{q}) = \phi$

We prove by contradiction. Suppose $b \neq st$ where $b \in \mathfrak{p}' B$ $s \in A - \mathfrak{p}'$, $t \in B - \mathfrak{q}$

$\mathrm{Frac}(A) = K \hookrightarrow L = \mathrm{Frac}(B)$ B integral over $A \Rightarrow b$ is integral over $\mathfrak{p}'$

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \cap & & \cap \\ K & \hookrightarrow & L \end{array}$$

$\Rightarrow$ minimal poly of b over k lie in $\sqrt{P'} = P'$

the non-leading coeff of

$t \in B - q \Rightarrow t$ integral over $A \Rightarrow$ the non-leading coeff of minimal poly of t over k lie in A

Write the minimal poly of t over k as $g(x) = x^r + v_1 x^{r-1} + \cdots + v_r$ with $v_i \in A$

Then the minimal poly of $b = st$ over k is

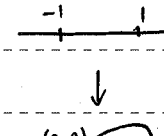
$f(x) = x^r + u_1 x^{r-1} + \cdots + u_r$ where $u_i = v_i s^i$

$u_i \in P'$, $s \in A - P' \Rightarrow v_r \in P'$ $\overset{t^r \in P'B}{\Rightarrow} t \in \sqrt{P'B}$ $\Rightarrow$

$P' \subseteq P$, $P = q^c \Rightarrow P'B \subseteq PB \subseteq q \Rightarrow \sqrt{P'B} \subseteq q \Rightarrow t \in q$ contradiction.

See also Eisenbud P289

e.g. k field char $k \neq 2$ $k[x,y]/(y^2 - x^3 - x^2) \hookrightarrow k[t]$ $t = \frac{y}{x}$



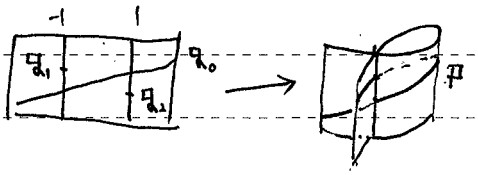
$x \mapsto t^2 - 1$

$y \mapsto t(t^2-1)$

Satisfies going down

$k[x,y,T]/(y^2 - x^3 - x^2) \hookrightarrow k[t,T]$ does not satisfy going down

Eisenbud P239



§4. Krull dimension

Def. Let A be a non-zero ring. The **Krull dimension** of A is the supremum of ~~the~~ lengths of chains of ~~direc~~ distinct prime ideals in A

$\dim A = \sup\{r : \exists \text{ a chain of prime ideals } P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_r\}$

If $A = 0$ then we define $\dim A = -\infty$

If $A \neq 0$ then $\dim A \geq 0$ or $+\infty$

e.g. $\dim A = 0 \Leftrightarrow A \neq 0$ and every prime ideal of A is maximal

k field $\dim k = 0$ $\quad \forall n \geq 2$ $\dim \mathbb{Z}/n\mathbb{Z} = 0$

Let A be a ring and let $m_1, \cdots, m_k$ $(k \geq 1)$ be maximal ideals of A (not necessarily distinct). Then $\dim A/m_1 \cdots m_k = 0$

$m_1 \cdots m_k \subseteq m_1 \Rightarrow A/m_1 \cdots m_k \neq 0$ $\quad m_1 \cdots m_k \subseteq P \Rightarrow m_i \subseteq P$ for some i

$\Rightarrow P = m_i$ for some i $\Rightarrow A/m_1 \cdots m_k$ has at most k prime ideals $m_i/m_1 \cdots m_k$ each of which is maximal

In particular, if m is maximal and $k \geq 1$ then $\dim A/m^k = 0$

e.g. Let A be a PID that is not a field. Then $\dim A = 1$

$A = \mathbb{Z}$, $k[x]$ (k, field)

Prop. Let A be a ring. Then

(1) $\dim A + 1 \leq \dim A[x] \leq 2\dim A + 1$

(2)* If A is Noetherian, then $\dim A[x] = \dim A + 1$

pf (1). Let $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_r$ be a prime chain in A

Then $P_0[x] \subsetneq P_1[x] \subsetneq \cdots \subsetneq P_r[x] \subsetneq (P_r, x)$ is a prime chain in $A[x]$ $\Rightarrow \dim A[x] \geq \dim A + 1$

Recall $A[x]/P[x] \simeq (A/P)[x]$

$A[x]/(P,x) \simeq A/P$

Recall that if $f: A \to B$ is a ring map and $P \in \operatorname{Spec} A$ then there is a inclusion-preserving bijection of sets $\{Q \in \operatorname{Spec} B : Q^c = P\} \xleftrightarrow{1:1} \operatorname{Spec} k(P) \underset{A}{\otimes} B$

$A_P/PA_P \underset{A}{\otimes} B = B_P/PB_P$

Set $B = A[x]$

$\{Q \in \operatorname{Spec} A[x] : Q \cap A = P\} \xleftrightarrow{1:1} \operatorname{Spec} k(P) \underset{A}{\otimes} A[x] = \operatorname{Spec} k(P)[x]$

$\dim k(P)[x] = 1 \Rightarrow$ There does not exist prime ideals $Q_0 \subsetneq Q_1 \subsetneq Q_2$ in $A[x]$ s.t. $Q_0 \cap A = Q_1 \cap A = Q_2 \cap A = P \Rightarrow \dim A[x] \leq 2\dim A + 1$

e.g. $\dim \mathbb{Z}[x] = 2$ $\quad 0 \subsetneq (x) \subsetneq (x, 2) \Rightarrow \dim \mathbb{Z}[x] \geq 2$

($\operatorname{ht} P = 1 \Rightarrow P = (p)$ or $(p(x))$ where p is prime, $p(x)$ is a primitive integral poly irreducible over $\mathbb{Q}$)

$\dim k[x_1, \cdots, x_n] = n$ $\quad 0 \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \cdots x_n) \Rightarrow \dim k[x_1, \cdots, x_n] \geq n$

$\dim k[x_1,\cdots,x_n,\cdots]=+\infty$

Nagata showed that there exists a Noetherian domain of infinite dimension

Prop. If $A\subseteq B$ is an integral extension of rings, then $\dim A=\dim B$

pf. Let $P_0\subsetneq P_1\subsetneq\cdots\subsetneq P_r$ be a prime chain in A

Lying over $\Rightarrow \exists\, Q_0\in \operatorname{Spec} B$ s.t. $Q_0^c=P_0$

Going up $\Rightarrow \exists$ a prime chain $Q_0\subseteq Q_1\subseteq\cdots\subseteq Q_r$ in B s.t. $Q_i^c=P_i$.

$P_i\subsetneq P_{i+1}\Rightarrow Q_i\subsetneq Q_{i+1}$ Hence $\dim B\geq\dim A$

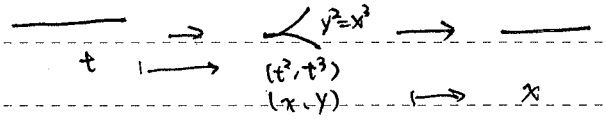
Let $Q_0\subsetneq Q_1\subsetneq\cdots\subsetneq Q_r$ be a prime chain in B

Then $Q_0^c\subseteq Q_1^c\subseteq\cdots\subseteq Q_r^c$ is a prime chain in A

Incomparability $\Rightarrow Q_i^c\subsetneq Q_{i+1}^c$ Hence $\dim A\geq\dim B$

e.g. $k[x]\xhookrightarrow[\text{integral}]{A=} k[x,y]/(y^2-x^3)\xhookrightarrow[\text{Normalization}]{\text{is}} k[t]$ $\quad\dim A=1$

$k[x]\oplus k[x]\bar{y}$ $\quad\bar{y}$ integral over $k[x]$



d integer $\dim \mathbb{Z}[\sqrt{d}]=1$ because $\mathbb{Z}\subseteq\mathbb{Z}[\sqrt{d}]$ is an integral extension

§5. Transcendence basis, transcendence degree

Def. Let k be a field $\{x_i\}_{i\in I}$ a set of indeterminates

Let $k[\{x_i\}_{i\in I}]$ be the polynomial algebra in the x_i over k

Let $k(\{x_i\}_{i\in I})$ be the fraction field of $k[\{x_i\}_{i\in I}]$

A purely transcendental extension of k is any extension K/k s.t. K is isomorphic to $k(\{x_i\}_{i\in I})$ for some set I

Let K/k be a field extension and B be a subset of K. We say B is algebraically independent over k if the k-algebra homomorphism $\sigma: k[\{x_b\}_{b\in B}]\to K$, $x_b\mapsto b$ is injective. That is for any $n\geq 1$ any nonzero poly $f(t_1,\cdots t_n)$ in $k[x_{t_1},\cdots,x_{t_n}]$ and any subset $\{b_1,\cdots b_n\}$ of B we have $f(b_1,\cdots,b_n)\neq 0$ in K.

If B is algebraically indep over k, then σ can be extended to a k-embedding $k(\{x_b\}_{b\in B}) \hookrightarrow K$ which induces a k-isom from $k(\{x_b\}_{b\in B})$ to the subfield $k(B)$ of K ~~generated~~ by B over k.

In particular, $k(B)/k$ is a purely transcendental extension

Let K/k be a field extension. A transcendence basis of K over k is a subset B of K s.t. B is algebraically indep over k and K is algebraic over $k(B)$ $\quad k \subseteq k(B) \subseteq K$

Thm Let K/k be a field extension. Then $\exists$ a transcendence basis of K over k exists; Moreover, any two transcendence basis of K/k have the same cardinality

Pf. Observe that a transcendence basis of K/k is the same thing as a maximal algebraically independent subset of K over k.

Use Zorn's lemma to show that it exists.

The transcendence degree of K over k, denoted by $\mathrm{tr.deg}_k K$ is the cardinality of any transcendence basis of K/k

e.g. $\mathrm{tr.deg}_k K = 0 \Leftrightarrow K/k$ is algebraic

$\mathrm{tr.deg}_k K \geq 1 \Leftrightarrow K/k$ is transcendental

e.g. Let k be a field x an indeterminate. Then $k(x)/k$ is a purely transcendental extension of transcendence degree 1 with transcendence basis x

Since x^2 is transcendental over k and $k(x)$ is algebraic over $k(x^2)$ x^2 is also a transcendence basis of $k(x)/k$

Recall that $\forall f(x) \in k(x)\backslash k$ $f(x)$ is transcendental over k and x is algebraic over $k(f(x))$ $\quad k \subsetneq k(f(x)) \subsetneq k(x) \Rightarrow f(x)$ is a transcendence basis of $k(x)/k$

e.g. $f(x) = \frac{x^2}{x+1}$ $\quad x^2 - f(x)x - f(x) = 0$

e.g. $\mathbb{Q}(x,\sqrt{2})/\mathbb{Q}$ $\quad \mathbb{Q} \subseteq \mathbb{Q}(x) \subseteq \mathbb{Q}(x,\sqrt{2}) \Rightarrow x$ is a transcendence basis of $\mathbb{Q}(x,\sqrt{2})/\mathbb{Q}$

$\mathrm{trdeg}_{\mathbb{Q}}\mathbb{Q}(x,\sqrt{2})=1$ Claim. $\mathbb{Q}(x,\sqrt{2})$ is not a purely transcendental extension of $\mathbb{Q}$

If it is then $\mathbb{Q}(x,\sqrt{2})\simeq\mathbb{Q}(t)$ Since $\sqrt{2}\in\mathbb{Q}(x,\sqrt{2})\setminus\mathbb{Q}$ and $\sqrt{2}$ is algebraic over $\mathbb{Q}$. We have a contradiction.

Let $A=\mathbb{C}[x,y]/(y^2-x^3+x)$ $K=\mathrm{Frac}(A)$

$\mathbb{C}[x]\hookrightarrow\mathbb{C}[x,y]/(y^2-x^3+x)$ integral

is

$\mathbb{C}[x]\oplus\mathbb{C}[x]\bar{y}$

$\mathbb{C}\hookrightarrow\mathbb{C}(x)\hookrightarrow K=\mathrm{Frac}(\mathbb{C}[x,y]/(y^2-x^3+x))$ algebraic

$\mathrm{trdeg}_{\mathbb{C}}K=1$ $\mathbb{C}$ alg clsd $\Rightarrow\forall f\in K\setminus\mathbb{C}$ f is transcendental over $\mathbb{C}$

$K/\mathbb{C}$ is not purely transcendental

Noether's normalization lemma. Let k be a field and let $A\neq 0$ be a f.g k-algebra. Then $\exists\, y_1,\cdots,y_d\in A$ alg indep over k s.t. A is module finite over $k[y_1,\cdots,y_d]$ ($\Leftrightarrow$ integral) (finite type + integral $\Leftrightarrow$ finite) ($d=\dim A$)

Thm 1. Let k be a field and let $\mathfrak{a}\subsetneq k[x_1,\cdots,x_n]$ be a proper ideal of $k[x_1,\cdots,x_n]$

Then $\exists\, y_1,\cdots,y_n\in k[x_1,\cdots,x_n]$ alg ~~closed~~ indep over k s.t.

(1) $k[x_1,\cdots,x_n]$ is module finite over $k[y_1,\cdots,y_n]$

(2) $\mathfrak{a}\cap k[y_1,\cdots,y_n]=(y_1,\cdots,y_h)$ for some $h\in\{0,1,\cdots,n\}$

Another pf. of Zariski's lemma using NNL.

~~$\exists y_1,\cdots$~~ (k field A f.g k-algebra. Then A field $\Rightarrow A/k$ finite field ext.)

$\exists\, y_1,\cdots,y_d\in A$ alg indep over k s.t. A is integral over $k[y_1,\cdots,y_d]$

A field $\Rightarrow k[y_1,\cdots,y_d]$ field $\Rightarrow d=0\Rightarrow A$ is finite over k.

Thm 1. $\Rightarrow$ NNL. $A=k[x_1,\cdots,x_n]/\mathfrak{a}$ $k[y_1,\cdots,y_n]\hookrightarrow k[x_1,\cdots,x_n]$ finite

$\Rightarrow k[y_1,\cdots,y_n]/\mathfrak{a}\cap k[y_1,\cdots,y_n]\hookrightarrow k[x_1,\cdots,x_n]/\mathfrak{a}$ finite

$k[\bar{y}_{h+1},\cdots,\bar{y}_n]$ $\quad$ A

MEMO NO: _____
DATE: / /

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e.g. $A=k[x,y]/(y^2-x^3)$ $u=\bar{x}$, $V=\bar{y}$ $\Rightarrow$ u (V) transcendental over k and finite over $k[u]$ ($k(V)$)

$k[u] \hookrightarrow k[x,y]/(y^2-x^3)$

$\mathfrak{a} = (y^2-x^3) \subseteq k[x,y]$

$(y') \subseteq k[x,y']$ $y'=y^2-x^3$

$(x,y) \in \mathbb{A}_k^2 \to y^2-x^3=0$ projection onto x-axis

$\downarrow$

$(x, y^2-x^3) \in \mathbb{A}_k^2 \to y=0$

pf. of Thm1 By induction on n

$n=1$ If $\mathfrak{a}=0$ then set $y=x$, $h=0$

If $\mathfrak{a}=(f(x)) \neq 0$, then set $y=f(x)$ $h=1$ $(\mathfrak{a} \cap k[y] = (y))$

Suppose $n>1$ and thm holds for $n-1$

If $\mathfrak{a}=0$ then set $y_i = x_i$ $(1 \le i \le n)$ $h=0$

If $\mathfrak{a} \neq 0$ then choose any nonzero poly $f(x_1, \cdots, x_n) \in \mathfrak{a}$

Claim1. For suitable choices of l_i $(2 \le i \le n)$, the ring $k[x_1, \cdots, x_n]$ is module finite over subring $k[f, x_2 - x_1^{l_2}, \cdots, x_n - x_1^{l_n}]$

($\Leftrightarrow$ integral)

Claim2. Set $y_1 = f$, $t_i = x_i - x_1^{l_i}$ $(2 \le i \le n)$ $C = k[y_1, t_2, \cdots, t_n]$

Then $y_1, t_2, \cdots, t_n$ alg indep over k and $k[x_1, \cdots, x_n]$ is module finite over $k[y_1, t_2, \cdots, t_n]$

pf. $k[y_1, t_2, \cdots, t_n] \hookrightarrow k[x_1, \cdots, x_n]$ finite

$\Rightarrow$ $\mathrm{Frac}(k[y_1, t_2, \cdots, t_n]) \hookrightarrow \mathrm{Frac}(k[x_1, \cdots, x_n]) = k(x_1, \cdots, x_n)$ algebraic (k)

$\Rightarrow$ $\mathrm{trdeg}_k \mathrm{Frac}(k[y_1, t_2, \cdots, t_n]) = \mathrm{trdeg}_k k(x_1, \cdots, x_n) = n$

$\Rightarrow$ $y_1, t_2, \cdots, t_n$ alg indep $\mathfrak{a} \cap k[t_2, \cdots, t_n]$

$y_1 C \subseteq \mathfrak{a} \cap C \Rightarrow \mathfrak{a} \cap C = y_1 C \oplus (\mathfrak{a} \cap C \cap k[t_2, \cdots, t_n])$

$(k[y_1, t_2, \cdots, t_n] = (y_1) \oplus k[t_2, \cdots, t_n])$

$\mathfrak{a}$ proper in $k[x_1, \cdots, x_n] \Rightarrow \mathfrak{a} \cap k[t_2, \cdots, t_n]$ proper in $k[t_2, \cdots, t_n]$

Apply induction hypothesis to $(k[t_2, \cdots, t_n], \mathfrak{a} \cap k[t_2, \cdots, t_n])$

$\Rightarrow \exists\, y_2, \cdots, y_n \in k[t_2, \cdots, t_n]$ alg indep over k s.t. $k[t_2, \cdots, t_n]$ is module finite over $k[y_2, \cdots, y_n]$ and $\mathfrak{a} \cap k[y_2, \cdots, y_n] = (y_2, \cdots, y_h)$ for some h

check that $y_1,\cdots,y_n$ alg indep over k

$k[x_1,\cdots x_n]$ is module finite over $k[y_1,\cdots,y_n]$

$\mathfrak{a}\cap k[y_1,\cdots,y_n]=(y_1,\cdots,y_h)$

Strengthened NNL: Let k be a field, A a f.g k-algebra

$\mathfrak{a}_1\subsetneq\mathfrak{a}_2\subsetneq\cdots\mathfrak{a}_r$ a chain of proper ideals in A

Then $\exists\ y_1,\cdots y_d\in A$ alg indep over k s.t.

(1) A is module finite over $k[y_1,\cdots,y_d]$

(2) $\forall i=1,\cdots,r\quad \mathfrak{a}_i\cap k[y_1,\cdots,y_d]=(y_1,\cdots,y_{h_i})$ for some h_i

$(\mathfrak{a}_i\subseteq\mathfrak{a}_{i+1}\Rightarrow(y_1,\cdots y_{h_i})\subseteq(y_1,\cdots,y_{h_{i+1}})\Rightarrow h_i\le h_{i+1})$

Thm2 Let k be a field and let $\mathfrak{a}_1\subsetneq\cdots\subsetneq\mathfrak{a}_r$ be a chain of proper ideals in $k[x_1,\cdots,x_n]$ Then $\exists y_1,\cdots,y_d\in k[x_1,\cdots x_n]$ alg indep over k s.t.

(1) $k[x_1,\cdots x_n]$ is module finite over $k[y_1,\cdots,y_n]$

(2) $\mathfrak{a}_i\cap k[x_1,y_1,\cdots,y_n]=(y_1,\cdots,y_{h_i})$ for some $h_i\in\{0,1,\cdots,n\}$

Thm2 $\Rightarrow$ ~~tut~~ SNNL

pf. of Thm2 By induction on r $r=1$ (thm1)

Apply induction hypothesis to $\mathfrak{a}_1\subsetneq\cdots\subsetneq\mathfrak{a}_{r-1}$ in $k[x_1,\cdots,x_n]$

$\Rightarrow\exists\ y_1,\cdots y_{h_{r-1}}\ t_{h_{r-1}+1},\cdots,t_n\in k[x_1,\cdots,x_n]$ alg indep over k

s.t. $k[x_1,\cdots,x_n]$ is module finite over $k[y_1,\cdots,y_{h_{r-1}},t_{h_{r-1}+1},\cdots,t_n]$

$\forall$ ~~i~~ $i=1,\cdots,r-1\quad \mathfrak{a}_i\cap k[y_1,\cdots,y_{h_{r-1}},t_{h_{r-1}+1},\cdots,t_n]=(y_1,\cdots,y_{h_i})$

$\mathfrak{a}_{r-1}\subsetneq\mathfrak{a}_r\Rightarrow(y_1,\cdots,y_{h_{r-1}})\subsetneq\mathfrak{a}_r\cap k[\cdots]\Rightarrow\mathfrak{a}_r\cap k[\cdots]=(y_1,\cdots,y_{h_{r-1}})$

Apply Thm1 to $\left(k[t_{h_{r-1}+1},\cdots,t_n],\ \mathfrak{a}_r\cap k[t_{h_{r-1}+1},\cdots,t_n]\right)$ $\oplus(\mathfrak{a}_r\cap k[t_{h_{r-1}+1},\cdots,t_n])$

Thm3. Let k be a field. Then $\dim k[x_1,\cdots,x_n]=n$

pf. It suffices to show $\dim k[x_1,\cdots,x_n]\le n$

Let $P_0\subsetneq P_1\subsetneq\cdots\subsetneq P_r$ be a prime chain in $k[x_1,\cdots,x_n]$

Thm2 $\Rightarrow\exists\ y_1,\cdots,y_n\in k[x_1,\cdots,x_n]$ s.t. (1) $k[x_1,\cdots,x_n]$ is integral over $k[y_1,\cdots,y_n]$

alg indep over k

(2) $P_i \cap k[y_1, \cdots, y_n] = (y_1, \cdots, y_{h_i})$ $i = 0, 1, \cdots, r$

Incomparability $\Rightarrow$ $(y_1, \cdots, y_{h_i}) \subsetneq (y_1, \cdots, y_{h_{i+1}}) \Rightarrow h_i < h_{i+1}$

$0 \le h_0 < h_1 < \cdots < h_r \le n \Rightarrow r \le n$

~~Corr.~~ Rmk. Rmk. $k[x_1, \cdots, x_n]/\mathfrak{o}_i$ integral over $k[y_1, \cdots, y_n]/\mathfrak{o}_i \cap k[y_1, \cdots, y_n] \Rightarrow n - h_i = \dim k[x_1, \cdots, x_n]/\mathfrak{o}_i$

Thm4 Let k be a field A a f.g. k-algebra

If A is an integral domain, then $\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A)$

pf. NNL $\Rightarrow$ $\exists y_1, \cdots, y_d \in A$ alg indep over k s.t.

A is module finite over $k[y_1, \cdots, y_d]$ (integral)

$\Rightarrow \operatorname{Frac}(A)$ is algebraic over $k(y_1, \cdots, y_d)$

$\Rightarrow \operatorname{trdeg}_k \operatorname{Frac}(A) = \operatorname{trdeg}_k k(y_1, \cdots, y_d) = d$

On the other hand, $\dim A = \dim k[y_1, \cdots, y_d] = d$

Hence $\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A)$

~~Thm5~~ Thm5. Let k be a field, A a f.g. k-algebra If A is an integral domain, ~~then~~ and $\dim A = d$. Then every maximal chain of prime ideals in A has length d.

pf. Let $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_r$ be a maximal chain of prime ideals in A

SNNL $\Rightarrow$ $\exists x_1, \cdots, x_d \in A$ alg indep over k s.t. A is integral over $k[y_1, \cdots, y_d]$ and $P_i \cap k[y_1, \cdots, y_d] = (y_1, \cdots, y_{h_i})$ for some h_i

A integral domain $\Rightarrow P_0 = 0 \Rightarrow h_0 = 0$

P_r maximal + Maximality $\Rightarrow (y_1, \cdots, y_{h_r})$ is maximal in $k[y_1, \cdots, y_d]$

$\Rightarrow h_r = d$

Incomparability $\Rightarrow h_i < h_{i+1}$ Altogether, we have $0 = h_0 < h_1 < \cdots < h_r = d$

Goal $r = d$ If $r < d$ then $\exists i \in \{0, \cdots, r-1\}$ s.t. $h_i + 1 < h_{i+1}$

$P_i \subsetneq P_{i+1} \subseteq A$ ← $k[y_1,\cdots y_d]$

$(y_1,\cdots y_{h_i}) \subsetneq (y_1,\cdots y_{h_{i+1}}) \subseteq k[y_1,\cdots y_d]$

$\leq (y_1,\cdots y_{h_i+1})$ ∪ℵ

$k[y_1,\cdots y_d]/(y_1,\cdots y_{h_i}) \subseteq A/P_i$ integral ext of integral domains

$k[\overline{y_{h_i+1}},\cdots,\overline{y_d}]$ integrally closed

Going down $Q/P_i \subsetneq P_{i+1}/P_i \subseteq A/P_i$

$\Rightarrow \exists Q$:

$(y_1,\cdots y_{h_i+1})/(y_1,\cdots y_{h_i}) \subsetneq (y_1,\cdots y_{h_{i+1}})/(y_1,\cdots y_{h_i}) \subseteq k[y_1,\cdots y_d]/(y_1,\cdots y_{h_i})$

$\overline{0}$

$P_i \subsetneq Q \subsetneq P_{i+1}$ contradiction

Rmk. Thm 4 fails if A is not an integral domain

$A = k[x,y,z]/(xz,yz)$ $(x,y)/(xz,yz) \subsetneq (x,y,z)/(xz,yz)$ maximal

$\dim A = 2$ $(z)/(xz,yz) \subsetneq (y,z)/(xz,yz) \subsetneq (x,y,z)/(xz,yz)$

